

Supporting Information File for the Ocean-TP Manuscript

Document scope. This file is prepared as supplementary explanatory material for manuscript submission. It records supporting analyses, fitting and evaluation results, equation definitions and term-level interpretations that provide additional context for the main text. All displayed numerical values, tabulated metrics and incremental changes presented in this file are taken from the fitting and evaluation records used for the manuscript. Additional supporting sections can be appended as the manuscript develops.

1 Supplementary Methods: Evaluation Metrics

For all quantitative analyses, T_i denotes the observed temperature, $\hat{T}_i$ denotes the reconstructed or predicted temperature, $e_i = \hat{T}_i - T_i$ denotes the residual error, and N denotes the number of quality-controlled T-P samples in the evaluated subset. Temperature errors are reported in °C. The primary accuracy metrics were mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination (R^2):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |e_i|, \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}, \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N e_i^2}{\sum_{i=1}^N (T_i - \bar{T})^2}, \quad (3)$$

where $\bar{T}$ is the mean observed temperature within the evaluated subset. These metrics were computed after converting all predictions back to physical temperature units.

For the Top- K analytical structure search, candidate functions were ranked using a weighted mean absolute error (WMAE),

$$\text{WMAE} = \frac{\sum_{i=1}^N w_i |e_i|}{\sum_{i=1}^N w_i}, \quad (4)$$

where w_i is the depth-regime sampling weight assigned to sample i . WMAE was used only for basis selection under uneven depth sampling. The reported benchmark, seasonal, sparse-anchor and prediction results use the unweighted MAE, RMSE and R^2 metrics unless otherwise stated. The vertical-gradient residual diagnostics reported in the seasonal analysis summarize the magnitude of the gradient-consistency diagnostic $\mathcal{R}_{\text{grad}}$ used to monitor residual smoothness, with “Grad. resid. mean” denoting mean($|\mathcal{R}_{\text{grad}}|$) and “Grad. resid. 95%” denoting its 95th percentile.

For depth-resolved analyses, metrics were computed within 1 m depth bins or within a depth window $\mathcal{D}$ by restricting the evaluated subset to samples whose pressure fell inside that range. When 1 m bins were aggregated into a broader depth window, sample counts were used as weights. For a depth-bin metric m_b and bin sample count n_b , the count-weighted MAE is

$$\text{MAE}_{\text{cw}}(\mathcal{D}) = \frac{\sum_{b \in \mathcal{D}} n_b \text{MAE}_b}{\sum_{b \in \mathcal{D}} n_b}. \quad (5)$$

Count-weighted RMSE was computed as the corresponding root of the sample-count-weighted squared bin RMSE. The double-exponential depth-decay fit was evaluated using the R^2 between the fitted curve and the 1 m MAE profile.

For chronological prediction analyses, daily metrics were computed from all samples observed on each date. If $\mathcal{S}_d$ is the set of samples on day d , daily MAE, RMSE and R^2 were computed by applying the same formulas to $\mathcal{S}_d$. The 7-day running mean was calculated as the arithmetic mean of daily MAE values within the corresponding seven-day window, and the optimal 7-day interval was the consecutive window with the lowest mean daily MAE. Stable days were defined by $\text{MAE} \leq 0.10^\circ\text{C}$ and $\text{RMSE} \leq 0.20^\circ\text{C}$. Error distributions were summarized using the mean, median, interquartile range and 95th percentile as appropriate. To test whether daily prediction error was driven by sampling density, Pearson’s correlation coefficient was computed between daily MAE and daily sample count.

2 Analytical Complexity Path

Supporting-file note. This section summarizes the candidate-form comparison used to select a parsimonious pressure-coordinate analytical backbone for Ocean-TP. The reported values are organized as MAE / RMSE / R^2 , and the displayed equations record the fitted coefficient or basis form associated with each comparison entry. If a later manuscript version uses different coefficient precision, it should cite the final fitted coefficient record.

2.1 Naming Rule and Model Catalogue

Model names use a short code that records the dominant functional form and active-term count. The term count k includes the intercept.

Table 1: Catalogue of candidate analytical structures.

k	Code	Representative name	Data basis	Metrics	Notes
var.	SAMP-PF	Sampled direct polyfit	Qualitative profile check	Full-data bias, unstable extrapolation	Based on $\leq 100k$ direct polyfit and included as a qualitative profile reference.
1	CONST1	Constant-only baseline	Evaluation metric	0.6420 / 0.8865 / 0.0000	No pressure-coordinate structure.
2	LIN2	Linear pressure trend	Evaluation metric	0.3375 / 0.5530 / 0.6110	Broad monotonic cooling only.
3	POLY2	Quadratic polynomial	Evaluation metric	0.2118 / 0.3915 / 0.8050	Captures only coarse vertical curvature.
4	MIX4S	Four-term mixed seed	Evaluation metric	0.1736 / 0.3442 / 0.8492	First compact nonlinear mixed form.
5	POLY4	Quartic polynomial	Evaluation metric	0.1481 / 0.3120 / 0.8761	Improves curvature but remains polynomial.
6	ZPOLY5	Standardized fifth-order polynomial	Evaluation metric	0.0533 / 0.1172 / 0.9825	Strong normalized polynomial anchor.
7	ZPOLY6	Standardized sixth-order polynomial	Evaluation metric	0.0540 / 0.1172 / 0.9825	Slightly worse than ZPOLY5 in MAE.
8	CHEB8	Chebyshev T_0 – T_7	Metric and equation record	0.0548 / 0.1164 / 0.9828	Orthogonal polynomial basis.
8	RELU8	Piecewise ReLU spline	Metric and equation record	0.0541 / 0.1160 / 0.9829	Thresholded piecewise linear basis.
8	SIG8	Sigmoid-transition basis	Metric and equation record	0.0520 / 0.1127 / 0.9838	Smooth transition basis.
8	MIX8A	Mixed features, version A	Evaluation metric	0.0513 / 0.1121 / 0.9840	Standardized mixed polynomial-log form.
8	MIX8B	Mixed features, version B	Evaluation metric	0.05107 / 0.11189 / 0.98407	Current recommended eight-term backbone.
9	PL9	Over-extended polynomial-log form	Evaluation metric	0.05128 / 0.11212 / 0.98402	Extra polynomial-log interaction does not improve.
10	SIG10	Over-parameterized sigmoid-transition form	Evaluation metric	0.05098 / 0.11178 / 0.98411	Slightly better MAE, but higher complexity.

2.2 Formula Notation and Term-Level Interpretation

The equations below should be read as analytical pressure-coordinate approximations fitted or evaluated against the observed temperature profile. Each candidate uses a small set of basis terms to represent different parts of the vertical structure, and the comparison is based on both evaluation error and parsimony. The coefficient signs and magnitudes are therefore interpreted as fitted contributions within the complete formula, not as isolated causal effects.

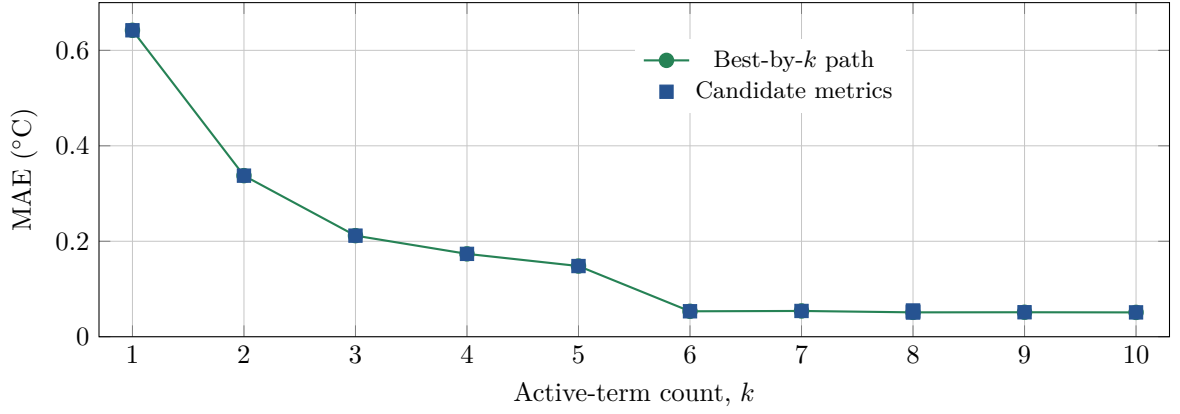


Figure 1: Complexity-path summary. Anchor metrics cluster around the eight-term structures, while the 10-term alternative is only marginally better than MIX8B.

Symbol or term	Definition	Interpretation in the analytical profile
$\hat{T}(P)$	Fitted temperature at pressure P	Model output used to reconstruct the temperature–pressure relationship.
P	Pressure in dbar	Primary vertical coordinate. Larger P corresponds to deeper water.
k	Active-term count including the intercept	Complexity index used to compare candidate analytical structures.
x, z	Standardized pressure coordinates	Center and scale pressure to improve numerical conditioning for polynomial and mixed polynomial terms.
u	Rescaled pressure coordinate on approximately $[-1, 1]$	Coordinate used by the Chebyshev basis to reduce high-order polynomial instability.
$\ln(P)$ and $\sqrt{P}$	Smooth nonlinear pressure terms	Capture broad vertical curvature that cannot be represented by a purely linear trend.
$\exp(-P/\tau)$	Exponential decay term with scale τ	Represents rapidly weakening near-surface or intermediate-depth contributions.
$1/(P + c)$	Shifted reciprocal pressure term	Adds strong shallow-pressure curvature while remaining bounded by the positive offset c .
$(P - \kappa)_+$	ReLU hinge with knot κ	Allows a slope change after a prescribed pressure threshold.
$L_{\kappa,s}(P)$	Sigmoid transition centered at κ with smoothness s	Represents a smooth layer transition such as thermocline-like curvature.
$T_i(u)$	i th Chebyshev polynomial mode	Provides numerically stable global polynomial modes over the rescaled pressure interval.

In this catalogue, polynomial models mainly test how far global curvature alone can explain the profile. Mixed-feature models combine broad background terms with decay and reciprocal terms, making them more suitable for profiles with shallow curvature and deep stabilization. Piecewise and sigmoid-transition models test whether explicit layer-transition terms improve the fit enough to justify the additional complexity.

2.3 Equation Definitions

Throughout, P is pressure in dbar. For standardized polynomial forms, $x = (P - 1641.2335)/990.7445$. For orthogonal and transition-basis equations, $u = 2(P - 100)/(4500 - 100) - 1$, $(a)_+ = \max(a, 0)$ and $L_{\kappa,s}(P) = \{1 + \exp[-(P - \kappa)/s]\}^{-1}$.

CONST1, $k = 1$.

$$\hat{T}_{\text{CONST1}}(P) = 2.204.$$

Explanation. This is the no-structure baseline. It represents only the mean thermal level and ignores all pressure-dependent stratification, so it defines the lower bound for useful profile reconstruction.

LIN2, $k = 2$.

$$\hat{T}_{\text{LIN2}}(P) = 4.400 - 7.05 \times 10^{-4}P.$$

Explanation. The linear trend adds a monotonic cooling-with-pressure component. It can describe the broad decrease of temperature with depth, but it cannot represent mixed-layer curvature, thermocline transition or deep asymptotic behaviour.

POLY2, $k = 3$.

$$\hat{T}_{\text{POLY2}}(P) = 5.12742481 - 2.64606596 \times 10^{-3}P + 5.35595691 \times 10^{-7}P^2.$$

Explanation. The quadratic term adds a single global curvature to the pressure trend. This improves on the linear model but forces the whole water column to share one smooth curvature, which is too rigid for layered ocean thermal structure.

MIX4S, $k = 4$.

$$\hat{T}_{\text{MIX4S}}(P) = 7.23196288 - 0.83859309 \ln(P) + 0.02043606\sqrt{P} + 1.68598290 \exp(-P/1000).$$

Explanation. This seed model introduces physically interpretable nonlinear pressure components. The logarithmic and square-root terms describe broad profile curvature, while the exponential decay term gives the model a first mechanism for upper-ocean attenuation.

POLY4, $k = 5$.

$$\begin{aligned} \hat{T}_{\text{POLY4}}(P) = & 5.05404204 - 2.26920822 \times 10^{-3}P - 3.96617497 \times 10^{-8}P^2 \\ & + 3.37214976 \times 10^{-10}P^3 - 6.71345918 \times 10^{-14}P^4. \end{aligned}$$

Explanation. The quartic polynomial increases flexibility by adding higher-order global curvature. It can reduce bias relative to low-order polynomials, but the basis remains purely algebraic and does not separately encode near-surface transitions or deep stabilization.

ZPOLY5, $k = 6$.

$$\begin{aligned} \hat{T}_{\text{ZPOLY5}}(P) = & 2.20411614 - 0.86369338x + 0.41811779x^2 - 0.08912643x^3 \\ & - 0.00667122x^4 + 0.00535626x^5. \end{aligned}$$

Explanation. Standardization improves numerical conditioning and allows the fifth-order polynomial to fit the main profile curvature more effectively. Its strong performance shows that normalization matters, although the representation is still a flexible empirical curve rather than a structured pressure-coordinate decomposition.

ZPOLY6, $k = 7$.

$$\begin{aligned} \hat{T}_{\text{ZPOLY6}}(P) = & 2.20278856 - 0.87714145x + 0.42640403x^2 - 0.06258684x^3 \\ & - 0.01749463x^4 - 0.00448305x^5 + 0.00410846x^6. \end{aligned}$$

Explanation. The sixth-order standardized polynomial adds one more curvature degree of freedom. Its similar or slightly worse MAE compared with ZPOLY5 indicates that adding polynomial order alone does not guarantee better profile reconstruction.

CHEB8, $k = 8$.

$$\begin{aligned} \hat{T}_{\text{CHEB8}}(P) = & 2.206 - 0.835T_1(u) + 0.421T_2(u) - 0.090T_3(u) \\ & - 0.008T_4(u) + 0.006T_5(u) - 0.003T_6(u) + 0.002T_7(u). \end{aligned}$$

Explanation. The Chebyshev basis represents the full-depth profile with orthogonal polynomial modes. This improves numerical stability over raw high-order polynomials, but the terms remain global modes and are not directly tied to specific ocean-layer transitions.

RELU8, $k = 8$.

$$\begin{aligned} \hat{T}_{\text{RELU8}}(P) = & 5.18 - 1.05 \times 10^{-3}P + 1.80 \times 10^{-3}(P - 200)_+ \\ & - 1.20 \times 10^{-3}(P - 500)_+ + 4.50 \times 10^{-4}(P - 1000)_+ \\ & - 1.80 \times 10^{-4}(P - 1800)_+ + 8.00 \times 10^{-5}(P - 2800)_+ \\ & - 3.00 \times 10^{-5}(P - 3800)_+. \end{aligned}$$

Explanation. The ReLU spline form divides the pressure axis into piecewise linear regimes. It is useful for approximating layer breaks and slope changes, but its knots impose sharp structural changes that may not reflect smooth thermodynamic transitions.

SIG8, $k = 8$.

$$\begin{aligned}\hat{T}_{\text{SIG8}}(P) = & 5.05 - 6.20 \times 10^{-4}P - 1.12L_{250,70}(P) - 0.75L_{500,120}(P) \\ & - 0.43L_{900,180}(P) - 0.26L_{1500,260}(P) - 0.12L_{2500,400}(P) - 0.07L_{3500,550}(P).\end{aligned}$$

Explanation. The sigmoid-transition basis replaces sharp knots with smooth pressure transitions. It is well suited to mixed-layer and thermocline gradients, but several transition terms can become redundant when the profile is already smooth at depth.

MIX8A, $k = 8$.

$$\begin{aligned}\hat{T}_{\text{MIX8A}}(P) = & 72.5243001 + 130.024309z - 1.45894798z^2 + 0.148251154z^3 \\ & + 23.8341486 \ln(P) - 6.13065083\sqrt{P} + \frac{2610.76502}{P} - 9.36352770z \ln(P). \\ z = & \frac{P - 1641.2335}{990.7445}.\end{aligned}$$

Explanation. MIX8A combines standardized polynomial terms with logarithmic, square-root, reciprocal and interaction components. It captures both normalized background curvature and pressure-dependent nonlinear structure, but the cross term and $1/P$ component make the expression less stable near shallow pressures.

MIX8B, $k = 8$, **current recommended backbone**.

$$\begin{aligned}\hat{T}_{\text{MIX8B}}(P) = & -356.873 + 0.0125047P + 62.2025 \ln(P) - 3.35576\sqrt{P} \\ & + 20.0563 \exp(-P/500) + 14.8832 \exp(-P/1200) \\ & + \frac{10011.3}{P + 50} + \frac{7576.48}{P + 500}.\end{aligned}$$

Explanation. MIX8B separates the profile into complementary pressure-coordinate components. The linear, logarithmic and square-root terms describe the broad vertical background; the two exponential terms capture rapid upper and intermediate decay; and the reciprocal terms constrain near-surface curvature while allowing deep-pressure stabilization.

PL9, $k = 9$.

$$\begin{aligned}\hat{T}_{\text{PL9}}(P) = & 2.214 - 0.872x + 0.421x^2 - 0.061x^3 - 0.019x^4 \\ & - 0.004x^5 + 0.004x^6 + 0.012 \ln(P) - 0.006x \ln(P).\end{aligned}$$

Explanation. PL9 extends the standardized polynomial family with logarithmic interaction terms. It tests whether extra mixed polynomial-log flexibility can improve the profile fit, but the additional interaction mainly increases structural complexity and may duplicate curvature already captured by the lower-order terms.

SIG10, $k = 10$.

$$\begin{aligned}\hat{T}_{\text{SIG10}}(P) = & 4.820 - 6.4 \times 10^{-4}P - 0.180 \ln(P) + 0.036\sqrt{P} \\ & + 0.42L_{250,80}(P) + 0.31L_{500,120}(P) - 0.22L_{850,180}(P) \\ & + 0.15L_{1300,250}(P) - 0.11L_{2200,350}(P) + 0.08L_{3300,500}(P).\end{aligned}$$

Explanation. SIG10 uses multiple smooth transition functions distributed across the pressure column. It may slightly improve numerical accuracy by fitting local transition regions, but the gain is small relative to the extra terms, making it less attractive as a parsimonious analytical backbone.

2.4 Suggested Reporting Logic

The comparison shows that low-order polynomials are insufficient, whereas compact nonlinear pressure-coordinate structures reduce errors to approximately 0.052–0.055°C. Among all eight-term candidates, MIX8B gives the strongest performance. The additional candidate fits serve as a complexity-path sensitivity check: a 10-term transition model may be numerically similar or slightly better, but the gain is marginal relative to the additional complexity. This supports presenting MIX8B as the parsimonious analytical backbone for Ocean-TP.

3 Mixed-B Term-Count Evolution

Experimental-result note. This section records the Mixed-B term-count evolution obtained from fitting and evaluation. The plotted values, tabulated metrics and incremental changes summarize the sequential improvement along the Mixed-B term-count path. Here k counts all active algebraic terms, including the intercept. The first-term point is the intercept-only null baseline; when the intercept is fitted as the sample mean, $R^2 = 0$ by definition. The first pressure-dependent term enters at $k = 2$, and the eighth-term point is the current Mixed-B anchor: MAE = 0.05107°C, RMSE = 0.11189°C and $R^2 = 0.98407$.

3.1 Experimental Term-Count Metric Path

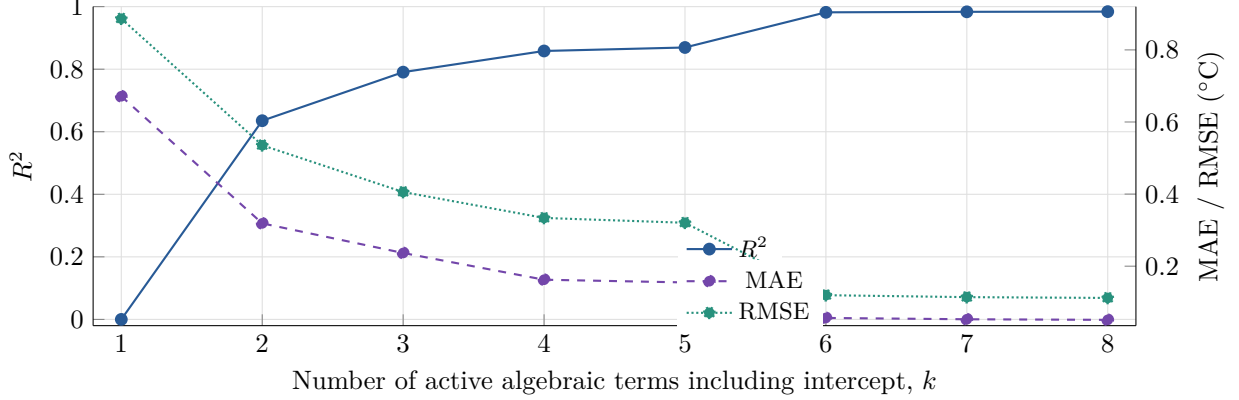


Figure 2: Term-count evolution of the Mixed-B analytical backbone. The first point is the intercept-only null baseline. Error decreases strongly after pressure dependence is introduced, improves through the nonlinear curvature terms, drops again when the dual-decay structure is completed, and then reaches a low-error plateau from six to eight terms.

Table 2: Term-by-term evolution of the Mixed-B equation. The term count k includes the intercept.

k	Newly added basis term	Basis composition after adding the term	MAE (°C)	RMSE (°C)	R^2
1	Intercept-only null baseline	$c_0^{(1)}$	0.67142	0.88659	0.00000
2	Linear pressure, P	$c_0^{(2)} + c_1^{(2)}P$	0.31876	0.53542	0.63529
3	Logarithmic curvature, $\ln P$	$c_0^{(3)} + c_1^{(3)}P + c_2^{(3)}\ln P$	0.23658	0.40573	0.79057
4	Square-root curvature, $\sqrt{P}$	$c_0^{(4)} + c_1^{(4)}P + c_2^{(4)}\ln P + c_3^{(4)}\sqrt{P}$	0.16243	0.33381	0.85824
5	Fast exponential decay, $e^{-P/500}$	$\hat{T}_4(P) + c_4^{(5)}e^{-P/500}$	0.15492	0.32041	0.86939
6	Slow exponential decay, $e^{-P/1200}$	$\hat{T}_5(P) + c_5^{(6)}e^{-P/1200}$	0.05638	0.11952	0.98183
7	Near-surface reciprocal, $(P + 50)^{-1}$	$\hat{T}_6(P) + c_6^{(7)}(P + 50)^{-1}$	0.05271	0.11409	0.98344
8	Stabilizing reciprocal, $(P + 500)^{-1}$	$\hat{T}_7(P) + c_7^{(8)}(P + 500)^{-1}$	0.05107	0.11189	0.98407

3.2 Incremental Error Reduction

Step	ΔMAE	ΔRMSE	ΔR^2	Interpretation
1→2	0.35266	0.35117	0.63529	Introducing pressure dependence gives the largest overall improvement because the model first moves beyond the intercept-only null baseline and acquires a vertical coordinate trend.
2→3	0.08218	0.12969	0.15528	Adding logarithmic curvature provides a substantial nonlinear improvement and captures broad upper-to-intermediate profile bending.
3→4	0.07415	0.07192	0.06767	The square-root term adds a second smooth curvature scale and further refines the profile.
4→5	0.00751	0.01340	0.01115	The first exponential term introduces fast pressure attenuation and gives a smaller transition refinement.
5→6	0.09854	0.20089	0.11244	The second exponential decay creates the second major low-error transition, indicating that a slower intermediate-depth scale is essential in the fitted path.
6→7	0.00367	0.00543	0.00161	The near-surface reciprocal acts mainly as a local correction after the dominant structure is already captured.
7→8	0.00164	0.00220	0.00063	The stabilizing reciprocal gives a final small refinement and completes the compact Mixed-B backbone.

3.3 Term-by-Term Interpretation

$k = 1$: intercept. The first term provides the intercept-only null baseline and acts as the reference state for the pressure-coordinate profile. Because this model predicts only the sample mean temperature, its R^2 is zero by definition. It cannot represent vertical stratification, so its error is substantially larger than all pressure-dependent candidates.

$k = 2$: linear pressure term. Adding P introduces the first-order cooling tendency with pressure. This term captures the broad vertical trend but remains too simple for thermocline curvature.

$k = 3$: logarithmic term. The addition of $\ln P$ produces a strong nonlinear improvement in the complexity path. It gives the equation a compact way to represent nonlinear curvature across the upper and intermediate water column, explaining the substantial increase in R^2 and the reduction in both MAE and RMSE.

$k = 4$: square-root term. The $\sqrt{P}$ component adds a second smooth curvature scale. Together with $\ln P$, it refines the background profile shape without introducing abrupt transitions.

$k = 5$ and $k = 6$: dual exponential decays. The $e^{-P/500}$ and $e^{-P/1200}$ terms describe fast and slower attenuation with pressure. In the fitted path, the second exponential term marks the transition into the low-error regime, indicating that multi-scale decay is essential for recovering the dominant vertical thermal structure.

$k = 7$ and $k = 8$: reciprocal stabilization. The $(P + 50)^{-1}$ and $(P + 500)^{-1}$ terms provide localized near-surface curvature while remaining bounded at depth. Their incremental gains are smaller than the earlier terms, but they complete the compact Mixed-B backbone and improve full-profile coherence.

3.4 Final Mixed-B Equation and Reporting Logic

$$\begin{aligned}\hat{T}_8^B(P) = & -356.873 + 0.0125047P + 62.2025 \ln(P) - 3.35576\sqrt{P} \\ & + 20.0563 \exp(-P/500) + 14.8832 \exp(-P/1200) \\ & + \frac{10011.3}{P + 50} + \frac{7576.48}{P + 500}.\end{aligned}$$

This final form can be described as an explicit pressure-coordinate backbone with three functional roles: broad background structure from P , $\ln P$ and $\sqrt{P}$; multi-scale decay from the two exponential terms; and near-surface plus deep-pressure stabilization from the reciprocal terms. The Mixed-B basis can therefore be reported as a term-count path from an intercept-only null baseline to a compact eight-term pressure-coordinate representation. Errors decrease strongly after pressure dependence is introduced, continue improving through the nonlinear curvature terms, enter a low-error regime once the dual exponential terms are included, and then receive smaller but consistent refinements from the reciprocal components.

4 Baseline Metrics for Result 1

Complete-data note. This section reports the complete baseline metric record supporting the final benchmark comparison in Result 1. The row-level source catalogue uses the same 10k evaluation split, with 8,000 training samples and 2,000 test samples. The source record contains one Mixed-B-only regression-fit analytical control and 298 data-driven benchmark configurations. For clarity, *Ocean-TP optimized Mixed-B* denotes the complete optimized reference result reported in the main manuscript, and is used here as the representative Ocean-TP model result. By contrast, *Mixed-B-only regression fit* denotes the analytical control obtained by an ordinary regression fit using only the Mixed-B functional form. The full Ocean-TP optimized Mixed-B result is shown as a manuscript-level reference in the representative visualization, but is not counted as one of the 298 conventional benchmark configurations. Rows preserve the source metric-record order, and the category labels are reported in English as analytical control, classical regression and deep regression. The metrics are displayed to five decimal places for consistency with the preceding tables.

4.1 Representative Model-Family Visualization

The manuscript Table 2 reports the Ocean-TP optimized Mixed-B result together with the best-performing representative entry from each major model family. Figure 3 visualizes those representative entries by MAE, including the full Ocean-TP optimized Mixed-B reference and the Mixed-B-only regression-fit analytical control, while the complete row-level metric catalogue is retained in Table 3.

4.2 Interpretive Discussion of Representative Model Families

The representative benchmark should be read as a family-level stress test rather than as a search for a single numerical winner. The 298 data-driven configurations cover linear, robust, polynomial, kernel, neighborhood, tree-ensemble, boosting, probabilistic and shallow neural-network regressors. The Mixed-B-only row is included as an analytical control that fits the Ocean-TP selected basis as a conventional regression model. Therefore, the key comparison is not only whether a model can reach a low MAE, but also what type of structure is required to do so and whether the resulting representation remains interpretable.

Near-frontier smooth nonlinear models. GaussianProcess, KernelRidge, SVR, NuSVR, Nystroem RBF + Ridge and KNN all enter the leading error band when configured with smooth radial-basis or local-neighborhood structure. Their representative MAE values lie close to the Mixed-B-only regression control, mostly between approximately 0.055 and 0.057°C. This indicates that the dominant pressure-temperature relationship in the 10k split is smooth enough for kernel and local interpolation methods to recover effectively. At the same time, these methods remain primarily algorithmic function approximators: they can match the curve but do not directly expose a compact pressure-coordinate equation. The Mixed-B-only result is therefore important because it reaches the same accuracy band with an explicit eight-term structure rather than through an implicit kernel expansion or local averaging rule.

Tree and ensemble models. ExtraTrees, HistGradientBoosting, GradientBoosting, RandomForest, VotingRegressor, StackingRegressor, AdaBoost, Bagging and NGBoost form a second group of strong nonlinear learners. The best tree-ensemble entries also remain within the narrow high-skill band, with representative MAE values near 0.056–0.060°C. This confirms that the benchmark is not biased toward one modeling family: several independent algorithmic routes can approximate the same vertical profile. However, the ensemble models distribute the reconstruction across many trees, boosting stages or stacked components. Their accuracy therefore supports the existence of a learnable nonlinear pressure structure, but it does not provide the same direct symbolic decomposition into background trend, multi-scale decay and reciprocal stabilization that the Mixed-B equation provides.

Polynomial and spline baselines. The polynomial and spline models help separate smooth global curvature from more structured physical representation. Spline regression and the best polynomial representative retain reasonable accuracy, but their role is mainly to show that generic smoothness alone is not the whole explanation. The complete catalogue shows that polynomial performance is degree-sensitive: moderate polynomial order can be competitive, whereas over-extended orders degrade markedly. This behavior is consistent with the interpretation from Section 2: adding algebraic flexibility alone does not guarantee a better reconstruction. The Mixed-B basis improves this situation by combining algebraic curvature with exponential and reciprocal pressure terms, giving the equation a more targeted way to represent upper-ocean transition and deep-pressure stabilization.

Linear and robust statistical regressors. Linear Regression, BayesianRidge, ElasticNet, Lasso, LinearSVR, HuberRegressor, TheilSenRegressor and RANSACRegressor occupy a clearly higher-error regime. Their representative MAE values remain around 0.27–0.30°C for the linear or robust-linear cases, far above the leading nonlinear group. This gap is useful because it shows that the benchmark difficulty is not mainly caused by a few outliers or by noise sensitivity. Robust estimators can reduce the influence of anomalous points, but they do not solve the core structural problem: the vertical temperature profile is nonlinear in pressure and contains curvature regimes that cannot be represented by a single global slope. The poor performance of these baselines therefore strengthens the claim that pressure-coordinate nonlinearity is essential.

Neighborhood-scale sensitivity. The nearest-neighbor family illustrates the importance of scale choice. KNN with a well-selected neighborhood reaches the leading band, while RadiusNeighbors is much more sensitive to the selected radius and remains outside the top group in the representative comparison. This contrast is useful because it shows

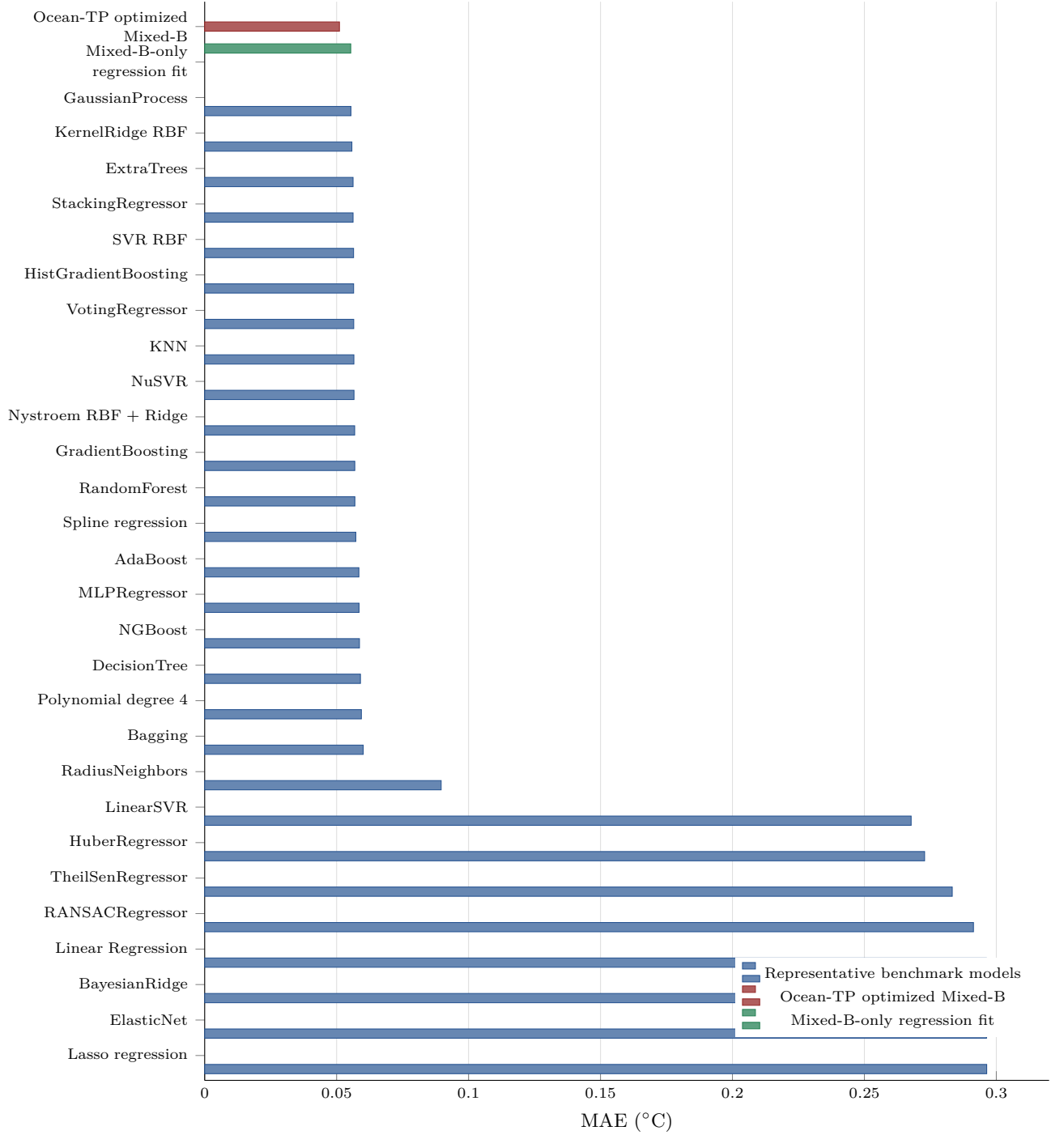


Figure 3: Representative baseline MAE comparison for the Result 1 benchmark. The plotted entries correspond to the condensed manuscript benchmark table, which retains the best-performing member of each major model family from the broader 298-configuration benchmark catalogue, together with the full Ocean-TP optimized Mixed-B reference and the Mixed-B-only regression control.

that local methods can be very accurate only when the locality scale is well matched to the density and curvature of the pressure-temperature data. The Mixed-B-only control avoids this explicit neighborhood-scale tuning by representing the profile through fixed analytical basis functions.

Neural and feature-mapped baselines. The MLP regressor representative reaches a competitive MAE near 0.0585°C, indicating that a shallow neural network can learn the smooth one-dimensional mapping with reasonable accuracy. The complete catalogue also records a FourierFeatures+Ridge entry with much poorer performance, which shows that adding a feature map does not automatically improve the reconstruction unless the representation is well matched to the target structure. For this benchmark, the neural and feature-mapped baselines are therefore best interpreted as flexible comparators, not as the central explanation of the observed accuracy. Ocean-TP uses the selected Mixed-B structure as the optimized pressure-coordinate representation before any subsequent constrained refinement.

Implication for the Result 1 claim. The central message of this benchmark is not that data-driven models are weak. On the contrary, several modern nonlinear regressors achieve very similar reconstruction errors. The key point is that the Mixed-B-only regression control lies inside the same leading performance band while remaining compact, explicit and physically interpretable. This gives the Result 1 benchmark a stronger interpretation: Ocean-TP does not rely on arbitrary curve fitting to reach the benchmark frontier. It first identifies a low-dimensional analytical pressure-coordinate backbone, then reserves later model components for constrained refinement rather than replacing the physical structure with an opaque predictor.

4.3 Complete Baseline Metric Catalogue

Table 3: Complete baseline metrics for the representative 10k Result 1 benchmark evaluation. All rows use an 8,000 / 2,000 train-test split.

No.	Category	Model	MAE	RMSE	R^2
1	Analytical control	Mixed-B-only regression fit	0.05535	0.10756	0.98469
2	Classical regression	Linear Regression	0.29625	0.37017	0.81868
3	Classical regression	Polynomial Regression (deg=5)	0.07097	0.13194	0.97697
4	Classical regression	SVR (RBF)	0.05870	0.11422	0.98274
5	Classical regression	Random Forest	0.06250	0.12158	0.98044
6	Classical regression	Spline Regression (B-spline, 10 knots)	0.05727	0.11110	0.98367
7	Classical regression	Nystroem RBF + Ridge (Kriging approx)	0.05684	0.11077	0.98376
8	Classical regression	Nystroem RBF(1024) + Ridge	0.05684	0.11077	0.98376
9	Classical regression	KNN (k=25, distance)	0.06507	0.12890	0.97802
10	Classical regression	Decision Tree (max_depth=18)	0.06406	0.12527	0.97923
11	Classical regression	ExtraTrees	0.06512	0.12891	0.97801
12	Classical regression	ElasticNet	0.29634	0.37015	0.81870
13	Classical regression	KernelRidge (RBF)	0.05688	0.11093	0.98372
14	Classical regression	HistGradientBoosting	0.05661	0.10873	0.98436
15	Classical regression	GradientBoosting	0.05688	0.10999	0.98399
16	Classical regression	AdaBoost (Tree base)	0.05843	0.11422	0.98274
17	Classical regression	Bagging (Tree base)	0.06008	0.11593	0.98222
18	Classical regression	BayesianRidge	0.29626	0.37017	0.81868
19	Classical regression	HuberRegressor	0.27282	0.40239	0.78574
20	Classical regression	Lasso	0.29636	0.37014	0.81871
21	Classical regression	LinearSVR	0.26772	0.43354	0.75129
22	Classical regression	NuSVR (RBF)	0.05717	0.11434	0.98270
23	Classical regression	SVR (poly, deg=3)	0.61084	0.74108	0.27327
24	Classical regression	SGDRegressor	0.29823	0.37001	0.81884

No.	Category	Model	MAE	RMSE	R^2
25	Classical regression	RANSACRegressor	0.29134	0.52866	0.63018
26	Classical regression	TheilSenRegressor	0.28330	0.48744	0.68560
27	Classical regression	RadiusNeighbors (radius=1.0)	0.25219	0.34235	0.84491
28	Classical regression	KernelRidge (poly, deg=3)	0.05978	0.11810	0.98155
29	Classical regression	GaussianProcess (RBF, subsample=2k)	0.05568	0.10815	0.98452
30	Classical regression	GaussianProcess (Matern $\nu=1.5$, subsample=2k)	0.05542	0.10814	0.98452
31	Classical regression	GaussianProcess (RationalQuadratic, subsample=2k)	0.05554	0.10805	0.98455
32	Classical regression	VotingRegressor (LR+RF+SVR)	0.11628	0.16437	0.96425
33	Classical regression	StackingRegressor (RF+SVR -> Ridge)	0.05807	0.11263	0.98321
34	Classical regression	VotingRegressor (SVR+HGB+KernelRidge)	0.05648	0.10973	0.98407
35	Classical regression	StackingRegressor (SVR+HGB+KernelRidge→Ridge)	0.05623	0.10842	0.98445
36	Classical regression	Polynomial Regression (deg=2)	0.08221	0.13376	0.97632
37	Classical regression	Polynomial Regression (deg=3)	0.05978	0.11810	0.98155
38	Classical regression	Polynomial Regression (deg=4)	0.05938	0.11803	0.98157
39	Classical regression	Polynomial Regression (deg=6)	0.13487	0.19623	0.94905
40	Classical regression	Polynomial Regression (deg=7)	0.21484	0.28749	0.89063
41	Classical regression	Polynomial Regression (deg=8)	0.29138	0.37408	0.81483
42	Classical regression	Polynomial Regression (deg=9)	0.36236	0.45100	0.73085
43	Classical regression	Polynomial Regression (deg=10)	0.42747	0.52025	0.64185
44	Classical regression	Polynomial Regression (deg=11)	0.48949	0.58117	0.55306
45	Classical regression	Polynomial Regression (deg=12)	0.53703	0.63110	0.47298
46	Classical regression	SVR (RBF,C=0.5,gamma=scale,eps=0.01)	0.05766	0.12166	0.98041
47	Classical regression	SVR (RBF,C=0.5,gamma=scale,eps=0.05)	0.05922	0.12081	0.98069
48	Classical regression	SVR (RBF,C=0.5,gamma=0.001,eps=0.01)	0.26184	0.39930	0.78902
49	Classical regression	SVR (RBF,C=0.5,gamma=0.001,eps=0.05)	0.26356	0.39071	0.79800
50	Classical regression	SVR (RBF,C=0.5,gamma=0.005,eps=0.01)	0.10184	0.17411	0.95989
51	Classical regression	SVR (RBF,C=0.5,gamma=0.005,eps=0.05)	0.10944	0.17834	0.95791
52	Classical regression	SVR (RBF,C=0.5,gamma=0.01,eps=0.01)	0.08467	0.14934	0.97049
53	Classical regression	SVR (RBF,C=0.5,gamma=0.01,eps=0.05)	0.08547	0.14648	0.97161
54	Classical regression	SVR (RBF,C=1.0,gamma=scale,eps=0.01)	0.05736	0.11994	0.98096
55	Classical regression	SVR (RBF,C=1.0,gamma=scale,eps=0.05)	0.05954	0.11910	0.98123
56	Classical regression	SVR (RBF,C=1.0,gamma=0.001,eps=0.01)	0.25010	0.39067	0.79804
57	Classical regression	SVR (RBF,C=1.0,gamma=0.001,eps=0.05)	0.25087	0.38456	0.80431
58	Classical regression	SVR (RBF,C=1.0,gamma=0.005,eps=0.01)	0.08889	0.15672	0.96750
59	Classical regression	SVR (RBF,C=1.0,gamma=0.005,eps=0.05)	0.09111	0.15501	0.96820
60	Classical regression	SVR (RBF,C=1.0,gamma=0.01,eps=0.01)	0.08080	0.14254	0.97312
61	Classical regression	SVR (RBF,C=1.0,gamma=0.01,eps=0.05)	0.08167	0.14050	0.97388
62	Classical regression	SVR (RBF,C=5.0,gamma=scale,eps=0.01)	0.05681	0.11601	0.98219

No.	Category	Model	MAE	RMSE	R^2
63	Classical regression	SVR (RBF,C=5.0,gamma=scale,eps=0.05)	0.05875	0.11548	0.98235
64	Classical regression	SVR (RBF,C=5.0,gamma=0.001,eps=0.01)	0.18268	0.27946	0.89666
65	Classical regression	SVR (RBF,C=5.0,gamma=0.001,eps=0.05)	0.18470	0.27564	0.89946
66	Classical regression	SVR (RBF,C=5.0,gamma=0.005,eps=0.01)	0.08004	0.14054	0.97386
67	Classical regression	SVR (RBF,C=5.0,gamma=0.005,eps=0.05)	0.08066	0.13845	0.97464
68	Classical regression	SVR (RBF,C=5.0,gamma=0.01,eps=0.01)	0.06728	0.12696	0.97867
69	Classical regression	SVR (RBF,C=5.0,gamma=0.01,eps=0.05)	0.07108	0.12832	0.97821
70	Classical regression	SVR (RBF,C=10.0,gamma=scale,eps=0.01)	0.05657	0.11431	0.98271
71	Classical regression	SVR (RBF,C=10.0,gamma=scale,eps=0.05)	0.05870	0.11422	0.98274
72	Classical regression	SVR (RBF,C=10.0,gamma=0.001,eps=0.01)	0.10831	0.17924	0.95749
73	Classical regression	SVR (RBF,C=10.0,gamma=0.001,eps=0.05)	0.11694	0.18471	0.95485
74	Classical regression	SVR (RBF,C=10.0,gamma=0.005,eps=0.01)	0.07711	0.13642	0.97537
75	Classical regression	SVR (RBF,C=10.0,gamma=0.005,eps=0.05)	0.07818	0.13503	0.97587
76	Classical regression	SVR (RBF,C=10.0,gamma=0.01,eps=0.01)	0.06204	0.12220	0.98024
77	Classical regression	SVR (RBF,C=10.0,gamma=0.01,eps=0.05)	0.06643	0.12385	0.97970
78	Classical regression	SVR (RBF,C=20.0,gamma=scale,eps=0.01)	0.05642	0.11304	0.98309
79	Classical regression	SVR (RBF,C=20.0,gamma=scale,eps=0.05)	0.05863	0.11272	0.98319
80	Classical regression	SVR (RBF,C=20.0,gamma=0.001,eps=0.01)	0.08998	0.15790	0.96701
81	Classical regression	SVR (RBF,C=20.0,gamma=0.001,eps=0.05)	0.09356	0.15786	0.96703
82	Classical regression	SVR (RBF,C=20.0,gamma=0.005,eps=0.01)	0.07251	0.13121	0.97722
83	Classical regression	SVR (RBF,C=20.0,gamma=0.005,eps=0.05)	0.07455	0.13129	0.97719
84	Classical regression	SVR (RBF,C=20.0,gamma=0.01,eps=0.01)	0.06056	0.12026	0.98086
85	Classical regression	SVR (RBF,C=20.0,gamma=0.01,eps=0.05)	0.06383	0.12117	0.98057
86	Classical regression	NuSVR (RBF,nu=0.25,C=1.0)	0.06015	0.11884	0.98131
87	Classical regression	NuSVR (RBF,nu=0.25,C=5.0)	0.05969	0.11554	0.98233
88	Classical regression	NuSVR (RBF,nu=0.25,C=10.0)	0.05976	0.11400	0.98280
89	Classical regression	NuSVR (RBF,nu=0.5,C=1.0)	0.05779	0.11959	0.98107
90	Classical regression	NuSVR (RBF,nu=0.5,C=5.0)	0.05733	0.11561	0.98231
91	Classical regression	NuSVR (RBF,nu=0.5,C=10.0)	0.05717	0.11434	0.98270
92	Classical regression	NuSVR (RBF,nu=0.75,C=1.0)	0.05739	0.11995	0.98096
93	Classical regression	NuSVR (RBF,nu=0.75,C=5.0)	0.05681	0.11599	0.98220
94	Classical regression	NuSVR (RBF,nu=0.75,C=10.0)	0.05658	0.11427	0.98272
95	Classical regression	LinearSVR (C=0.1,eps=0.01)	0.26777	0.43585	0.74863
96	Classical regression	LinearSVR (C=0.1,eps=0.05)	0.26776	0.43345	0.75140
97	Classical regression	LinearSVR (C=1.0,eps=0.01)	0.26773	0.43506	0.74955
98	Classical regression	LinearSVR (C=1.0,eps=0.05)	0.26772	0.43354	0.75129
99	Classical regression	LinearSVR (C=10.0,eps=0.01)	0.26782	0.43756	0.74665
100	Classical regression	LinearSVR (C=10.0,eps=0.05)	0.26776	0.43354	0.75128

No.	Category	Model	MAE	RMSE	R^2
101	Classical regression	RandomForest (n=100,depth=None,leaf=1)	0.06277	0.12230	0.98021
102	Classical regression	RandomForest (n=100,depth=None,leaf=2)	0.06103	0.11812	0.98154
103	Classical regression	RandomForest (n=100,depth=12,leaf=1)	0.06021	0.11621	0.98213
104	Classical regression	RandomForest (n=100,depth=12,leaf=2)	0.05955	0.11474	0.98258
105	Classical regression	RandomForest (n=100,depth=24,leaf=1)	0.06276	0.12227	0.98022
106	Classical regression	RandomForest (n=100,depth=24,leaf=2)	0.06102	0.11812	0.98154
107	Classical regression	RandomForest (n=200,depth=None,leaf=1)	0.06260	0.12206	0.98029
108	Classical regression	RandomForest (n=200,depth=None,leaf=2)	0.06075	0.11752	0.98173
109	Classical regression	RandomForest (n=200,depth=12,leaf=1)	0.06008	0.11593	0.98222
110	Classical regression	RandomForest (n=200,depth=12,leaf=2)	0.05935	0.11427	0.98272
111	Classical regression	RandomForest (n=200,depth=24,leaf=1)	0.06258	0.12200	0.98030
112	Classical regression	RandomForest (n=200,depth=24,leaf=2)	0.06075	0.11751	0.98173
113	Classical regression	RandomForest (n=300,depth=None,leaf=1)	0.06250	0.12158	0.98044
114	Classical regression	RandomForest (n=300,depth=None,leaf=2)	0.06064	0.11711	0.98185
115	Classical regression	RandomForest (n=300,depth=12,leaf=1)	0.06006	0.11572	0.98228
116	Classical regression	RandomForest (n=300,depth=12,leaf=2)	0.05929	0.11397	0.98281
117	Classical regression	RandomForest (n=300,depth=24,leaf=1)	0.06248	0.12153	0.98046
118	Classical regression	RandomForest (n=300,depth=24,leaf=2)	0.06063	0.11711	0.98185
119	Classical regression	ExtraTrees (n=100,depth=None,leaf=1)	0.06514	0.12890	0.97801
120	Classical regression	ExtraTrees (n=100,depth=None,leaf=2)	0.06073	0.11724	0.98181
121	Classical regression	ExtraTrees (n=100,depth=12,leaf=1)	0.05754	0.11189	0.98343
122	Classical regression	ExtraTrees (n=100,depth=12,leaf=2)	0.05689	0.11019	0.98393
123	Classical regression	ExtraTrees (n=100,depth=24,leaf=1)	0.06504	0.12880	0.97805
124	Classical regression	ExtraTrees (n=100,depth=24,leaf=2)	0.06072	0.11722	0.98182
125	Classical regression	ExtraTrees (n=200,depth=None,leaf=1)	0.06513	0.12893	0.97800
126	Classical regression	ExtraTrees (n=200,depth=None,leaf=2)	0.06114	0.11742	0.98176
127	Classical regression	ExtraTrees (n=200,depth=12,leaf=1)	0.05753	0.11195	0.98342
128	Classical regression	ExtraTrees (n=200,depth=12,leaf=2)	0.05737	0.11031	0.98390
129	Classical regression	ExtraTrees (n=200,depth=24,leaf=1)	0.06504	0.12880	0.97805
130	Classical regression	ExtraTrees (n=200,depth=24,leaf=2)	0.06114	0.11741	0.98176
131	Classical regression	ExtraTrees (n=300,depth=None,leaf=1)	0.06512	0.12891	0.97801
132	Classical regression	ExtraTrees (n=300,depth=None,leaf=2)	0.06089	0.11726	0.98180
133	Classical regression	ExtraTrees (n=300,depth=12,leaf=1)	0.05745	0.11171	0.98349
134	Classical regression	ExtraTrees (n=300,depth=12,leaf=2)	0.05711	0.11029	0.98390
135	Classical regression	ExtraTrees (n=300,depth=24,leaf=1)	0.06503	0.12878	0.97806
136	Classical regression	ExtraTrees (n=300,depth=24,leaf=2)	0.06090	0.11724	0.98181
137	Classical regression	HistGradientBoosting (depth=4,lr=0.05,iter=200)	0.05644	0.10848	0.98443
138	Classical regression	HistGradientBoosting (depth=4,lr=0.05,iter=300)	0.05652	0.10860	0.98439

No.	Category	Model	MAE	RMSE	R^2
139	Classical regression	HistGradientBoosting (depth=4,lr=0.1,iter=200)	0.05659	0.10867	0.98437
140	Classical regression	HistGradientBoosting (depth=4,lr=0.1,iter=300)	0.05661	0.10870	0.98436
141	Classical regression	HistGradientBoosting (depth=6,lr=0.05,iter=200)	0.05656	0.10864	0.98438
142	Classical regression	HistGradientBoosting (depth=6,lr=0.05,iter=300)	0.05659	0.10870	0.98436
143	Classical regression	HistGradientBoosting (depth=6,lr=0.1,iter=200)	0.05660	0.10872	0.98436
144	Classical regression	HistGradientBoosting (depth=6,lr=0.1,iter=300)	0.05661	0.10873	0.98436
145	Classical regression	HistGradientBoosting (depth=8,lr=0.05,iter=200)	0.05658	0.10869	0.98437
146	Classical regression	HistGradientBoosting (depth=8,lr=0.05,iter=300)	0.05661	0.10872	0.98436
147	Classical regression	HistGradientBoosting (depth=8,lr=0.1,iter=200)	0.05661	0.10873	0.98436
148	Classical regression	HistGradientBoosting (depth=8,lr=0.1,iter=300)	0.05662	0.10873	0.98436
149	Classical regression	KernelRidge (RBF,alpha=0.01,gamma=0.5)	0.05816	0.11457	0.98263
150	Classical regression	KernelRidge (RBF,alpha=0.01,gamma=1.0)	0.05688	0.11093	0.98372
151	Classical regression	KernelRidge (RBF,alpha=0.01,gamma=2.0)	0.05602	0.10859	0.98440
152	Classical regression	KernelRidge (RBF,alpha=0.005,gamma=0.5)	0.05788	0.11368	0.98290
153	Classical regression	KernelRidge (RBF,alpha=0.005,gamma=1.0)	0.05670	0.11022	0.98393
154	Classical regression	KernelRidge (RBF,alpha=0.005,gamma=2.0)	0.05578	0.10812	0.98453
155	Classical regression	KernelRidge (poly,deg=2,alpha=0.01)	0.08221	0.13376	0.97632
156	Classical regression	KernelRidge (poly,deg=2,alpha=0.005)	0.08221	0.13376	0.97632
157	Classical regression	KernelRidge (poly,deg=3,alpha=0.01)	0.05978	0.11810	0.98155
158	Classical regression	KernelRidge (poly,deg=3,alpha=0.005)	0.05978	0.11810	0.98155
159	Classical regression	KernelRidge (poly,deg=4,alpha=0.01)	0.05938	0.11803	0.98157
160	Classical regression	KernelRidge (poly,deg=4,alpha=0.005)	0.05938	0.11803	0.98157
161	Classical regression	KNN (k=15,distance)	0.06508	0.12890	0.97801
162	Classical regression	KNN (k=15,uniform)	0.05714	0.11011	0.98396
163	Classical regression	KNN (k=25,distance)	0.06507	0.12890	0.97802
164	Classical regression	KNN (k=25,uniform)	0.05708	0.10939	0.98417
165	Classical regression	KNN (k=35,distance)	0.06502	0.12884	0.97803
166	Classical regression	KNN (k=35,uniform)	0.05673	0.10892	0.98430
167	Classical regression	DecisionTree (depth=10,leaf=1)	0.05925	0.11616	0.98215
168	Classical regression	DecisionTree (depth=10,leaf=2)	0.05903	0.11411	0.98277
169	Classical regression	DecisionTree (depth=14,leaf=1)	0.06302	0.12386	0.97970
170	Classical regression	DecisionTree (depth=14,leaf=2)	0.06215	0.12028	0.98086
171	Classical regression	DecisionTree (depth=18,leaf=1)	0.06406	0.12527	0.97923
172	Classical regression	DecisionTree (depth=18,leaf=2)	0.06284	0.12182	0.98036
173	Classical regression	DecisionTree (depth=22,leaf=1)	0.06505	0.12869	0.97808
174	Classical regression	DecisionTree (depth=22,leaf=2)	0.06304	0.12229	0.98021
175	Classical regression	RadiusNeighbors (r=0.5)	0.08961	0.17243	0.96066
176	Classical regression	RadiusNeighbors (r=1.0)	0.25219	0.34235	0.84491

No.	Category	Model	MAE	RMSE	R^2
177	Classical regression	RadiusNeighbors (r=2.0)	0.57570	0.68867	0.37242
178	Classical regression	ElasticNet (alpha=0.001,l1=0.1)	0.29634	0.37015	0.81870
179	Classical regression	ElasticNet (alpha=0.001,l1=0.2)	0.29634	0.37015	0.81870
180	Classical regression	ElasticNet (alpha=0.001,l1=0.5)	0.29635	0.37015	0.81870
181	Classical regression	ElasticNet (alpha=0.0005,l1=0.1)	0.29629	0.37016	0.81869
182	Classical regression	ElasticNet (alpha=0.0005,l1=0.2)	0.29630	0.37016	0.81869
183	Classical regression	ElasticNet (alpha=0.0005,l1=0.5)	0.29630	0.37016	0.81869
184	Classical regression	Nystroem RBF(n=256,gamma=0.5) + Ridge	0.05800	0.11436	0.98269
185	Classical regression	Nystroem RBF(n=256,gamma=1.0) + Ridge	0.05684	0.11077	0.98376
186	Classical regression	Nystroem RBF(n=512,gamma=0.5) + Ridge	0.05800	0.11436	0.98269
187	Classical regression	Nystroem RBF(n=512,gamma=1.0) + Ridge	0.05684	0.11077	0.98376
188	Classical regression	Nystroem RBF(n=1024,gamma=0.5) + Ridge	0.05800	0.11436	0.98269
189	Classical regression	Nystroem RBF(n=1024,gamma=1.0) + Ridge	0.05684	0.11077	0.98376
190	Classical regression	SVR (sigmoid,C=0.5,gamma=0.001,eps=0.01)	0.28648	0.40217	0.78598
191	Classical regression	SVR (sigmoid,C=0.5,gamma=0.001,eps=0.05)	0.28613	0.40080	0.78743
192	Classical regression	SVR (sigmoid,C=0.5,gamma=0.005,eps=0.01)	0.26837	0.42702	0.75871
193	Classical regression	SVR (sigmoid,C=0.5,gamma=0.005,eps=0.05)	0.26877	0.42106	0.76540
194	Classical regression	SVR (sigmoid,C=0.5,gamma=0.01,eps=0.01)	0.26901	0.43549	0.74904
195	Classical regression	SVR (sigmoid,C=0.5,gamma=0.01,eps=0.05)	0.26911	0.43058	0.75468
196	Classical regression	SVR (sigmoid,C=0.5,gamma=0.05,eps=0.01)	0.42566	1.00021	-0.32379
197	Classical regression	SVR (sigmoid,C=0.5,gamma=0.05,eps=0.05)	0.42607	0.99639	-0.31369
198	Classical regression	SVR (sigmoid,C=1.0,gamma=0.001,eps=0.01)	0.27173	0.41639	0.77057
199	Classical regression	SVR (sigmoid,C=1.0,gamma=0.001,eps=0.05)	0.27322	0.40756	0.78020
200	Classical regression	SVR (sigmoid,C=1.0,gamma=0.005,eps=0.01)	0.26811	0.43255	0.75242
201	Classical regression	SVR (sigmoid,C=1.0,gamma=0.005,eps=0.05)	0.26818	0.42812	0.75747
202	Classical regression	SVR (sigmoid,C=1.0,gamma=0.01,eps=0.01)	0.27020	0.44153	0.74204
203	Classical regression	SVR (sigmoid,C=1.0,gamma=0.01,eps=0.05)	0.27022	0.43761	0.74659
204	Classical regression	SVR (sigmoid,C=1.0,gamma=0.05,eps=0.01)	0.63365	1.73569	-2.98640
205	Classical regression	SVR (sigmoid,C=1.0,gamma=0.05,eps=0.05)	0.63090	1.71610	-2.89695
206	Classical regression	SVR (sigmoid,C=5.0,gamma=0.001,eps=0.01)	0.26782	0.43153	0.75358
207	Classical regression	SVR (sigmoid,C=5.0,gamma=0.001,eps=0.05)	0.26790	0.42699	0.75874
208	Classical regression	SVR (sigmoid,C=5.0,gamma=0.005,eps=0.01)	0.26930	0.44026	0.74351
209	Classical regression	SVR (sigmoid,C=5.0,gamma=0.005,eps=0.05)	0.26928	0.43707	0.74722
210	Classical regression	SVR (sigmoid,C=5.0,gamma=0.01,eps=0.01)	0.28023	0.47539	0.70095
211	Classical regression	SVR (sigmoid,C=5.0,gamma=0.01,eps=0.05)	0.28022	0.47243	0.70466
212	Classical regression	SVR (sigmoid,C=10.0,gamma=0.001,eps=0.01)	0.26776	0.43429	0.75042
213	Classical regression	SVR (sigmoid,C=10.0,gamma=0.001,eps=0.05)	0.26778	0.43021	0.75509
214	Classical regression	SVR (sigmoid,C=10.0,gamma=0.005,eps=0.01)	0.27086	0.44539	0.73750

No.	Category	Model	MAE	RMSE	R^2
215	Classical regression	SVR (sigmoid,C=10.0,gamma=0.005,eps=0.05)	0.27084	0.44245	0.74096
216	Classical regression	SVR (sigmoid,C=10.0,gamma=0.01,eps=0.01)	0.29291	0.51747	0.64567
217	Classical regression	SVR (sigmoid,C=10.0,gamma=0.01,eps=0.05)	0.29287	0.51451	0.64972
218	Classical regression	SVR (sigmoid,C=20.0,gamma=0.001,eps=0.01)	0.26780	0.43535	0.74921
219	Classical regression	SVR (sigmoid,C=20.0,gamma=0.001,eps=0.05)	0.26778	0.43228	0.75273
220	Classical regression	SVR (sigmoid,C=20.0,gamma=0.005,eps=0.01)	0.27400	0.45549	0.72547
221	Classical regression	SVR (sigmoid,C=20.0,gamma=0.005,eps=0.05)	0.27397	0.45239	0.72919
222	Classical regression	SVR (sigmoid,C=20.0,gamma=0.01,eps=0.01)	0.31860	0.60677	0.51281
223	Classical regression	SVR (sigmoid,C=20.0,gamma=0.01,eps=0.05)	0.31852	0.60376	0.51764
224	Classical regression	KNN (k=5,uniform,p=1)	0.06082	0.11706	0.98187
225	Classical regression	KNN (k=5,uniform,p=2)	0.06082	0.11706	0.98187
226	Classical regression	KNN (k=5,distance,p=1)	0.06561	0.12942	0.97784
227	Classical regression	KNN (k=5,distance,p=2)	0.06561	0.12942	0.97784
228	Classical regression	KNN (k=10,uniform,p=1)	0.05877	0.11319	0.98305
229	Classical regression	KNN (k=10,uniform,p=2)	0.05877	0.11319	0.98305
230	Classical regression	KNN (k=10,distance,p=1)	0.06506	0.12883	0.97804
231	Classical regression	KNN (k=10,distance,p=2)	0.06506	0.12883	0.97804
232	Classical regression	KNN (k=20,uniform,p=1)	0.05721	0.10949	0.98414
233	Classical regression	KNN (k=20,uniform,p=2)	0.05721	0.10949	0.98414
234	Classical regression	KNN (k=20,distance,p=1)	0.06507	0.12889	0.97802
235	Classical regression	KNN (k=20,distance,p=2)	0.06507	0.12889	0.97802
236	Classical regression	KNN (k=30,uniform,p=1)	0.05679	0.10952	0.98413
237	Classical regression	KNN (k=30,uniform,p=2)	0.05679	0.10952	0.98413
238	Classical regression	KNN (k=30,distance,p=1)	0.06502	0.12884	0.97803
239	Classical regression	KNN (k=30,distance,p=2)	0.06502	0.12884	0.97803
240	Classical regression	KNN (k=40,uniform,p=1)	0.05654	0.10882	0.98433
241	Classical regression	KNN (k=40,uniform,p=2)	0.05654	0.10882	0.98433
242	Classical regression	KNN (k=40,distance,p=1)	0.06499	0.12882	0.97804
243	Classical regression	KNN (k=40,distance,p=2)	0.06499	0.12882	0.97804
244	Classical regression	KNN (k=50,uniform,p=1)	0.05654	0.10958	0.98411
245	Classical regression	KNN (k=50,uniform,p=2)	0.05654	0.10958	0.98411
246	Classical regression	KNN (k=50,distance,p=1)	0.06500	0.12885	0.97803
247	Classical regression	KNN (k=50,distance,p=2)	0.06500	0.12885	0.97803
248	Classical regression	RandomForest (n=400,depth=None,leaf=1)	0.06255	0.12173	0.98039
249	Classical regression	RandomForest (n=400,depth=None,leaf=2)	0.06066	0.11718	0.98183
250	Classical regression	RandomForest (n=400,depth=None,leaf=4)	0.05897	0.11358	0.98293
251	Classical regression	RandomForest (n=400,depth=8,leaf=1)	0.05712	0.11018	0.98394
252	Classical regression	RandomForest (n=400,depth=8,leaf=2)	0.05694	0.10971	0.98407

No.	Category	Model	MAE	RMSE	R^2
253	Classical regression	RandomForest (n=400,depth=8,leaf=4)	0.05700	0.10964	0.98409
254	Classical regression	RandomForest (n=400,depth=12,leaf=1)	0.06008	0.11581	0.98225
255	Classical regression	RandomForest (n=400,depth=12,leaf=2)	0.05932	0.11405	0.98279
256	Classical regression	RandomForest (n=400,depth=12,leaf=4)	0.05858	0.11248	0.98326
257	Classical regression	RandomForest (n=400,depth=16,leaf=1)	0.06194	0.11995	0.98096
258	Classical regression	RandomForest (n=400,depth=16,leaf=2)	0.06037	0.11646	0.98205
259	Classical regression	RandomForest (n=400,depth=16,leaf=4)	0.05890	0.11341	0.98298
260	Classical regression	RandomForest (n=400,depth=24,leaf=1)	0.06254	0.12168	0.98041
261	Classical regression	RandomForest (n=400,depth=24,leaf=2)	0.06066	0.11717	0.98183
262	Classical regression	RandomForest (n=400,depth=24,leaf=4)	0.05897	0.11358	0.98293
263	Classical regression	RandomForest (n=400,depth=32,leaf=1)	0.06255	0.12173	0.98039
264	Classical regression	RandomForest (n=400,depth=32,leaf=2)	0.06066	0.11718	0.98183
265	Classical regression	RandomForest (n=400,depth=32,leaf=4)	0.05897	0.11358	0.98293
266	Classical regression	RandomForest (n=600,depth=None,leaf=1)	0.06251	0.12167	0.98041
267	Classical regression	RandomForest (n=600,depth=None,leaf=2)	0.06062	0.11706	0.98187
268	Classical regression	RandomForest (n=600,depth=None,leaf=4)	0.05893	0.11348	0.98296
269	Classical regression	RandomForest (n=600,depth=8,leaf=1)	0.05709	0.11018	0.98394
270	Classical regression	RandomForest (n=600,depth=8,leaf=2)	0.05691	0.10969	0.98408
271	Classical regression	RandomForest (n=600,depth=8,leaf=4)	0.05697	0.10961	0.98410
272	Classical regression	RandomForest (n=600,depth=12,leaf=1)	0.06008	0.11582	0.98225
273	Classical regression	RandomForest (n=600,depth=12,leaf=2)	0.05930	0.11399	0.98281
274	Classical regression	RandomForest (n=600,depth=12,leaf=4)	0.05855	0.11240	0.98328
275	Classical regression	RandomForest (n=600,depth=16,leaf=1)	0.06194	0.11998	0.98095
276	Classical regression	RandomForest (n=600,depth=16,leaf=2)	0.06036	0.11641	0.98207
277	Classical regression	RandomForest (n=600,depth=16,leaf=4)	0.05888	0.11333	0.98300
278	Classical regression	RandomForest (n=600,depth=24,leaf=1)	0.06250	0.12163	0.98043
279	Classical regression	RandomForest (n=600,depth=24,leaf=2)	0.06062	0.11705	0.98187
280	Classical regression	RandomForest (n=600,depth=24,leaf=4)	0.05893	0.11348	0.98296
281	Classical regression	RandomForest (n=600,depth=32,leaf=1)	0.06251	0.12167	0.98041
282	Classical regression	RandomForest (n=600,depth=32,leaf=2)	0.06062	0.11706	0.98187
283	Classical regression	RandomForest (n=600,depth=32,leaf=4)	0.05893	0.11348	0.98296
284	Classical regression	ExtraTrees (n=400,depth=None,leaf=1)	0.06512	0.12891	0.97801
285	Classical regression	ExtraTrees (n=400,depth=None,leaf=2)	0.06080	0.11720	0.98183
286	Classical regression	ExtraTrees (n=400,depth=None,leaf=4)	0.05765	0.11075	0.98377
287	Classical regression	ExtraTrees (n=400,depth=8,leaf=1)	0.05623	0.10821	0.98451
288	Classical regression	ExtraTrees (n=400,depth=8,leaf=2)	0.05689	0.10881	0.98433
289	Classical regression	ExtraTrees (n=400,depth=8,leaf=4)	0.05677	0.10868	0.98437
290	Classical regression	ExtraTrees (n=400,depth=12,leaf=1)	0.05745	0.11173	0.98348

No.	Category	Model	MAE	RMSE	R^2
291	Classical regression	ExtraTrees (n=400,depth=12,leaf=2)	0.05697	0.11016	0.98394
292	Classical regression	ExtraTrees (n=400,depth=12,leaf=4)	0.05653	0.10892	0.98430
293	Classical regression	ExtraTrees (n=400,depth=16,leaf=1)	0.06161	0.12200	0.98030
294	Classical regression	ExtraTrees (n=400,depth=16,leaf=2)	0.05917	0.11440	0.98268
295	Classical regression	NGBoost (Normal)	0.05865	0.11653	0.98203
296	Deep regression	MLPRegressor (64x64)	0.05901	0.11926	0.98118
297	Deep regression	MLPRegressor (256x128)	0.05919	0.11022	0.98392
298	Deep regression	MLPRegressor (128x64)	0.05850	0.11910	0.98123
299	Deep regression	FourierFeatures+Ridge	0.72718	0.86687	0.00563

5 Seasonal Mixed-B Equation Records for Result 2

Supporting-file note. This section records the four season-specific Mixed-B analytical equations used for the seasonal stability analysis in Result 2. The four equations share the same eight-term pressure-coordinate basis and differ only in the fitted coefficient vector. Throughout, P denotes pressure in dbar.

5.1 Season 1: January 22–May 27

$$\begin{aligned}\hat{T}_{S1}(P) = & 0.575963 - 0.000714P + 0.310251 \ln P + 0.002999\sqrt{P} \\ & + 2.356074e^{-P/500} + 1.290324e^{-P/1200} + \frac{0.041274}{P+50} + \frac{0.005494}{P+500}.\end{aligned}\quad (6)$$

5.2 Season 2: May 27–July 21

$$\begin{aligned}\hat{T}_{S2}(P) = & 0.594587 - 0.000761P + 0.320774 \ln P + 0.002806\sqrt{P} \\ & + 2.325574e^{-P/500} + 1.302425e^{-P/1200} + \frac{0.043476}{P+50} + \frac{0.005258}{P+500}.\end{aligned}\quad (7)$$

5.3 Season 3: July 21–October 20

$$\begin{aligned}\hat{T}_{S3}(P) = & 0.543880 - 0.000983P + 0.307154 \ln P + 0.016287\sqrt{P} \\ & + 2.113262e^{-P/500} + 1.212209e^{-P/1200} + \frac{0.012660}{P+50} + \frac{0.003176}{P+500}.\end{aligned}\quad (8)$$

5.4 Season 4: October 20–January 22

$$\begin{aligned}\hat{T}_{S4}(P) = & 0.518960 - 0.001025P + 0.298059 \ln P + 0.020408\sqrt{P} \\ & + 2.111670e^{-P/500} + 1.193199e^{-P/1200} + \frac{0.002496}{P+50} + \frac{0.002885}{P+500}.\end{aligned}\quad (9)$$

5.5 Seasonal Reconstruction-Error Box-Plot Statistics

Table 4 reports the supplementary summary statistics corresponding to the seasonal reconstruction-error box plot in the main manuscript. The date ranges use month-day notation, and the whisker, quartile, median and mean values are reported on the same error scale as the plotted distribution. The accompanying R^2 values provide the season-specific goodness-of-fit reference associated with the four Mixed-B seasonal records.

Table 4: Supplementary seasonal box-plot statistics for the Result 2 reconstruction-error analysis.

Season	Date range	Lower whisker	Q1	Median	Mean	Q3	Upper whisker	R^2	R^2 label
S1	01-22–05-27	-0.2476	-0.0978	0.0487	0.0512	0.1984	0.3031	0.9934	0.99
S2	05-27–07-21	-0.3027	-0.1186	0.0617	0.0594	0.2236	0.3489	0.9733	0.97
S3	07-21–10-20	-0.1987	-0.0813	0.0411	0.0392	0.1685	0.2527	0.9908	0.99
S4	10-20–01-22	-0.2214	-0.0876	0.0513	0.0495	0.1918	0.2779	0.9932	0.99

The seasonal box-plot summaries are consistent with the Result 2 interpretation: Season 2 has the broadest error spread and the lowest fitted R^2 , whereas Season 1 and Season 4 retain tighter high-skill reconstruction behavior. Season 3 remains close to the winter-transition seasons in R^2 while showing a slightly narrower central error interval than Season 1 and Season 4.

5.6 Monthly–Seasonal–Annual Equation Comparison

Figure 4 and Tables 5–6 provide the supplementary data record for the Result 2 comparison among monthly, seasonal and annual equation parameterizations. The monthly equation uses month-specific Mixed-B coefficients, the seasonal equation uses the coefficient set assigned to the corresponding season, and the annual equation uses a single full-year coefficient set. All MAE values are reported in $^{\circ}\text{C}$.

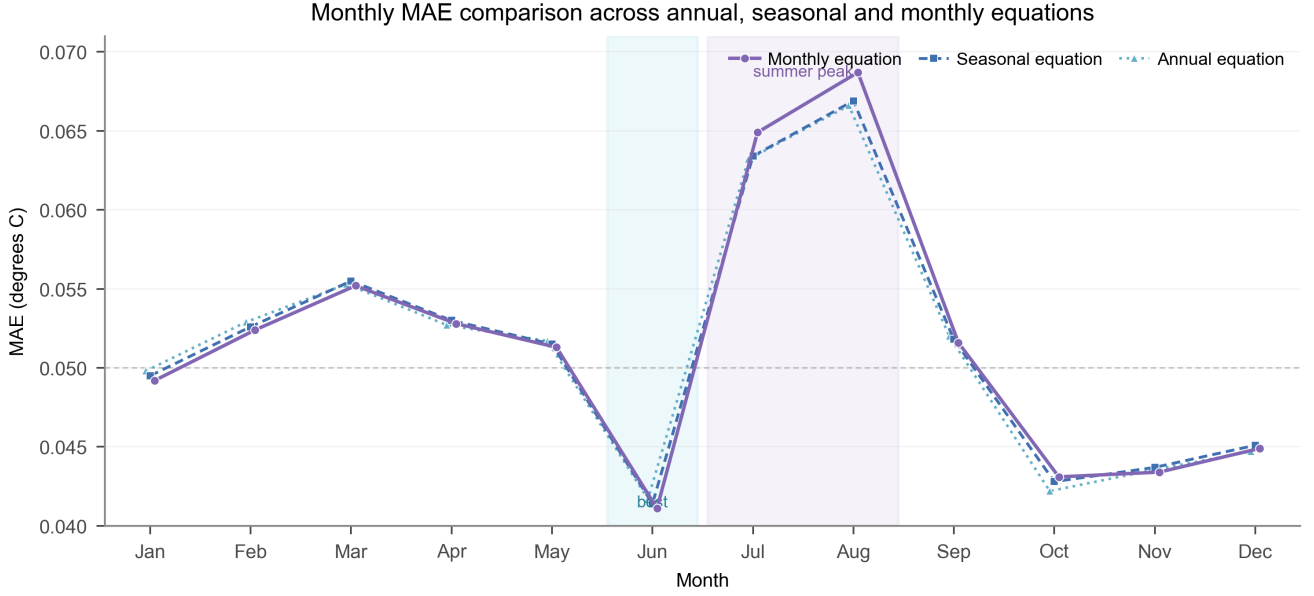


Figure 4: Monthly MAE comparison across annual, seasonal and monthly equation parameterizations for Result 2. The dashed reference line marks $\text{MAE} = 0.05^{\circ}\text{C}$, while the shaded intervals identify the June low-error month and the July–August summer peak.

Table 5: Source table data corresponding to the monthly MAE comparison across monthly, seasonal and annual equations in Result 2.

Month	Season	Monthly model	Seasonal model	Annual model
Jan	S1	0.0492	0.0495	0.0498
Feb	S1	0.0524	0.0526	0.0529
Mar	S1	0.0552	0.0555	0.0553
Apr	S1	0.0528	0.0530	0.0527
May	S1	0.0513	0.0515	0.0517
Jun	S2	0.0411	0.0414	0.0417
Jul	S2	0.0649	0.0634	0.0632
Aug	S3	0.0687	0.0669	0.0666
Sep	S3	0.0516	0.0518	0.0520
Oct	S3	0.0431	0.0428	0.0422
Nov	S4	0.0434	0.0437	0.0435
Dec	S4	0.0449	0.0451	0.0447

The comparison shows that the monthly and seasonal parameterizations follow closely aligned MAE trajectories for most months, while the annual equation provides a stable single-parameter-set reference. June forms the lowest-error month in this record, whereas July and August mark the summer error peak associated with stronger seasonal stratification.

Table 6: Monthly Mixed-B coefficient records corresponding to the month-specific equation path. The coefficient columns follow the basis order 1, P , $\ln P$, $\sqrt{P}$, $e^{-P/500}$, $e^{-P/1200}$, $(P + 50)^{-1}$ and $(P + 500)^{-1}$.

Month	Season	MAE	c_0	c_P	$c_{\ln P}$	$c_{\sqrt{P}}$	c_{e500}	c_{e1200}	c_{50}^{-1}	c_{500}^{-1}
Jan	S1	0.0492	0.57433100	-0.00070910	0.30951660	0.00332540	2.34111400	1.28270800	0.04045800	0.00534440
Feb	S1	0.0524	0.57548300	-0.00071256	0.31003500	0.00309500	2.35167400	1.28808400	0.04103400	0.00545000
Mar	S1	0.0552	0.57682700	-0.00071659	0.31063980	0.00282620	2.36399400	1.29435600	0.04170600	0.00557320
Apr	S1	0.0528	0.57649100	-0.00071558	0.31048860	0.00289340	2.36091400	1.29278800	0.04153800	0.00554240
May	S1	0.0513	0.57668300	-0.00071616	0.31057500	0.00285500	2.36267400	1.29368400	0.04163400	0.00556000
Jun	S2	0.0411	0.59158700	-0.00075200	0.31942400	0.00340600	2.29807400	1.28842500	0.04197600	0.00498300
Jul	S2	0.0649	0.59758700	-0.00077000	0.32212400	0.00220600	2.35307400	1.31642500	0.04497600	0.00553300
Aug	S3	0.0687	0.54767500	-0.00099438	0.30886175	0.01552800	2.14804950	1.22991900	0.01455750	0.00352388
Sep	S3	0.0516	0.54422500	-0.00098403	0.30730925	0.01621800	2.11642450	1.21381900	0.01283250	0.00320762
Oct	S3	0.0431	0.53974000	-0.00097058	0.30529100	0.01711500	2.07531200	1.19288900	0.01059000	0.00279650
Nov	S4	0.0434	0.51693500	-0.00101893	0.29714775	0.02081300	2.09310750	1.18374900	0.00148350	0.00269937
Dec	S4	0.0449	0.52098500	-0.00103108	0.29897025	0.02000300	2.13023250	1.20264900	0.00350850	0.00307063

5.7 Daily Prediction Records for Result 2.4

This subsection records the day-by-day prediction-error source values for Result 2.4. The sample-count column reports the corrected count after subtracting 10,000 from entries whose supplied count exceeded 10,000. The corrected record spans 2025-01-01–2025-05-25, contains 145 daily entries and totals 1,354,997 samples. The arithmetic mean daily MAE is 0.060607°C and the sample-weighted mean MAE is 0.060774°C. The lowest daily MAE is 0.007079°C on 2025-01-20, and the highest daily MAE is 0.102812°C on 2025-05-15.

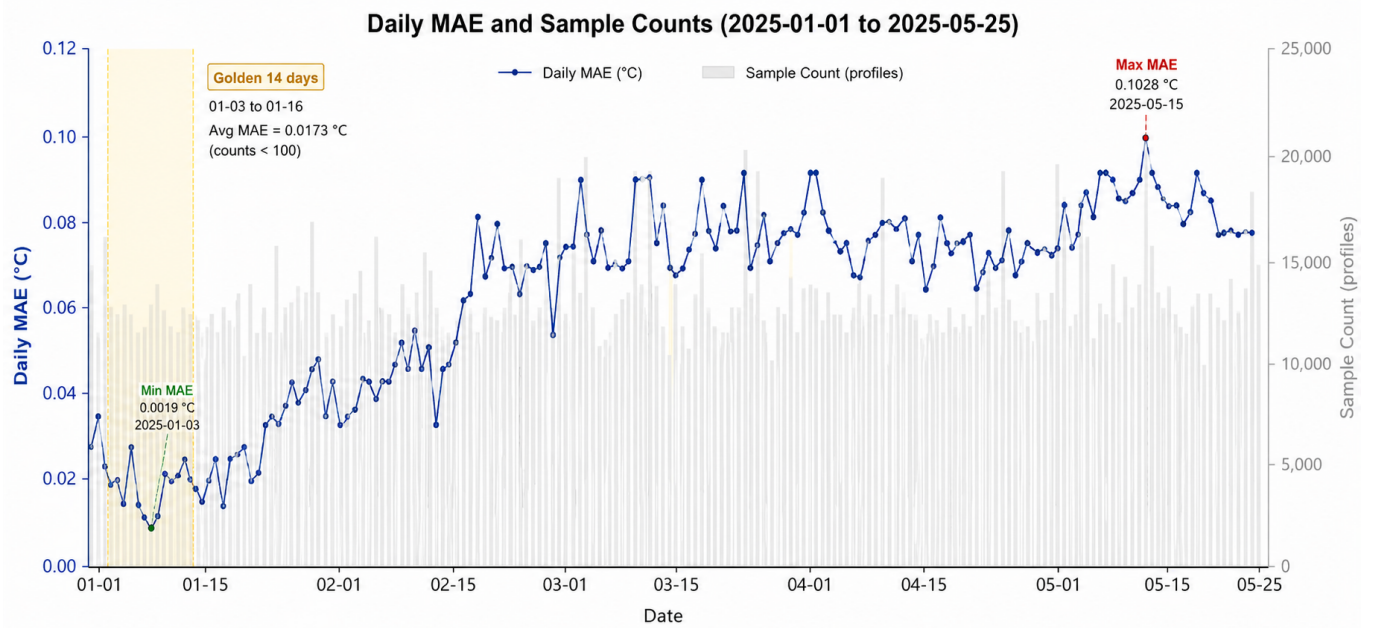


Figure 5: Daily MAE and sample-count visualization for the Result 2.4 prediction record from 2025-01-01 to 2025-05-25. The corrected tabular sample counts used for support-file lookup are reported in Tables 7 and 8.

Table 7: Monthly summary of the corrected daily prediction records for Result 2.4. The weighted MAE uses the corrected daily sample counts.

Month	Days	Sample count	Mean MAE	Weighted MAE	Min MAE	Max MAE
2025-01	31	287257	0.020521	0.020564	0.007079	0.035898
2025-02	28	261991	0.047487	0.047492	0.019342	0.082274
2025-03	31	288916	0.077832	0.078013	0.039598	0.100475
2025-04	30	282145	0.073097	0.073136	0.056453	0.083768
2025-05	25	234688	0.088658	0.088731	0.071020	0.102812
Total	145	1354997	0.060607	0.060774	0.007079	0.102812

Table 8: Complete corrected daily prediction source table for Result 2.4 in compact three-block layout. Each block lists date, MAE and corrected sample count.

Date	MAE	n	Date	MAE	n	Date	MAE	n
2025-01-01	0.018980	8694	2025-01-02	0.027031	9584	2025-01-03	0.009779	9170
2025-01-04	0.012029	9600	2025-01-05	0.018997	9394	2025-01-06	0.017661	8369
2025-01-07	0.021985	9564	2025-01-08	0.014047	9246	2025-01-09	0.022825	9565
2025-01-10	0.012962	8835	2025-01-11	0.022851	8720	2025-01-12	0.019711	9537
2025-01-13	0.015170	9032	2025-01-14	0.020290	9575	2025-01-15	0.014215	9209
2025-01-16	0.020308	8844	2025-01-17	0.014368	9561	2025-01-18	0.019278	9199
2025-01-19	0.025191	9550	2025-01-20	0.007079	9215	2025-01-21	0.024577	9004
2025-01-22	0.021783	9549	2025-01-23	0.031040	9287	2025-01-24	0.022052	9582
2025-01-25	0.026362	9399	2025-01-26	0.033309	8963	2025-01-27	0.035898	9578
2025-01-28	0.019023	9461	2025-01-29	0.030743	9566	2025-01-30	0.016628	9466
2025-01-31	0.019990	8939	2025-02-01	0.019342	9581	2025-02-02	0.030099	9429
2025-02-03	0.026974	9539	2025-02-04	0.025222	9490	2025-02-05	0.031809	8948
2025-02-06	0.030179	9581	2025-02-07	0.038389	9463	2025-02-08	0.030959	9537
2025-02-09	0.041171	9435	2025-02-10	0.032850	8821	2025-02-11	0.038421	9551
2025-02-12	0.019517	9180	2025-02-13	0.032697	9587	2025-02-14	0.031135	9265
2025-02-15	0.039664	8975	2025-02-16	0.054980	9527	2025-02-17	0.056656	9352
2025-02-18	0.082274	9578	2025-02-19	0.060704	9264	2025-02-20	0.070220	8711
2025-02-21	0.080407	9556	2025-02-22	0.063748	9350	2025-02-23	0.068189	9565
2025-02-24	0.055364	9337	2025-02-25	0.064026	8950	2025-02-26	0.064199	9527
2025-02-27	0.064169	9303	2025-02-28	0.076265	9589	2025-03-01	0.039598	8773
2025-03-02	0.068737	8974	2025-03-03	0.072830	9586	2025-03-04	0.067790	9230
2025-03-05	0.094991	9515	2025-03-06	0.072224	9459	2025-03-07	0.072113	8822
2025-03-08	0.077749	9589	2025-03-09	0.064917	9138	2025-03-10	0.064027	9536
2025-03-11	0.065908	9187	2025-03-12	0.066929	8744	2025-03-13	0.091907	9552
2025-03-14	0.093450	9484	2025-03-15	0.092467	9583	2025-03-16	0.064677	9138
2025-03-17	0.100475	9031	2025-03-18	0.078064	9586	2025-03-19	0.087952	9267
2025-03-20	0.093914	9548	2025-03-21	0.064053	9246	2025-03-22	0.082968	8905
2025-03-23	0.072737	9588	2025-03-24	0.067365	9297	2025-03-25	0.075249	9567
2025-03-26	0.074609	9443	2025-03-27	0.081439	9016	2025-03-28	0.090441	9575
2025-03-29	0.080968	9476	2025-03-30	0.097658	9578	2025-03-31	0.094597	9483
2025-04-01	0.080776	9034	2025-04-02	0.081737	9615	2025-04-03	0.075576	9464
2025-04-04	0.075354	9543	2025-04-05	0.064313	9467	2025-04-06	0.060316	8912
2025-04-07	0.077279	9615	2025-04-08	0.073870	9549	2025-04-09	0.080140	9561
2025-04-10	0.081026	9511	2025-04-11	0.079554	9000	2025-04-12	0.083768	9570
2025-04-13	0.064670	9474	2025-04-14	0.076561	9568	2025-04-15	0.056453	9148
2025-04-16	0.073213	8951	2025-04-17	0.081870	9570	2025-04-18	0.068767	9384
2025-04-19	0.080080	9561	2025-04-20	0.078762	9435	2025-04-21	0.071650	9009
2025-04-22	0.072860	9577	2025-04-23	0.058648	9475	2025-04-24	0.067416	9574
2025-04-25	0.065181	9466	2025-04-26	0.076377	9030	2025-04-27	0.078777	9560
2025-04-28	0.062358	9464	2025-04-29	0.068339	9555	2025-04-30	0.077218	9503
2025-05-01	0.075498	8837	2025-05-02	0.075541	9610	2025-05-03	0.071020	9377
2025-05-04	0.086197	9565	2025-05-05	0.080553	9441	2025-05-06	0.076339	8818
2025-05-07	0.093489	9584	2025-05-08	0.087121	9427	2025-05-09	0.099484	9577
2025-05-10	0.102212	9470	2025-05-11	0.096702	9008	2025-05-12	0.090740	9509
2025-05-13	0.095042	9437	2025-05-14	0.098386	9623	2025-05-15	0.102812	9479
2025-05-16	0.089950	9006	2025-05-17	0.090370	9569	2025-05-18	0.090335	9461
2025-05-19	0.087623	9578	2025-05-20	0.084832	9304	2025-05-21	0.087876	9019
2025-05-22	0.095653	9580	2025-05-23	0.090977	9399	2025-05-24	0.090340	9562
2025-05-25	0.077353	9448						

Fixed-equation daily stability record for 2023–2024. The 2023–2024 fixed-equation daily stability record is compiled here as a companion daily evaluation record for the 2025 Result 2.4 prediction table. It evaluates the fixed equation over the daily profile sets preceding the 2025 prediction window, so the 2023–2024 stability record and the 2025 prediction record are grouped under the same support-file subsection. The record spans 2023-01-01–2024-12-31, contains 731 daily entries and totals 8,247,086 profile samples. Across the complete record, the sample-weighted MAE is 0.048157°C, the sample-weighted RMS RMSE is 0.090166°C, and the day-mean R^2 is 0.993586. All 731 daily entries satisfy the stability threshold $\text{MAE} \leq 0.10^\circ\text{C}$ and $\text{RMSE} \leq 0.20^\circ\text{C}$. The lowest daily MAE is 0.0292°C on 2023-12-28, and the highest daily MAE is 0.0707°C on 2023-05-15.

Table 9: Annual summary of the fixed-equation daily stability evaluation record for 2023–2024.

Year	Days	Sample count	Weighted MAE	RMS RMSE	Mean R^2	Stable days
2023	365	4083163	0.049001	0.091616	0.993471	365
2024	366	4163923	0.047330	0.088721	0.993702	366
Total	731	8247086	0.048157	0.090166	0.993586	731

Table 10: Quarterly summary of sample density and fixed-equation stability metrics for 2023–2024.

Period	Days	Mean n	Mean 1 m bins	Spacing (m)	Weighted MAE	RMS RMSE	Mean R^2
2023-Q1	90	11270	3665.9	1.0550	0.043390	0.084329	0.993976
2023-Q2	91	11734	3688.8	1.0484	0.052655	0.095618	0.993192
2023-Q3	92	11134	3605.8	1.0726	0.054728	0.099033	0.992876
2023-Q4	92	10618	3583.7	1.0792	0.044826	0.086158	0.993848
2024-Q1	91	11271	3653.1	1.0587	0.042038	0.078931	0.994280
2024-Q2	91	11700	3632.9	1.0646	0.048210	0.090462	0.993681
2024-Q3	92	11314	3536.1	1.0937	0.052959	0.097610	0.992926
2024-Q4	92	11225	3669.0	1.0541	0.046004	0.086665	0.993925

Table 11: Segment-level summary of the fixed-equation daily stability evaluation record.

Segment	Days	Sample count	Weighted MAE	RMS RMSE	Mean R^2	Stable days
2023 full year	365	4083163	0.049001	0.091616	0.993471	365
2024 H1	182	2090349	0.045182	0.085000	0.993981	182
2024 Jul-Oct	123	1388422	0.052470	0.096840	0.992983	123
2024 Nov-Dec	61	685152	0.043467	0.082408	0.994318	61

Table 12: Monthly lookup summary for the fixed-equation daily stability evaluation record.

Period	Days	Samples	W-MAE	RMS-RMSE	Mean R^2	Period	Days	Samples	W-MAE	RMS-RMSE	Mean R^2
2023-01	31	341331	0.040675	0.080783	0.994029	2024-01	31	341403	0.039999	0.075295	0.994429
2023-02	28	313223	0.043155	0.084122	0.994000	2024-02	29	326131	0.041806	0.078579	0.994134
2023-03	31	359707	0.046172	0.087734	0.993900	2024-03	31	358110	0.044193	0.082553	0.994268
2023-04	30	351343	0.049648	0.092237	0.993517	2024-04	30	349804	0.045998	0.086388	0.994130
2023-05	31	365410	0.052660	0.096119	0.992958	2024-05	31	365943	0.048801	0.091513	0.993423
2023-06	30	350997	0.055659	0.098380	0.993110	2024-06	30	348958	0.049808	0.093300	0.993500
2023-07	31	353546	0.056661	0.100684	0.992581	2024-07	31	369818	0.051997	0.096368	0.993000
2023-08	31	343963	0.055171	0.101114	0.992890	2024-08	31	328975	0.054008	0.099287	0.992981
2023-09	30	326810	0.052169	0.094933	0.993167	2024-09	30	342095	0.052991	0.097316	0.992793
2023-10	31	330004	0.048158	0.089689	0.993700	2024-10	31	347534	0.051005	0.094498	0.993152
2023-11	30	316918	0.044656	0.085580	0.993913	2024-11	30	335165	0.045001	0.085292	0.993993
2023-12	31	329911	0.041657	0.083051	0.993932	2024-12	31	349987	0.041997	0.079548	0.994632

Table 13: Complete fixed-equation daily stability source table for 2023–2024 in compact three-block layout. Each block lists date, MAE, RMSE, R^2 and sample count.

Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n
2023-01-01	0.0450	0.0773	0.9954	10959	2023-01-02	0.0433	0.0917	0.9927	10812	2023-01-03	0.0387	0.0919	0.9951	10549
2023-01-04	0.0392	0.0793	0.9937	11109	2023-01-05	0.0327	0.0740	0.9939	10241	2023-01-06	0.0339	0.0792	0.9926	10556
2023-01-07	0.0312	0.0656	0.9937	11781	2023-01-08	0.0346	0.0807	0.9934	11174	2023-01-09	0.0351	0.0689	0.9932	11349
2023-01-10	0.0339	0.0713	0.9973	10696	2023-01-11	0.0352	0.0847	0.9945	11471	2023-01-12	0.0356	0.0845	0.9920	10619
2023-01-13	0.0379	0.0792	0.9952	10386	2023-01-14	0.0396	0.0813	0.9938	11144	2023-01-15	0.0426	0.0712	0.9948	11371
2023-01-16	0.0412	0.0791	0.9949	11370	2023-01-17	0.0367	0.0797	0.9933	11051	2023-01-18	0.0408	0.0691	0.9941	10335
2023-01-19	0.0396	0.0798	0.9936	10157	2023-01-20	0.0369	0.0715	0.9945	11248	2023-01-21	0.0345	0.0852	0.9953	11029
2023-01-22	0.0366	0.0727	0.9936	11254	2023-01-23	0.0377	0.0765	0.9955	11924	2023-01-24	0.0433	0.0849	0.9934	11308
2023-01-25	0.0484	0.0856	0.9929	10650	2023-01-26	0.0522	0.0945	0.9944	11087	2023-01-27	0.0425	0.0767	0.9944	11094
2023-01-28	0.0466	0.0873	0.9933	11715	2023-01-29	0.0543	0.0986	0.9940	10833	2023-01-30	0.0530	0.0767	0.9937	11435
2023-01-31	0.0565	0.0949	0.9927	10624	2023-02-01	0.0388	0.0773	0.9949	11052	2023-02-02	0.0348	0.0829	0.9953	11041
2023-02-03	0.0456	0.0835	0.9953	11008	2023-02-04	0.0412	0.0829	0.9942	11323	2023-02-05	0.0348	0.0863	0.9943	11322
2023-02-06	0.0301	0.0596	0.9944	11282	2023-02-07	0.0364	0.0827	0.9952	11056	2023-02-08	0.0438	0.0824	0.9948	11420
2023-02-09	0.0436	0.0800	0.9936	11380	2023-02-10	0.0472	0.0964	0.9943	10719	2023-02-11	0.0515	0.0893	0.9921	10731
2023-02-12	0.0586	0.0958	0.9915	11050	2023-02-13	0.0516	0.0933	0.9942	10869	2023-02-14	0.0493	0.0899	0.9935	10387
2023-02-15	0.0518	0.0892	0.9931	11022	2023-02-16	0.0473	0.0809	0.9953	10356	2023-02-17	0.0402	0.0781	0.9951	11560
2023-02-18	0.0444	0.0828	0.9920	11438	2023-02-19	0.0449	0.0969	0.9937	10919	2023-02-20	0.0466	0.0979	0.9942	11364
2023-02-21	0.0440	0.0793	0.9947	11264	2023-02-22	0.0457	0.0838	0.9937	11518	2023-02-23	0.0477	0.0783	0.9930	10991
2023-02-24	0.0412	0.0804	0.9934	11332	2023-02-25	0.0353	0.0835	0.9930	11414	2023-02-26	0.0319	0.0636	0.9941	11588
2023-02-27	0.0397	0.0866	0.9946	12095	2023-02-28	0.0431	0.0722	0.9945	11722	2023-03-01	0.0443	0.0820	0.9942	11318
2023-03-02	0.0485	0.0892	0.9939	11637	2023-03-03	0.0512	0.0901	0.9943	11539	2023-03-04	0.0443	0.0923	0.9955	12008
2023-03-05	0.0376	0.0699	0.9942	11479	2023-03-06	0.0424	0.0774	0.9940	11924	2023-03-07	0.0444	0.0926	0.9941	11122
2023-03-08	0.0438	0.0827	0.9932	11588	2023-03-09	0.0430	0.0789	0.9940	12013	2023-03-10	0.0456	0.0939	0.9945	11463
2023-03-11	0.0462	0.0889	0.9952	11437	2023-03-12	0.0476	0.0966	0.9944	11403	2023-03-13	0.0480	0.1005	0.9934	11616
2023-03-14	0.0576	0.0851	0.9937	11786	2023-03-15	0.0521	0.0890	0.9938	12006	2023-03-16	0.0511	0.0935	0.9925	11281
2023-03-17	0.0503	0.0897	0.9932	11502	2023-03-18	0.0485	0.0998	0.9943	11560	2023-03-19	0.0419	0.0769	0.9946	11666
2023-03-20	0.0434	0.0821	0.9944	11174	2023-03-21	0.0398	0.0758	0.9951	11781	2023-03-22	0.0396	0.0867	0.9933	11395
2023-03-23	0.0421	0.0794	0.9950	11639	2023-03-24	0.0460	0.0967	0.9933	11307	2023-03-25	0.0435	0.0828	0.9932	12364
2023-03-26	0.0485	0.1015	0.9937	11526	2023-03-27	0.0523	0.0928	0.9910	12451	2023-03-28	0.0472	0.0879	0.9937	11651
2023-03-29	0.0424	0.0766	0.9953	11590	2023-03-30	0.0476	0.0889	0.9923	10977	2023-03-31	0.0477	0.0862	0.9931	11467
2023-04-01	0.0524	0.0816	0.9922	12406	2023-04-02	0.0436	0.0765	0.9940	11728	2023-04-03	0.0453	0.0900	0.9949	11115
2023-04-04	0.0544	0.0886	0.9940	10533	2023-04-05	0.0520	0.0930	0.9938	11233	2023-04-06	0.0524	0.1023	0.9944	11489
2023-04-07	0.0509	0.0923	0.9940	10981	2023-04-08	0.0442	0.0963	0.9952	11521	2023-04-09	0.0555	0.0995	0.9934	11464
2023-04-10	0.0486	0.0864	0.9932	11751	2023-04-11	0.0498	0.0868	0.9937	12167	2023-04-12	0.0520	0.1052	0.9935	11619
2023-04-13	0.0468	0.0935	0.9939	11783	2023-04-14	0.0535	0.0943	0.9943	11983	2023-04-15	0.0507	0.0984	0.9924	11949
2023-04-16	0.0536	0.1052	0.9927	12073	2023-04-17	0.0564	0.0951	0.9926	11475	2023-04-18	0.0540	0.0917	0.9932	12785
2023-04-19	0.0557	0.0982	0.9927	11593	2023-04-20	0.0526	0.1022	0.9930	11970	2023-04-21	0.0465	0.0874	0.9954	11263
2023-04-22	0.0434	0.0945	0.9922	12323	2023-04-23	0.0467	0.0894	0.9935	12661	2023-04-24	0.0497	0.0897	0.9926	11821
2023-04-25	0.0500	0.0904	0.9921	11648	2023-04-26	0.0503	0.0890	0.9948	11165	2023-04-27	0.0498	0.0930	0.9940	11749
2023-04-28	0.0444	0.0796	0.9932	11125	2023-04-29	0.0432	0.0855	0.9952	12583	2023-04-30	0.0408	0.0785	0.9937	11680
2023-05-01	0.0524	0.1009	0.9921	12326	2023-05-02	0.0524	0.0900	0.9944	12018	2023-05-03	0.0494	0.0959	0.9928	11620
2023-05-04	0.0557	0.0902	0.9935	12032	2023-05-05	0.0531	0.0889	0.9929	12348	2023-05-06	0.0487	0.0951	0.9926	12021
2023-05-07	0.0518	0.0960	0.9921	11677	2023-05-08	0.0544	0.0982	0.9928	11495	2023-05-09	0.0503	0.0912	0.9923	12174
2023-05-10	0.0572	0.1075	0.9932	11479	2023-05-11	0.0530	0.1035	0.9936	12073	2023-05-12	0.0555	0.0912	0.9938	12067
2023-05-13	0.0605	0.1050	0.9923	11616	2023-05-14	0.0640	0.1068	0.9909	11686	2023-05-15	0.0707	0.1137	0.9911	12221
2023-05-16	0.0552	0.1007	0.9921	11955	2023-05-17	0.0675	0.0992	0.9935	11351	2023-05-18	0.0548	0.0923	0.9931	11497
2023-05-19	0.0521	0.1048	0.9925	12603	2023-05-20	0.0463	0.1036	0.9931	11132	2023-05-21	0.0466	0.0805	0.9940	11479
2023-05-22	0.0455	0.0852	0.9941	10971	2023-05-23	0.0518	0.0910	0.9930	11340	2023-05-24	0.0549	0.1056	0.9930	11568
2023-05-25	0.0542	0.0965	0.9922	11891	2023-05-26	0.0503	0.0935	0.9931	11517	2023-05-27	0.0505	0.0849	0.9914	11005
2023-05-28	0.0445	0.0937	0.9943	12175	2023-05-29	0.0443	0.0904	0.9952	11980	2023-05-30	0.0483	0.0814	0.9935	12457
2023-05-31	0.0482	0.0851	0.9931	11636	2023-06-01	0.0559	0.0959	0.9915	10858	2023-06-02	0.0606	0.1080	0.9921	11143
2023-06-03	0.0569	0.1018	0.9937	11672	2023-06-04	0.0524	0.0890	0.9944	10650	2023-06-05	0.0465	0.0857	0.9965	11867
2023-06-06	0.0451	0.0957	0.9946	12233	2023-06-07	0.0505	0.0992	0.9920	12424	2023-06-08	0.0512	0.1003	0.9922	11601
2023-06-09	0.0526	0.1030	0.9942	11893	2023-06-10	0.0564	0.0856	0.9925	11870	2023-06-11	0.0578	0.1026	0.9943	11372
2023-06-12	0.0665	0.1100	0.9922	11740	2023-06-13	0.0656	0.1035	0.9911	12499	2023-06-14	0.0670	0.1136	0.9915	11575
2023-06-15	0.0659	0.0937	0.9917	12138	2023-06-16	0.0595	0.1033	0.9920	12464	2023-06-17	0.0606	0.1014	0.9936	11170
2023-06-18	0.0596	0.1063	0.9941	11798	2023-06-19	0.0594	0.1096	0.9928	12112	2023-06-20	0.0584	0.1071	0.9927	11234
2023-06-21	0.0578	0.1040	0.9935	11593	2023-06-22	0.0480	0.1030	0.9943	11490	2023-06-23	0.0540	0.0977	0.9941	11949
2023-06-24	0.0539	0.0838	0.9926	12232	2023-06-25	0.0483	0.0834	0.9922	11548	2023-06-26	0.0500	0.0886	0.9935	11657
2023-06-27	0.0475	0.0900	0.9929	11786	2023-06-28	0.0582	0.0878	0.9930	11323	2023-06-29	0.0532	0.0928	0.9940	11970
2023-06-30	0.0559	0.0942	0.9935	11190	2023-07-01	0.0625	0.0975	0.9931	11262	2023-07-02	0.0629	0.0985	0.9916	11256
2023-07-03	0.0663	0.1081	0.9910	11016	2023-07-04	0.0641	0.1055	0.9939	12125	2023-07-05	0.0603	0.1038	0.9935	11646
2023-07-06	0.0560	0.1018	0.9934	11972	2023-07-07	0.0584	0.1101	0.9915	11838	2023-07-08	0.0541	0.0898	0.9915	10955
2023-07-09	0.0486	0.1007	0.9940	11378	2023-07-10	0.0591	0.0981	0.9922	11466	2023-07-11	0.0538	0.0987	0.9905	11198
2023-07-12	0.0589	0.1045	0.9928	12051	2023-07-13	0.0479	0.0985	0.9939	11335	2023-07-14	0.0522	0.0885	0.9914	11800
2023-07-15	0.0642	0.0973	0.9903	11150	2023-07-16	0.0547	0.1026	0.9928	11340	2023-07-17	0.0516	0.1012	0.9936	11989
2023-07-18	0.0538	0.0957	0.9945	11377	2023-07-19	0.0549	0.1060	0.9924	10103	2023-07-20	0.0623	0.0987	0.9915	11392
2023-07-21	0.0612	0.1012	0.9938	11727	2023-07-22	0.0680	0.1005	0.9895	10801	2023-07-23	0.0610	0.1065	0.9919	11938
2023-07-24	0.0568	0.1060	0.992											

Table 13: Complete fixed-equation daily stability source table for 2023–2024, continued.

Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n
2023-10-19	0.0444	0.0827	0.9945	10880	2023-10-20	0.0488	0.0899	0.9935	10055	2023-10-21	0.0465	0.0847	0.9959	10850
2023-10-22	0.0469	0.0860	0.9947	11268	2023-10-23	0.0518	0.1011	0.9922	10398	2023-10-24	0.0516	0.1030	0.9931	10911
2023-10-25	0.0536	0.0877	0.9928	10134	2023-10-26	0.0590	0.0957	0.9925	10123	2023-10-27	0.0533	0.0998	0.9922	11494
2023-10-28	0.0472	0.0923	0.9946	10469	2023-10-29	0.0470	0.0781	0.9928	11210	2023-10-30	0.0477	0.0857	0.9929	10796
2023-10-31	0.0439	0.0806	0.9947	10948	2023-11-01	0.0460	0.0906	0.9943	10965	2023-11-02	0.0498	0.0852	0.9932	9934
2023-11-03	0.0422	0.0858	0.9930	10738	2023-11-04	0.0372	0.0668	0.9937	11155	2023-11-05	0.0372	0.0787	0.9943	10112
2023-11-06	0.0480	0.0903	0.9951	11392	2023-11-07	0.0474	0.0733	0.9936	10292	2023-11-08	0.0445	0.0705	0.9940	10490
2023-11-09	0.0452	0.0879	0.9927	9880	2023-11-10	0.0426	0.0792	0.9939	10294	2023-11-11	0.0325	0.0759	0.9962	10055
2023-11-12	0.0392	0.0766	0.9953	9828	2023-11-13	0.0480	0.0914	0.9950	10593	2023-11-14	0.0551	0.0939	0.9920	10303
2023-11-15	0.0492	0.1015	0.9941	11480	2023-11-16	0.0479	0.0977	0.9921	10457	2023-11-17	0.0541	0.0913	0.9937	11253
2023-11-18	0.0508	0.0936	0.9932	11054	2023-11-19	0.0501	0.1017	0.9913	9958	2023-11-20	0.0477	0.0854	0.9942	10342
2023-11-21	0.0478	0.0842	0.9936	11252	2023-11-22	0.0375	0.0670	0.9960	10247	2023-11-23	0.0435	0.0863	0.9931	10101
2023-11-24	0.0402	0.0920	0.9929	10551	2023-11-25	0.0395	0.0880	0.9945	10412	2023-11-26	0.0417	0.0824	0.9947	10843
2023-11-27	0.0455	0.0720	0.9937	10529	2023-11-28	0.0426	0.0822	0.9934	10837	2023-11-29	0.0435	0.0885	0.9954	10796
2023-11-30	0.0410	0.0714	0.9954	10837	2023-12-01	0.0409	0.0681	0.9937	10344	2023-12-02	0.0405	0.0829	0.9945	11192
2023-12-03	0.0492	0.0929	0.9943	10986	2023-12-04	0.0506	0.0987	0.9935	10520	2023-12-05	0.0451	0.0754	0.9930	10125
2023-12-06	0.0438	0.0921	0.9931	10910	2023-12-07	0.0449	0.0767	0.9936	11243	2023-12-08	0.0401	0.0925	0.9939	10258
2023-12-09	0.0473	0.0988	0.9929	9899	2023-12-10	0.0415	0.0941	0.9948	11308	2023-12-11	0.0436	0.085	0.9932	10554
2023-12-12	0.0430	0.0744	0.9937	11029	2023-12-13	0.0422	0.0794	0.9937	10585	2023-12-14	0.0407	0.0795	0.9947	10886
2023-12-15	0.0409	0.0908	0.9950	10056	2023-12-16	0.0459	0.0894	0.9927	10351	2023-12-17	0.0491	0.0902	0.9930	11225
2023-12-18	0.0432	0.0924	0.9927	11145	2023-12-19	0.0412	0.0748	0.9943	9989	2023-12-20	0.0392	0.0667	0.9921	10694
2023-12-21	0.0477	0.0914	0.9944	10567	2023-12-22	0.0375	0.0786	0.9941	10386	2023-12-23	0.0423	0.0735	0.9942	10709
2023-12-24	0.0459	0.0917	0.9951	11197	2023-12-25	0.0354	0.0857	0.9951	9773	2023-12-26	0.0320	0.0716	0.9947	10939
2023-12-27	0.0294	0.0635	0.9943	10730	2023-12-28	0.0372	0.0952	0.9952	10882	2023-12-29	0.0391	0.0889	0.9942	11422
2023-12-30	0.0414	0.0734	0.9952	10028	2023-12-31	0.0378	0.0717	0.9930	10336	2024-01-01	0.0394	0.0801	0.9944	10558
2024-01-02	0.0334	0.0683	0.9961	11321	2024-01-03	0.0358	0.0659	0.9939	11106	2024-01-04	0.0362	0.0769	0.9939	10380
2024-01-05	0.0374	0.0647	0.9945	10403	2024-01-06	0.0493	0.0776	0.9938	11335	2024-01-07	0.0460	0.0761	0.9928	10856
2024-01-08	0.0464	0.0729	0.9946	10604	2024-01-09	0.0461	0.0686	0.9935	11280	2024-01-10	0.0416	0.0649	0.9945	10450
2024-01-11	0.0391	0.0765	0.9959	11700	2024-01-12	0.0424	0.0720	0.9937	1178	2024-01-13	0.0487	0.0730	0.9926	11141
2024-01-14	0.0492	0.0781	0.9931	11245	2024-01-15	0.0473	0.0873	0.9926	11082	2024-01-16	0.0449	0.0659	0.9944	10203
2024-01-17	0.0436	0.0913	0.9944	10337	2024-01-18	0.0392	0.0789	0.9948	11293	2024-01-19	0.0408	0.0736	0.9942	10339
2024-01-20	0.0377	0.0732	0.9950	11124	2024-01-21	0.0375	0.0708	0.9940	11084	2024-01-22	0.0387	0.0740	0.9953	11054
2024-01-23	0.0388	0.0859	0.9947	12531	2024-01-24	0.0301	0.0737	0.9946	11672	2024-01-25	0.0374	0.0674	0.9945	11458
2024-01-26	0.0359	0.0678	0.9975	11204	2024-01-27	0.0383	0.0780	0.9949	11198	2024-01-28	0.0369	0.0754	0.9938	10924
2024-01-29	0.0366	0.0853	0.9937	11007	2024-01-30	0.0486	0.0741	0.9939	12250	2024-01-31	0.0340	0.0841	0.9936	10721
2024-02-01	0.0393	0.0657	0.9945	10881	2024-02-02	0.0395	0.0815	0.9937	11496	2024-02-03	0.0375	0.0699	0.9945	10882
2024-02-04	0.0430	0.0825	0.9940	11395	2024-02-05	0.0425	0.0691	0.9944	9882	2024-02-06	0.0436	0.0707	0.9947	10893
2024-02-07	0.0380	0.0689	0.9944	11570	2024-02-08	0.0368	0.0729	0.9939	10900	2024-02-09	0.0371	0.0655	0.9947	11751
2024-02-10	0.0413	0.0669	0.9936	11399	2024-02-11	0.0378	0.0733	0.9951	11291	2024-02-12	0.0381	0.0851	0.9954	11693
2024-02-13	0.0446	0.0791	0.9937	11723	2024-02-14	0.0485	0.0878	0.9937	11508	2024-02-15	0.0500	0.0977	0.9937	11300
2024-02-16	0.0434	0.0839	0.9948	11391	2024-02-17	0.0416	0.0792	0.9927	11495	2024-02-18	0.0526	0.0930	0.9928	11079
2024-02-19	0.0427	0.0944	0.9950	11311	2024-02-20	0.0418	0.0782	0.9939	10741	2024-02-21	0.0445	0.0868	0.9944	11895
2024-02-22	0.0438	0.0761	0.9922	11422	2024-02-23	0.0356	0.0789	0.9959	10464	2024-02-24	0.0423	0.0674	0.9933	12554
2024-02-25	0.0412	0.0862	0.9950	10637	2024-02-26	0.0420	0.0703	0.9941	10485	2024-02-27	0.0461	0.0891	0.9935	11815
2024-02-28	0.0389	0.0615	0.9949	11538	2024-02-29	0.0333	0.0794	0.9934	10740	2024-03-01	0.0446	0.0941	0.9943	11189
2024-03-02	0.0485	0.0935	0.9912	11212	2024-03-03	0.0404	0.0862	0.9937	11084	2024-03-04	0.0502	0.0953	0.9951	11351
2024-03-05	0.0569	0.1014	0.9917	11967	2024-03-06	0.0539	0.0916	0.9953	10656	2024-03-07	0.0504	0.0839	0.9932	12338
2024-03-08	0.0510	0.0863	0.9935	11403	2024-03-09	0.0449	0.0755	0.9915	11357	2024-03-10	0.0431	0.0861	0.9946	12038
2024-03-11	0.0496	0.0942	0.9930	10699	2024-03-12	0.0410	0.0795	0.9937	11673	2024-03-13	0.0432	0.0739	0.9965	11273
2024-03-14	0.0436	0.0831	0.9926	10666	2024-03-15	0.0400	0.0678	0.9955	11562	2024-03-16	0.0448	0.0778	0.9953	11287
2024-03-17	0.0450	0.0919	0.9952	12152	2024-03-18	0.0448	0.0777	0.9953	11460	2024-03-19	0.0362	0.0835	0.9949	12526
2024-03-20	0.0356	0.0769	0.9949	11953	2024-03-21	0.0416	0.0721	0.9936	11531	2024-03-22	0.0409	0.0650	0.9946	11633
2024-03-23	0.0429	0.0661	0.9931	11401	2024-03-24	0.0398	0.0685	0.9933	12012	2024-03-25	0.0418	0.0768	0.9948	12157
2024-03-26	0.0451	0.0853	0.9933	11014	2024-03-27	0.0421	0.0656	0.9939	12049	2024-03-28	0.0405	0.0883	0.9936	11802
2024-03-29	0.0365	0.0823	0.9961	11521	2024-03-30	0.0411	0.0900	0.9961	11706	2024-03-31	0.0458	0.0879	0.9960	11320
2024-04-01	0.0493	0.0961	0.9943	11185	2024-04-02	0.0484	0.0899	0.9933	10764	2024-04-03	0.0474	0.0843	0.9942	12119
2024-04-04	0.0511	0.1011	0.9924	12358	2024-04-05	0.0448	0.0833	0.9937	11688	2024-04-06	0.0458	0.0759	0.9930	11245
2024-04-07	0.0512	0.0763	0.9933	11140	2024-04-08	0.0486	0.0935	0.9939	12250	2024-04-09	0.0530	0.0901	0.9931	11541
2024-04-10	0.0547	0.0795	0.9922	10921	2024-04-11	0.0453	0.0885	0.9926	11783	2024-04-12	0.0431	0.0985	0.9938	12110
2024-04-13	0.0434	0.0835	0.9950	11928	2024-04-14	0.0420	0.0807	0.9941	11230	2024-04-15	0.0399	0.0697	0.9961	11574
2024-04-16	0.0416	0.0861	0.9942	11978	2024-04-17	0.0468	0.0747	0.9947	11656	2024-04-18	0.0531	0.0926	0.9931	12769
2024-04-19	0.0417	0.0892	0.9949	11630	2024-04-20	0.0407	0.0739	0.9974	11657	2024-04-21	0.0410	0.0766	0.9963	11534
2024-04-22	0.0410	0.0895	0.9954	10613	2024-04-23	0.0475	0.0925	0.9955	11800	2024-04-24	0.0453	0.0939	0.9949	11245
2024-04-25	0.0446	0.0811	0.9919	11393	2024-04-26	0.0422	0.0910	0.9954	12096	2024-04-27	0.0483	0.0931	0.9932	12436
2024-04-28	0.0412	0.0831	0.9938	12371	2024-04-29	0.0477	0.0756	0.9944	10999	2024-04-30	0.0491	0.0872	0.9936	11731
2024-05-01	0.0481	0.0851	0.9949	11923	2024-05-02	0.0559	0.1005	0.9920	12359	2024-05-03	0.0481	0.0927	0.9941	12327
2024-05-04	0.0491	0.0855	0.9926	11893	2024-05-05	0.0557	0.0915	0.9919	11364	2024-05-06	0.0544	0.0963	0.9933	12446
2024-05-07	0.0548	0.0888	0.9949	11792	2024-05-08	0.0486	0.0914	0.9943	11401	2024-05-09	0.0497	0.0940	0.9937	11068
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Table 13: Complete fixed-equation daily stability source table for 2023–2024, continued.

Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n	Date	MAE	RMSE	R^2	n
2024-08-08	0.0542	0.1012	0.9937	10837	2024-08-09	0.0582	0.1106	0.9935	10800	2024-08-10	0.0530	0.1027	0.9928	9511
2024-08-11	0.0542	0.0970	0.9940	9412	2024-08-12	0.0608	0.0990	0.9919	10365	2024-08-13	0.0642	0.1127	0.9911	11456
2024-08-14	0.0609	0.1125	0.9921	10584	2024-08-15	0.0576	0.1042	0.9931	9488	2024-08-16	0.0587	0.0986	0.9923	9537
2024-08-17	0.0545	0.1072	0.9920	11512	2024-08-18	0.0516	0.1056	0.9929	10054	2024-08-19	0.0465	0.0964	0.9945	10007
2024-08-20	0.0464	0.0895	0.9934	9519	2024-08-21	0.0463	0.0934	0.9935	10366	2024-08-22	0.0580	0.0916	0.9919	10453
2024-08-23	0.0620	0.0992	0.9929	10208	2024-08-24	0.0613	0.1050	0.9914	10435	2024-08-25	0.0570	0.1044	0.9920	11420
2024-08-26	0.0536	0.0905	0.9938	11080	2024-08-27	0.0486	0.1004	0.9936	11344	2024-08-28	0.0490	0.0861	0.9945	10208
2024-08-29	0.0420	0.0932	0.9942	10888	2024-08-30	0.0517	0.0924	0.9931	11496	2024-08-31	0.0503	0.0812	0.9927	11604
2024-09-01	0.0549	0.1091	0.9929	11722	2024-09-02	0.0517	0.0914	0.9925	11662	2024-09-03	0.0538	0.0952	0.9924	11783
2024-09-04	0.0537	0.0959	0.9927	11149	2024-09-05	0.0561	0.1013	0.9938	11184	2024-09-06	0.0385	0.1038	0.9946	11721
2024-09-07	0.0427	0.0906	0.9940	11570	2024-09-08	0.0436	0.0848	0.9944	11516	2024-09-09	0.0502	0.0908	0.9931	11146
2024-09-10	0.0498	0.1030	0.9931	11670	2024-09-11	0.0495	0.0901	0.9929	11748	2024-09-12	0.0472	0.1025	0.9933	12037
2024-09-13	0.0521	0.0941	0.9925	11157	2024-09-14	0.0524	0.0895	0.9934	11403	2024-09-15	0.0522	0.0912	0.9932	10545
2024-09-16	0.0577	0.0900	0.9923	11667	2024-09-17	0.0486	0.0803	0.9935	11378	2024-09-18	0.0521	0.0870	0.9917	10904
2024-09-19	0.0541	0.1097	0.9927	11945	2024-09-20	0.0552	0.0914	0.9915	11544	2024-09-21	0.0494	0.0994	0.9931	11671
2024-09-22	0.0614	0.0958	0.9915	11320	2024-09-23	0.0595	0.0920	0.9912	11101	2024-09-24	0.0531	0.1086	0.9941	12007
2024-09-25	0.0581	0.1000	0.9916	11670	2024-09-26	0.0538	0.1040	0.9940	11592	2024-09-27	0.0565	0.1084	0.9915	11307
2024-09-28	0.0598	0.1041	0.9924	10671	2024-09-29	0.0590	0.1043	0.9922	10465	2024-09-30	0.0656	0.1005	0.9917	10840
2024-10-01	0.0504	0.0935	0.9921	11107	2024-10-02	0.0579	0.0842	0.9917	11113	2024-10-03	0.0467	0.1019	0.9929	11372
2024-10-04	0.0528	0.0980	0.9955	11544	2024-10-05	0.0589	0.1038	0.9927	10548	2024-10-06	0.0656	0.1123	0.9914	11320
2024-10-07	0.0622	0.0925	0.9939	10644	2024-10-08	0.0552	0.0840	0.9930	11907	2024-10-09	0.0496	0.0933	0.9942	11530
2024-10-10	0.0576	0.0989	0.9927	11201	2024-10-11	0.0680	0.1202	0.9922	11614	2024-10-12	0.0637	0.1091	0.9913	12224
2024-10-13	0.0569	0.0928	0.9925	11130	2024-10-14	0.0608	0.1033	0.9910	10777	2024-10-15	0.0651	0.1032	0.9900	10683
2024-10-16	0.0594	0.0923	0.9935	11885	2024-10-17	0.0534	0.0990	0.9920	11677	2024-10-18	0.0404	0.0898	0.9940	11247
2024-10-19	0.0381	0.0902	0.9950	10800	2024-10-20	0.0414	0.0918	0.9940	11284	2024-10-21	0.0395	0.0809	0.9935	11635
2024-10-22	0.0460	0.0859	0.9944	11360	2024-10-23	0.0441	0.0935	0.9945	10716	2024-10-24	0.0333	0.0811	0.9951	10938
2024-10-25	0.0325	0.0781	0.9938	11790	2024-10-26	0.0450	0.0949	0.9933	10831	2024-10-27	0.0429	0.0860	0.9943	10580
2024-10-28	0.0451	0.0946	0.9929	11159	2024-10-29	0.0472	0.0960	0.9932	11672	2024-10-30	0.0538	0.0898	0.9925	10722
2024-10-31	0.0468	0.0772	0.9946	10524	2024-11-01	0.0452	0.0823	0.9940	10620	2024-11-02	0.0501	0.0852	0.9918	11372
2024-11-03	0.0437	0.0932	0.9948	11235	2024-11-04	0.0459	0.0969	0.9946	12026	2024-11-05	0.0499	0.0975	0.9928	11009
2024-11-06	0.0468	0.0851	0.9931	10592	2024-11-07	0.0466	0.0769	0.9935	10912	2024-11-08	0.0424	0.0819	0.9960	10822
2024-11-09	0.0492	0.0841	0.9933	10762	2024-11-10	0.0425	0.0774	0.9964	10290	2024-11-11	0.0450	0.0794	0.9927	11142
2024-11-12	0.0479	0.0833	0.9937	11667	2024-11-13	0.0502	0.0997	0.9935	11284	2024-11-14	0.0482	0.0829	0.9934	10574
2024-11-15	0.0441	0.0901	0.9957	12115	2024-11-16	0.0454	0.0746	0.9945	12173	2024-11-17	0.0432	0.0787	0.9930	10937
2024-11-18	0.0450	0.0953	0.9954	11819	2024-11-19	0.0459	0.0756	0.9937	10858	2024-11-20	0.0365	0.0789	0.9939	10998
2024-11-21	0.0439	0.0947	0.9959	11899	2024-11-22	0.0480	0.0837	0.9932	10653	2024-11-23	0.0387	0.0878	0.9933	10875
2024-11-24	0.0445	0.0878	0.9944	11412	2024-11-25	0.0398	0.0856	0.9952	11133	2024-11-26	0.0433	0.0815	0.9930	10373
2024-11-27	0.0432	0.0821	0.9945	11093	2024-11-28	0.0421	0.0805	0.9941	11561	2024-11-29	0.0428	0.0747	0.9941	11673
2024-11-30	0.0499	0.0897	0.9923	11286	2024-12-01	0.0440	0.0727	0.9931	12217	2024-12-02	0.0443	0.0899	0.9953	12055
2024-12-03	0.0478	0.0928	0.9954	11706	2024-12-04	0.0482	0.0836	0.9934	11389	2024-12-05	0.0473	0.0891	0.9941	10866
2024-12-06	0.0478	0.0943	0.9938	11825	2024-12-07	0.0368	0.0638	0.9953	11141	2024-12-08	0.0424	0.0756	0.9936	11271
2024-12-09	0.0366	0.0860	0.9968	11135	2024-12-10	0.0387	0.0752	0.9952	10861	2024-12-11	0.0378	0.0817	0.9950	11424
2024-12-12	0.0438	0.0841	0.9931	11343	2024-12-13	0.0384	0.0778	0.9956	10950	2024-12-14	0.0403	0.0645	0.9946	10106
2024-12-15	0.0472	0.0751	0.9943	10575	2024-12-16	0.0471	0.0906	0.9938	11597	2024-12-17	0.0431	0.0766	0.9951	12334
2024-12-18	0.0471	0.0861	0.9948	10264	2024-12-19	0.0451	0.0789	0.9946	11208	2024-12-20	0.0458	0.0919	0.9939	11689
2024-12-21	0.0482	0.0702	0.9954	11088	2024-12-22	0.0358	0.0792	0.9956	11443	2024-12-23	0.0328	0.0642	0.9957	11561
2024-12-24	0.0296	0.0649	0.9951	11715	2024-12-25	0.0387	0.0828	0.9948	11049	2024-12-26	0.0430	0.0879	0.9946	11386
2024-12-27	0.0433	0.0662	0.9933	11605	2024-12-28	0.0383	0.0677	0.9954	11461	2024-12-29	0.0407	0.0751	0.9942	10631
2024-12-30	0.0408	0.0881	0.9935	11634	2024-12-31	0.0411	0.0691	0.9952	10458					

5.8 Annual Error Distribution and R^2 Trend

Figure 6 and Table 14 provide the annual-scale source record for the 2015–2024 preview. The annual MAE, RMSE and R^2 columns summarize yearly evaluation metrics, while the box-plot statistics summarize the annual prediction-error distribution. The right-hand panel tracks the same annual R^2 sequence.

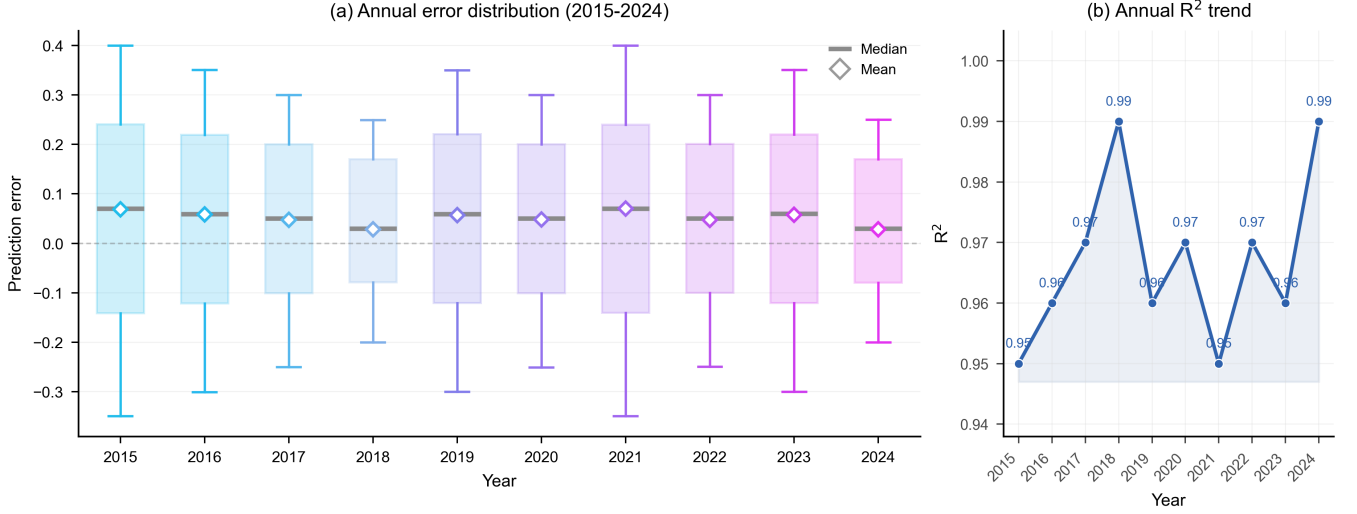


Figure 6: Annual prediction-error distribution and annual R^2 trend for the 2015–2024 preview. Panel (a) summarizes annual error distributions by whisker, quartile, median and mean values; panel (b) reports the annual R^2 sequence.

Table 14: Source table data corresponding to the annual evaluation metrics, prediction-error distribution and annual R^2 trend.

Year	MAE	RMSE	R^2	Lower whisker	Q1	Median	Mean marker	Q3	Upper whisker
2015	0.0685	0.1268	0.9766	-0.3494	-0.1412	0.0669	0.0685	0.2408	0.4003
2016	0.0582	0.1174	0.9812	-0.3006	-0.1214	0.0567	0.0582	0.2196	0.3507
2017	0.0476	0.1096	0.9854	-0.2502	-0.1007	0.0491	0.0476	0.2005	0.3001
2018	0.0284	0.1012	0.9916	-0.2004	-0.0792	0.0312	0.0284	0.1698	0.2495
2019	0.0567	0.1161	0.9827	-0.2997	-0.1205	0.0554	0.0567	0.2209	0.3502
2020	0.0478	0.1089	0.9861	-0.2508	-0.1011	0.0484	0.0478	0.2002	0.3004
2021	0.0699	0.1285	0.9758	-0.3491	-0.1406	0.0711	0.0699	0.2397	0.3998
2022	0.0475	0.1082	0.9868	-0.2496	-0.1003	0.0492	0.0475	0.2007	0.3000
2023	0.0576	0.1158	0.9824	-0.3003	-0.1208	0.0568	0.0576	0.2201	0.3505
2024	0.0285	0.0864	0.9921	-0.1998	-0.0797	0.0293	0.0285	0.1703	0.2502

Across the 2015–2024 sequence, the mean annual MAE is 0.05107°C , the mean annual RMSE is 0.11189°C and the mean annual R^2 is 0.98407. The annual R^2 values range from 0.9758 to 0.9921, with the highest values appearing in 2018 and 2024.

6 Depth-Resolved Reconstruction Accuracy

This section records the 1 m depth-resolved accuracy profile used to support the vertical reconstruction-error assessment. The complete source table is retained as `data/supporting/result2_depth_accuracy_1m.csv`; it contains 3867 depth levels from 159 to 4025 m and 27,107,729 depth-bin samples. The columns in the source table are depth, sample count, MAE, RMSE and the corresponding depth-bin R^2 .

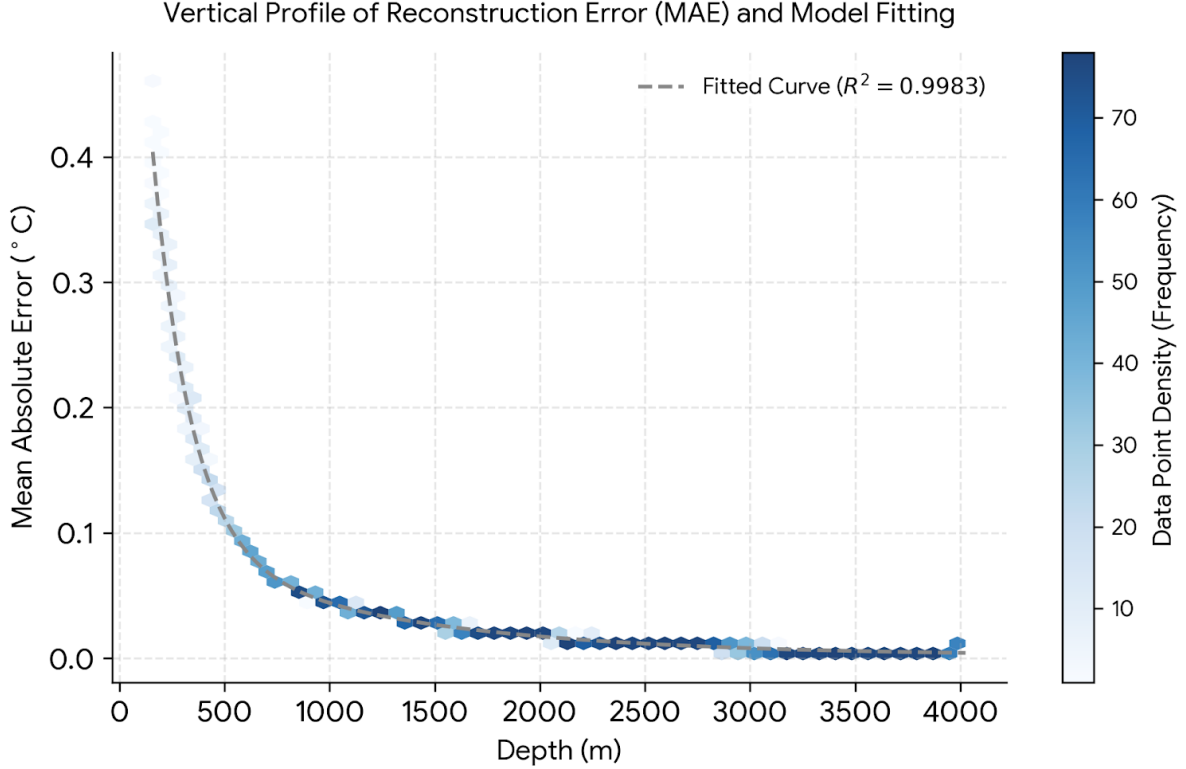


Figure 7: Vertical profile of reconstruction error and fitted depth-decay curve. The horizontal coordinate is depth, the vertical coordinate is MAE, and the point color represents the depth-bin data density. The dashed curve reports the fitted vertical error profile with $R^2 = 0.9983$.

Figure 7 shows a rapid reduction in reconstruction error from the upper-ocean depth range toward the deeper profile. Across the complete 1 m source record, the count-weighted MAE is 0.051075°C and the count-weighted RMSE is 0.065764°C . The largest depth-bin MAE occurs near 166 m, while the lowest MAE occurs near 3833 m.

Table 15: Depth-band summary of the 1 m vertical reconstruction-accuracy source records. MAE and RMSE are count-weighted within each depth band.

Depth band (m)	Number of 1 m bins	Sample count	MAE	RMSE
150–250	91	697476	0.3279	0.4463
250–500	250	2434715	0.1743	0.2339
500–750	250	2543831	0.0829	0.1040
750–1000	250	2672883	0.0518	0.0639
1000–1500	500	5415302	0.0348	0.0430
1500–2000	500	5416099	0.0221	0.0269
2000–2500	500	2243349	0.0148	0.0177
2500–3000	500	2090406	0.0094	0.0114
3000–3500	500	1880966	0.0063	0.0075
3500–4025	526	1712702	0.0052	0.0062

Table 16: Representative 1 m depth-bin records sampled from the complete vertical reconstruction-accuracy source table.

Depth (m)	Sample count	MAE	RMSE	Depth-bin R^2
160	564	0.4569	0.5564	-0.0003
200	7895	0.3358	0.4593	-0.0021
250	8928	0.2722	0.3804	-0.0002
300	9289	0.2203	0.3070	-0.0012
400	9966	0.1526	0.1997	-0.0009
500	10042	0.1103	0.1405	-0.0012
750	10523	0.0619	0.0764	-0.0000
1000	10791	0.0444	0.0549	-0.0245
1500	10761	0.0252	0.0315	-0.0516
2000	10890	0.0186	0.0223	-0.0565
2500	4406	0.0115	0.0140	-0.0050
3000	3972	0.0077	0.0093	-0.0033
3500	3452	0.0055	0.0064	-0.4156
4000	1000	0.0102	0.0105	-18.3876

The representative records and band summaries show the same depth-dependent error attenuation as the fitted curve in Figure 7. The depth-bin R^2 column is retained from the source table for traceability; in very deep low-variance bins it is more sensitive to local variance than MAE and RMSE, so the vertical profile interpretation is primarily based on MAE, RMSE and the fitted MAE-depth relation.

6.1 Complete 1 m Depth-Resolved Source Table

Table 17 compiles the complete 1 m source record into the PDF using a compact four-block landscape layout. Each table row contains up to four independent depth-bin records, which reduces page count while preserving direct lookup by depth.

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table in compact four-block layout. Each row contains up to four independent depth-bin records from `supporting_data/result2_depth_accuracy_1m.csv`.

Depth	<i>n</i>	MAE	RMSE	<i>R</i> ²	Depth	<i>n</i>	MAE	RMSE	<i>R</i> ²	Depth	<i>n</i>	MAE	RMSE	<i>R</i> ²	Depth	<i>n</i>	MAE	RMSE	<i>R</i> ²
159	117	0.37157	0.42636	-0.7273	160	564	0.45688	0.55635	-0.0003	161	2757	0.41363	0.52234	-0.0176	162	4755	0.37918	0.49003	-0.0332
163	5112	0.37805	0.48918	-0.0231	164	4143	0.35149	0.45679	-0.0044	165	4284	0.42910	0.54724	-0.0082	166	5751	0.46069	0.57831	-0.0487
167	7228	0.41707	0.53378	-0.0376	168	7424	0.39479	0.50629	-0.0200	169	6749	0.40058	0.51100	-0.0233	170	6473	0.38900	0.50230	-0.0127
171	7037	0.36708	0.48421	-0.0115	172	7692	0.34965	0.46851	-0.0099	173	7588	0.35097	0.47115	-0.0056	174	7175	0.35974	0.47781	-0.0049
175	7198	0.36170	0.48269	-0.0040	176	7213	0.36125	0.48050	-0.0031	177	7187	0.36193	0.48175	-0.0015	178	7238	0.36124	0.48134	-0.0012
179	7305	0.35849	0.48044	-0.0006	180	7252	0.35498	0.47528	-0.0007	181	7323	0.35773	0.47879	-0.0001	182	7371	0.35511	0.47669	-0.0002
183	7373	0.35070	0.47179	-0.0082	184	7448	0.34819	0.47000	-0.0000	185	7413	0.35020	0.47185	-0.0000	186	7483	0.34739	0.46933	-0.0002
187	7572	0.34433	0.46680	-0.0001	188	6996	0.34677	0.47013	-0.0002	189	7681	0.34327	0.46762	-0.0006	190	7738	0.34575	0.47075	-0.0012
191	7681	0.34557	0.46928	-0.0020	192	7708	0.34753	0.47153	-0.0013	193	7632	0.34645	0.47092	-0.0020	194	7745	0.34499	0.46867	-0.0021
195	7697	0.34259	0.46643	-0.0019	196	7734	0.34323	0.46543	-0.0017	197	7761	0.34209	0.46553	-0.0011	198	7825	0.33913	0.46238	-0.0018
199	7930	0.33750	0.46018	-0.0019	200	7895	0.33583	0.45926	-0.0021	201	7981	0.33811	0.46002	-0.0021	202	8041	0.33384	0.45656	-0.0025
203	8129	0.33057	0.45285	-0.0025	204	8155	0.33005	0.45301	-0.0031	205	8162	0.33104	0.45476	-0.0029	206	8241	0.32934	0.45462	-0.0041
207	8055	0.33232	0.45691	-0.0046	208	8133	0.33156	0.45513	-0.0041	209	8090	0.32816	0.45045	-0.0035	210	8203	0.32555	0.44846	-0.0044
212	8154	0.32565	0.44785	-0.0032	212	8137	0.32332	0.44417	-0.0033	213	8296	0.32277	0.44427	-0.0038	214	8232	0.32103	0.44201	-0.0030
215	8338	0.31991	0.44027	-0.0027	216	8294	0.31909	0.43936	-0.0028	217	8371	0.31572	0.43564	-0.0042	218	8416	0.31195	0.43201	-0.0031
219	8467	0.31354	0.43256	-0.0033	220	8537	0.31273	0.43185	-0.0033	221	8553	0.31043	0.43037	-0.0031	222	8498	0.31209	0.43365	-0.0049
223	8401	0.31155	0.43091	-0.0035	224	8504	0.31318	0.43232	-0.0032	225	8500	0.30872	0.42665	-0.0023	226	8508	0.30841	0.42657	-0.0027
227	8515	0.30539	0.42286	-0.0023	228	8461	0.30466	0.42228	-0.0023	229	8540	0.30488	0.42208	-0.0031	230	8688	0.30090	0.41544	-0.0022
231	8676	0.29837	0.41369	-0.0016	232	8602	0.29834	0.41400	-0.0023	233	8724	0.29476	0.40913	-0.0016	234	8676	0.29466	0.40895	-0.0015
235	8754	0.29280	0.40647	-0.0018	236	8735	0.29312	0.40784	-0.0018	237	8861	0.29261	0.40701	-0.0022	238	8853	0.29044	0.40318	-0.0015
239	8842	0.29105	0.40530	-0.0015	240	8759	0.28699	0.39929	-0.0017	241	8854	0.28654	0.39951	-0.0008	242	8736	0.28554	0.39812	-0.0007
243	8788	0.28281	0.39658	-0.0018	244	8774	0.28373	0.39608	-0.0018	245	8884	0.27637	0.38689	-0.0008	246	8854	0.27637	0.38689	-0.0008
247	8896	0.27667	0.38488	-0.0001	248	8897	0.27444	0.38329	-0.0003	249	8879	0.27353	0.38275	-0.0003	250	8928	0.27217	0.38038	-0.0002
251	8928	0.27329	0.38235	-0.0002	252	9006	0.27150	0.37911	-0.0001	253	9034	0.27100	0.37976	-0.0006	254	9046	0.26944	0.37628	-0.0001
255	9016	0.26846	0.37584	-0.0001	256	8988	0.26842	0.37586	-0.0002	257	8960	0.26753	0.37378	-0.0000	258	8922	0.26606	0.37340	-0.0001
259	9035	0.26333	0.36784	-0.0000	260	8994	0.26165	0.36584	-0.0000	261	8986	0.25977	0.36384	-0.0000	262	9056	0.25785	0.36168	-0.0000
263	9001	0.25755	0.36133	-0.0000	264	9066	0.25750	0.36104	-0.0000	265	9106	0.25424	0.35672	-0.0001	266	9077	0.25422	0.35682	-0.0000
267	9137	0.25538	0.35780	-0.0000	268	9067	0.25033	0.35187	-0.0000	269	9041	0.24957	0.35217	-0.0000	270	9215	0.25050	0.35192	-0.0000
271	9193	0.25130	0.35392	-0.0002	272	9130	0.24930	0.35293	-0.0012	273	9279	0.24536	0.34893	-0.0012	274	9115	0.24562	0.34552	-0.0011
275	9080	0.24469	0.34510	-0.0001	276	9118	0.24278	0.34134	-0.0002	277	9152	0.24116	0.34011	-0.0000	278	9168	0.23913	0.33674	-0.0002
279	9207	0.23887	0.33560	-0.0005	280	9232	0.23839	0.33512	-0.0005	281	9222	0.23900	0.33506	-0.0004	282	9316	0.23726	0.33240	-0.0005
283	9216	0.23612	0.33071	-0.0002	284	9152	0.23526	0.32963	-0.0006	285	9275	0.23625	0.33090	-0.0003	286	9295	0.23346	0.32663	-0.0005
287	9213	0.23139	0.32411	-0.0007	288	9300	0.23289	0.32569	-0.0005	289	9293	0.22974	0.32195	-0.0005	290	9204	0.23162	0.32375	-0.0008
291	9257	0.22852	0.31999	-0.0007	292	9350	0.22764	0.31893	-0.0007	293	9311	0.22645	0.31801	-0.0007	294	9307	0.22591	0.31577	-0.0010
295	9331	0.22343	0.31276	-0.0011	296	9365	0.22215	0.31108	-0.0006	297	9395	0.22198	0.30966	-0.0007	298	9299	0.22167	0.30882	-0.0011
299	9361	0.22252	0.30959	-0.0008	300	9289	0.22034	0.30702	-0.0012	301	9356	0.22026	0.30635	-0.0009	302	9334	0.21803	0.30415	-0.0006
303	9383	0.21802	0.30347	-0.0008	304	9437	0.21685	0.30102	-0.0012	305	9342	0.21665	0.30063	-0.0008	306	9341	0.21552	0.30041	-0.0008
307	9379	0.21654	0.30022	-0.0010	308	9331	0.21430	0.29698	-0.0008	309	9293	0.21426	0.29724	-0.0006	310	9381	0.21200	0.29426	-0.0008
311	9481	0.21156	0.29295	-0.0010	312	9411	0.20960	0.28976	-0.0008	313	9410	0.20910	0.28902	-0.0015	314	9432	0.20913	0.28816	-0.0013
315	9375	0.20984	0.28941	-0.0008	316	9378	0.20808	0.28656	-0.0011	317	9484	0.20684	0.28459	-0.0015	318	9461	0.20572	0.28399	-0.0011
319	9403	0.20484	0.28268	-0.0009	320	9458	0.20432	0.28142	-0.0010	321	9437	0.20468	0.28178	-0.0011	322	9456	0.20486	0.28050	-0.0014
323	9439	0.20383	0.27802	-0.0022	324	9737	0.20735	0.28077	-0.0042	325	10108	0.20999	0.28029	-0.0111	326	10146	0.21085	0.28105	-0.0147
327	9770	0.20466	0.27484	-0.0081	328	9832	0.20437	0.27633	-0.0057	329	9809	0.20154	0.27157	-0.0065	330	9868	0.20255	0.27199	-0.0064
331	9799	0.20102	0.27074	-0.0060	332	9795	0.20041	0.27003	-0.0050	333	9843	0.19947	0.26794	-0.0059	334	9808	0.19791	0.26675	-0.0057
335	9749	0.19745	0.26602	-0.0050	336	9842	0.19601	0.26379	-0.0054	337	9868	0.19587	0.26331	-0.0050	338	9824	0.19490	0.26209	-0.0043
339	9883	0.19384	0.26031	-0.0051	340	9880	0.19310	0.25933	-0.0043	341	9806	0.19267	0.25858	-0.0044	342	9878	0.19197	0.25710	-0.0054
343	9870	0.18968	0.25532	-0.0036	344	9832	0.18912	0.25422	-0.0039	345	9834	0.18803	0.25246	-0.0047	346	9841	0.18830	0.25285	-0.0044
347	9821	0.18780	0.25107	-0.0039	348	9796	0.18597	0.24909	-0.0031	349	9819	0.18499	0.24763	-0.0039	350	10265	0.18382	0.24597	-0.0013
351	9908	0.18508	0.24733	-0.0038	352	9891	0.18213	0.24379	-0.0029	353	9886	0.18346	0.24515	-0.0034	354	9908	0.18266	0.24368	-0.0034
355	9903	0.18089	0.24073	-0.0035	356	9878	0.18032	0.24036	-0.0031	357	9911	0.17829	0.23779	-0.0027	358	9963	0.18003	0.24023	-0.0024
359	9889	0.17691	0.23564	-0.0029	360	9902	0.17581	0.23439	-0.0027	361	9888	0.17644	0.23473	-0.0028	362	9947	0.17564	0.23388	-0.0035
363	9883	0.17423	0.23177	-0.0033	364	9913	0.17340	0.23081	-0.0030	365	9906	0.17297	0.22968	-0.0029	366	9935	0.17270	0.22991	-0.0026
367	9907	0.17218	0.22876	-0.0026	368	9973	0.17098	0.22705	-0.0028	369	9991	0.17061	0.22636	-0.0028	370	9967	0.17105	0.22687	-0.0020
371	9957	0.16887	0.22380	-0.0019	372	9899	0.16904	0.22356	-0.0029	373	9888	0.16781	0.22240	-0.0021	374	9836	0.16741	0.22229	-0.0014
375	9958	0.16637	0.22096	-0.0013	376	9895	0.16569	0.21902	-0.0019	377	9916	0.16494	0.21812	-0.0027	378	9877	0.16581	0.21981	-0.0016
379	9855	0.16435	0.21713	-0.0018	380	9933	0.16375	0.21626	-0.0023	381	9951	0.16205	0.21396	-0.0027	382	9960	0.16275	0.21538	-0.0016
383	9942	0.16090	0.21290	-0.0024	384	10010	0.16146	0.21304											

Table 17: Complete 1 m depth-resolved reconstruction accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
475	9997	0.12066	0.15446	-0.0013	476	10025	0.12012	0.15344	-0.0005	477	10075	0.11918	0.15226	-0.0008	478	10092	0.11837	0.15131	-0.0006
479	10046	0.11822	0.15117	-0.0006	480	10012	0.11783	0.15013	-0.0010	481	10104	0.11774	0.15023	-0.0010	482	10061	0.11711	0.14944	-0.0012
483	10008	0.11638	0.14831	-0.0006	484	9912	0.11640	0.14812	-0.0008	485	10043	0.11594	0.14748	-0.0007	486	10048	0.11541	0.14717	-0.0009
487	10056	0.11482	0.14611	-0.0006	488	10025	0.11408	0.14541	-0.0010	489	9965	0.11423	0.14555	-0.0012	490	9988	0.11415	0.14561	-0.0014
491	10065	0.11371	0.14514	-0.0020	492	10017	0.11294	0.14395	-0.0011	493	10146	0.11254	0.14339	-0.0013	494	10071	0.11205	0.14280	-0.0013
495	10074	0.11218	0.14334	-0.0010	496	10110	0.11133	0.14204	-0.0020	497	10088	0.11138	0.14196	-0.0015	498	10019	0.11057	0.14106	-0.0021
499	10036	0.11036	0.14045	-0.0014	500	10042	0.11033	0.14050	-0.0012	501	10093	0.10938	0.13942	-0.0012	502	10057	0.10978	0.13980	-0.0013
503	10050	0.10915	0.13897	-0.0019	504	10011	0.10874	0.13842	-0.0017	505	9938	0.10870	0.13840	-0.0018	506	10040	0.10814	0.13790	-0.0026
507	9986	0.10828	0.13757	-0.0022	508	10090	0.10814	0.13762	-0.0019	509	10145	0.10716	0.13650	-0.0024	510	10058	0.10671	0.13587	-0.0020
511	10044	0.10627	0.13557	-0.0027	512	10092	0.10623	0.13531	-0.0025	513	10134	0.10576	0.13468	-0.0024	514	10058	0.10531	0.13424	-0.0025
515	10068	0.10516	0.13369	-0.0020	516	9987	0.10444	0.13301	-0.0025	517	10042	0.10479	0.13331	-0.0026	518	10085	0.10415	0.13249	-0.0025
519	10077	0.10330	0.13137	-0.0027	520	10025	0.10358	0.13162	-0.0030	521	10001	0.10320	0.13129	-0.0030	522	9980	0.10233	0.13030	-0.0031
523	10038	0.10293	0.13110	-0.0032	524	10077	0.10229	0.13000	-0.0026	525	10053	0.10170	0.12932	-0.0032	526	10126	0.10170	0.12953	-0.0041
527	10094	0.10135	0.12898	-0.0032	528	10071	0.10096	0.12828	-0.0036	529	10123	0.10100	0.12851	-0.0038	530	10030	0.10114	0.12857	-0.0037
531	10059	0.10057	0.12759	-0.0031	532	10009	0.10048	0.12756	-0.0042	533	10075	0.09973	0.12666	-0.0031	534	10037	0.09933	0.12645	-0.0038
535	10058	0.09936	0.12648	-0.0039	536	10011	0.09928	0.12579	-0.0042	537	10126	0.09843	0.12494	-0.0044	538	10006	0.09863	0.12505	-0.0042
539	10003	0.09879	0.12527	-0.0043	540	10095	0.09809	0.12463	-0.0045	541	10084	0.09818	0.12449	-0.0048	542	10137	0.09784	0.12417	-0.0047
543	10140	0.09776	0.12403	-0.0057	544	10075	0.09727	0.12327	-0.0047	545	10079	0.09724	0.12324	-0.0051	546	10069	0.09712	0.12285	-0.0045
547	9994	0.09669	0.12245	-0.0044	548	10098	0.09622	0.12184	-0.0049	549	10059	0.09630	0.12194	-0.0056	550	10043	0.09587	0.12148	-0.0049
551	10013	0.09589	0.12132	-0.0062	552	10035	0.09521	0.12052	-0.0061	553	10003	0.09548	0.12059	-0.0055	554	10065	0.09528	0.12047	-0.0056
555	10063	0.09430	0.11936	-0.0052	556	10069	0.09480	0.11999	-0.0070	557	10063	0.09435	0.11913	-0.0059	558	10162	0.09451	0.11941	-0.0069
559	10127	0.09420	0.11904	-0.0063	560	10077	0.09422	0.11879	-0.0067	561	10060	0.09372	0.11850	-0.0068	562	10029	0.09370	0.11837	-0.0066
563	10040	0.09395	0.11808	-0.0073	564	10017	0.09322	0.11762	-0.0077	565	10096	0.09256	0.11704	-0.0071	566	10010	0.09252	0.11671	-0.0073
567	10024	0.09234	0.11626	-0.0066	568	10002	0.09219	0.11613	-0.0072	569	10087	0.09187	0.11548	-0.0060	570	10042	0.09171	0.11549	-0.0074
571	10052	0.09201	0.11583	-0.0078	572	10069	0.09110	0.11459	-0.0071	573	10066	0.09150	0.11500	-0.0081	574	10145	0.09122	0.11473	-0.0087
575	10074	0.09093	0.11448	-0.0081	576	10040	0.09075	0.11424	-0.0083	577	10106	0.09072	0.11424	-0.0089	578	10060	0.09034	0.11345	-0.0084
579	10059	0.09021	0.11349	-0.0088	580	10085	0.08978	0.11290	-0.0089	581	10082	0.08971	0.11291	-0.0090	582	10101	0.08981	0.11286	-0.0085
583	10016	0.08924	0.11196	-0.0079	584	10050	0.08937	0.11203	-0.0088	585	10019	0.08895	0.11165	-0.0105	586	10065	0.08888	0.11149	-0.0086
587	10042	0.08890	0.11158	-0.0099	588	10066	0.08839	0.11110	-0.0110	589	10124	0.08793	0.11032	-0.0097	590	10169	0.08897	0.11167	-0.0109
591	10105	0.08887	0.11078	-0.0103	592	10127	0.08810	0.11035	-0.0102	593	10094	0.08789	0.11027	-0.0101	594	10083	0.08774	0.11042	-0.0102
595	10089	0.08761	0.10998	-0.0106	596	10114	0.08738	0.10968	-0.0106	597	10085	0.08675	0.10937	-0.0115	598	10065	0.08693	0.10859	-0.0115
599	10049	0.08694	0.10878	-0.0125	600	10068	0.08609	0.10786	-0.0100	601	10012	0.08635	0.10811	-0.0120	602	10083	0.08634	0.10805	-0.0128
603	10057	0.08570	0.10746	-0.0127	604	10134	0.08623	0.10776	-0.0129	605	10104	0.08556	0.10695	-0.0107	606	10083	0.08617	0.10804	-0.0148
607	10155	0.08526	0.10674	-0.0113	608	10074	0.08532	0.10682	-0.0125	609	10074	0.08509	0.10629	-0.0121	610	10098	0.08508	0.10661	-0.0136
611	10155	0.08488	0.10584	-0.0127	612	10114	0.08503	0.10624	-0.0141	613	10008	0.08429	0.10526	-0.0137	614	10048	0.08437	0.10540	-0.0137
615	10009	0.08388	0.10467	-0.0138	616	10108	0.08375	0.10462	-0.0130	617	10126	0.08322	0.10399	-0.0138	618	10058	0.08315	0.10383	-0.0142
619	10143	0.08319	0.10397	-0.0145	620	10078	0.08309	0.10399	-0.0135	621	10156	0.08282	0.10352	-0.0135	622	10130	0.08270	0.10355	-0.0153
623	10134	0.08262	0.10361	-0.0148	624	10078	0.08242	0.10371	-0.0147	625	10141	0.08266	0.10275	-0.0141	626	10123	0.08262	0.10249	-0.0139
627	10088	0.08221	0.10241	-0.0141	628	10106	0.08153	0.10197	-0.0141	629	10011	0.08161	0.10189	-0.0145	630	10105	0.08110	0.10119	-0.0138
631	10093	0.08137	0.10163	-0.0131	632	10060	0.08149	0.10155	-0.0148	633	10105	0.08069	0.10064	-0.0124	634	10064	0.08082	0.10082	-0.0143
635	10059	0.08075	0.10089	-0.0158	636	10148	0.08026	0.10026	-0.0132	637	10123	0.08037	0.10025	-0.0148	638	10080	0.08014	0.10014	-0.0135
639	10178	0.08012	0.09988	-0.0127	640	10100	0.08005	0.10016	-0.0153	641	10077	0.07979	0.09959	-0.0141	642	10069	0.07936	0.09905	-0.0133
643	10108	0.07934	0.09902	-0.0136	644	10051	0.07893	0.09841	-0.0135	645	10128	0.07933	0.09916	-0.0147	646	10104	0.07869	0.09820	-0.0140
647	10053	0.07860	0.09807	-0.0125	648	10188	0.07833	0.09757	-0.0140	649	10128	0.07789	0.09714	-0.0130	650	10120	0.07810	0.09732	-0.0118
651	10109	0.07770	0.09782	-0.0126	652	10129	0.07723	0.09729	-0.0136	653	10055	0.07795	0.09784	-0.0093	654	10144	0.07744	0.09668	-0.0125
655	10132	0.07677	0.09554	-0.0127	656	10074	0.07733	0.09630	-0.0137	657	10135	0.07693	0.09582	-0.0121	658	10075	0.07660	0.09546	-0.0130
659	10094	0.07628	0.09504	-0.0125	660	10164	0.07644	0.09530	-0.0127	661	10114	0.07635	0.09529	-0.0141	662	10136	0.07589	0.09433	-0.0121
663	10189	0.07532	0.09379	-0.0123	664	10197	0.07554	0.09401	-0.0105	665	10129	0.07517	0.09363	-0.0123	666	10131	0.07477	0.09312	-0.0105
667	10153	0.07466	0.09298	-0.0117	668	10168	0.07475	0.09284	-0.0107	669	10192	0.07433	0.09239	-0.0095	670	10155	0.07461	0.09275	-0.0108
671	10147	0.07412	0.09208	-0.0091	672	10178	0.07451	0.09275	-0.0105	673	10149	0.07364	0.09169	-0.0100	674	10226	0.07379	0.09179	-0.0103
675	10162	0.07356	0.09148	-0.0097	676	10224	0.07323	0.09088	-0.0090	677	10241	0.07375	0.09157	-0.0100	678	10244	0.07321	0.09088	-0.0089
679	10269	0.07318	0.09048	-0.0093	680	10251	0.07263	0.09022	-0.0094	681	10159	0.07255	0.09042	-0.0088	682	10255	0.07232	0.08978	-0.0071
683	10209	0.07218	0.08956	-0.0088	684	10308	0.07244	0.08924	-0.0069	685	10303	0.07200	0.08935	-0.0075	686	10241	0.07190	0.08919	-0.0064
687	10268	0.07155	0.08890	-0.0081	688	10309	0.07157	0.08904	-0.0070	689	10329	0.07117	0.08844	-0.0074	690	10378	0.07119	0.08840	-0.0068
691	10322	0.07096	0.08793	-0.0063	692	10294	0.07052	0.08756	-0.0060	693	10374	0.07093	0.08806	-0.0069	694	10411	0.07028	0.08717	-0.0050
695	10447	0.06994	0.08686	-0.0053	696	10400	0.06950	0.08652	-0.0059	697	10339	0.06966	0.08646	-0.0062	698	10439	0.06968	0.08657	-0.0055

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2
847	10578	0.05329	0.06547	-0.0173	848	10706	0.05304	0.06528	-0.0172	849	10665	0.05311	0.06540	-0.0176	850	10691	0.05297	0.06527	-0.0177
851	10656	0.05287	0.06507	-0.0169	852	10714	0.05275	0.06493	-0.0193	853	10678	0.05248	0.06464	-0.0189	854	10623	0.05248	0.06453	-0.0205
855	10715	0.05241	0.06443	-0.0203	856	10850	0.05266	0.06459	-0.0246	857	10775	0.05244	0.06437	-0.0222	858	10782	0.05268	0.06462	-0.0230
859	10892	0.05234	0.06424	-0.0216	860	10791	0.05214	0.06401	-0.0233	861	10844	0.05237	0.06420	-0.0214	862	10746	0.05220	0.06395	-0.0236
863	10773	0.05218	0.06411	-0.0240	864	10763	0.05214	0.06395	-0.0230	865	10723	0.05202	0.06385	-0.0257	866	10761	0.05169	0.06360	-0.0243
867	10648	0.05214	0.06399	-0.0219	868	10720	0.05135	0.06326	-0.0238	869	10708	0.05167	0.06344	-0.0213	870	10680	0.05162	0.06352	-0.0236
871	10744	0.05177	0.06353	-0.0243	872	10684	0.05151	0.06329	-0.0244	873	10672	0.05147	0.06317	-0.0232	874	10732	0.05146	0.06322	-0.0248
875	10789	0.05121	0.06297	-0.0240	876	10680	0.05127	0.06300	-0.0241	877	10701	0.05116	0.06307	-0.0266	878	10797	0.05138	0.06330	-0.0242
879	10788	0.05115	0.06277	-0.0260	880	10688	0.05104	0.06264	-0.0287	881	10675	0.05101	0.06278	-0.0257	882	10772	0.05085	0.06248	-0.0259
883	10753	0.05114	0.06276	-0.0261	884	10814	0.05079	0.06232	-0.0264	885	10711	0.05071	0.06226	-0.0262	886	10716	0.05064	0.06222	-0.0274
887	10723	0.05039	0.06196	-0.0276	888	10771	0.05062	0.06220	-0.0287	889	10764	0.05058	0.06210	-0.0301	890	10795	0.05026	0.06176	-0.0285
891	10790	0.05020	0.06178	-0.0255	892	10787	0.05039	0.06182	-0.0294	893	10793	0.05023	0.06171	-0.0302	894	10758	0.05018	0.06171	-0.0276
895	10707	0.05013	0.06170	-0.0297	896	10720	0.05019	0.06174	-0.0311	897	10691	0.04989	0.06148	-0.0316	898	10760	0.04959	0.06114	-0.0287
899	10762	0.04993	0.06151	-0.0266	900	10763	0.04961	0.06108	-0.0300	901	10717	0.04946	0.06091	-0.0293	902	10771	0.04936	0.06083	-0.0301
903	10728	0.04944	0.06094	-0.0322	904	10781	0.04964	0.06115	-0.0296	905	10745	0.04915	0.06057	-0.0325	906	10824	0.04923	0.06071	-0.0319
907	10769	0.04912	0.06065	-0.0312	908	10807	0.04926	0.06056	-0.0318	909	10854	0.04919	0.06059	-0.0319	910	10851	0.04919	0.06063	-0.0342
911	10713	0.04916	0.06067	-0.0325	912	10802	0.04891	0.06035	-0.0332	913	10755	0.04889	0.06033	-0.0322	914	10777	0.04884	0.06023	-0.0306
915	10794	0.04892	0.06030	-0.0324	916	10769	0.04854	0.05998	-0.0316	917	10761	0.04842	0.05977	-0.0318	918	10756	0.04832	0.05962	-0.0313
919	10756	0.04842	0.05978	-0.0302	920	10773	0.04834	0.05972	-0.0334	921	10824	0.04824	0.05947	-0.0357	922	10806	0.04821	0.05947	-0.0365
923	10829	0.04813	0.05944	-0.0318	924	10830	0.04845	0.05972	-0.0333	925	10779	0.04830	0.05960	-0.0349	926	10821	0.04817	0.05949	-0.0348
927	10743	0.04817	0.05946	-0.0315	928	10782	0.04815	0.05936	-0.0372	929	10777	0.04787	0.05907	-0.0339	930	10781	0.04781	0.05899	-0.0336
931	10788	0.04782	0.05905	-0.0335	932	10800	0.04773	0.05888	-0.0333	933	10789	0.04768	0.05876	-0.0335	934	10786	0.04777	0.05881	-0.0343
935	10799	0.04787	0.05897	-0.0341	936	10837	0.04770	0.05893	-0.0347	937	10795	0.04751	0.05856	-0.0347	938	10806	0.04744	0.05851	-0.0349
939	10820	0.04756	0.05855	-0.0351	940	10852	0.04767	0.05877	-0.0377	941	10779	0.04734	0.05847	-0.0357	942	10833	0.04743	0.05854	-0.0344
943	10806	0.04741	0.05848	-0.0352	944	10801	0.04706	0.05810	-0.0346	945	10783	0.04693	0.05793	-0.0336	946	10783	0.04685	0.05790	-0.0361
947	10779	0.04695	0.05797	-0.0360	948	10819	0.04698	0.05804	-0.0350	949	10820	0.04668	0.05769	-0.0367	950	10879	0.04691	0.05783	-0.0357
951	10830	0.04654	0.05758	-0.0351	952	10823	0.04669	0.05770	-0.0370	953	10805	0.04674	0.05766	-0.0375	954	10832	0.04644	0.05739	-0.0350
955	10834	0.04651	0.05738	-0.0348	956	10774	0.04635	0.05726	-0.0354	957	10868	0.04671	0.05765	-0.0361	958	10817	0.04645	0.05732	-0.0327
959	10816	0.04644	0.05732	-0.0339	960	10816	0.04641	0.05732	-0.0343	961	10782	0.04612	0.05702	-0.0328	962	10737	0.04628	0.05715	-0.0340
963	10867	0.04641	0.05703	-0.0341	964	10902	0.04644	0.05723	-0.0343	965	10807	0.04618	0.05685	-0.0298	966	10883	0.04602	0.05693	-0.0347
967	10782	0.04605	0.05689	-0.0326	968	10803	0.04608	0.05689	-0.0328	969	10812	0.04597	0.05679	-0.0319	970	10849	0.04594	0.05678	-0.0296
971	10807	0.04570	0.05656	-0.0309	972	10822	0.04574	0.05656	-0.0309	973	10789	0.04581	0.05658	-0.0310	974	10852	0.04583	0.05659	-0.0290
975	10816	0.04579	0.05663	-0.0304	976	10839	0.04586	0.05663	-0.0293	977	10809	0.04545	0.05619	-0.0295	978	10747	0.04552	0.05628	-0.0299
979	10832	0.04562	0.05637	-0.0295	980	10803	0.04540	0.05606	-0.0277	981	10852	0.04530	0.05599	-0.0291	982	10816	0.04539	0.05604	-0.0289
983	10786	0.04525	0.05584	-0.0305	984	10860	0.04526	0.05593	-0.0304	985	10796	0.04518	0.05580	-0.0263	986	10914	0.04489	0.05546	-0.0279
987	10854	0.04481	0.05538	-0.0264	988	10787	0.04494	0.05554	-0.0279	989	10838	0.04486	0.05546	-0.0268	990	10811	0.04505	0.05563	-0.0293
991	10846	0.04498	0.05560	-0.0274	992	10782	0.04486	0.05545	-0.0248	993	10824	0.04460	0.05520	-0.0274	994	10845	0.04482	0.05546	-0.0259
995	10811	0.04476	0.05531	-0.0270	996	10811	0.04470	0.05531	-0.0247	997	10788	0.04451	0.05516	-0.0248	998	10760	0.04458	0.05511	-0.0249
999	10932	0.04444	0.05500	-0.0255	1000	10791	0.04437	0.05488	-0.0245	1001	10805	0.04417	0.05468	-0.0247	1002	10832	0.04410	0.05460	-0.0254
1003	10803	0.04438	0.05487	-0.0252	1004	10756	0.04427	0.05482	-0.0259	1005	10810	0.04410	0.05464	-0.0258	1006	10833	0.04423	0.05472	-0.0259
1007	10887	0.04397	0.05451	-0.0211	1008	10800	0.04394	0.05445	-0.0241	1009	10793	0.04375	0.05418	-0.0231	1010	10860	0.04384	0.05439	-0.0244
1011	10753	0.04397	0.05446	-0.0252	1012	10848	0.04380	0.05415	-0.0211	1013	10772	0.04382	0.05423	-0.0225	1014	10815	0.04399	0.05437	-0.0231
1015	10782	0.04372	0.05410	-0.0223	1016	10765	0.04369	0.05397	-0.0225	1017	10840	0.04346	0.05381	-0.0228	1018	10804	0.04358	0.05396	-0.0221
1019	10824	0.04352	0.05383	-0.0215	1020	10903	0.04361	0.05397	-0.0237	1021	10832	0.04355	0.05383	-0.0219	1022	10849	0.04332	0.05370	-0.0217
1023	10777	0.04347	0.05373	-0.0217	1024	10878	0.04359	0.05395	-0.0245	1025	10797	0.04357	0.05387	-0.0218	1026	10837	0.04331	0.05372	-0.0237
1027	10772	0.04313	0.05342	-0.0196	1028	10860	0.04287	0.05320	-0.0194	1029	10779	0.04304	0.05334	-0.0176	1030	10769	0.04294	0.05320	-0.0183
1031	10889	0.04264	0.05291	-0.0172	1032	10822	0.04263	0.05289	-0.0194	1033	10778	0.04289	0.05316	-0.0175	1034	10813	0.04279	0.05297	-0.0184
1035	10805	0.04273	0.05294	-0.0180	1036	10802	0.04267	0.05279	-0.0173	1037	10803	0.04268	0.05283	-0.0176	1038	10776	0.04245	0.05265	-0.0167
1039	10845	0.04255	0.05272	-0.0158	1040	10854	0.04250	0.05262	-0.0159	1041	10794	0.04259	0.05276	-0.0181	1042	10832	0.04219	0.05226	-0.0159
1043	10822	0.04237	0.05243	-0.0158	1044	10793	0.04220	0.05225	-0.0160	1045	10817	0.04233	0.05241	-0.0153	1046	10830	0.04233	0.05237	-0.0136
1047	10798	0.04219	0.05223	-0.0152	1048	10767	0.04197	0.05206	-0.0136	1049	10814	0.04208	0.05218	-0.0145	1050	10868	0.04189	0.05197	-0.0147
1051	10886	0.04211	0.05207	-0.0146	1052	10841	0.04175	0.05174	-0.0131	1053	10781	0.04176	0.05177	-0.0141	1054	10801	0.04181	0.05187	-0.0146
1055	10861	0.04203	0.05204	-0.0144	1056	10790	0.04173	0.05174	-0.0130	1057	10758	0.04169	0.05178	-0.0120	1058	10821	0.04160	0.05178	-0.0127
1059	10818	0.04169	0.05174	-0.0111	1060	10749	0.04164	0.05170	-0.0114	1061	10804	0.04161	0.05156	-0.0114	1062	10847	0.04137	0.05133	-0.0093
1063	10842	0.04142	0.05133	-0.0117	1064	10839	0.04152	0.05143	-0.0116	1065	10871	0.04135	0.05125	-0.0115	1066	10806	0.04135	0.05115	-0.0108
1067	10949	0.04124	0.05109	-0.0099	1068	10787	0.04134	0.05119	-0.0111	1069	10776	0.0							

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
1219	10833	0.03740	0.04570	-0.0025	1220	10838	0.03726	0.04552	-0.0029	1221	10858	0.03720	0.04548	-0.0028	1222	10850	0.03724	0.04548	-0.0031
1223	10876	0.03712	0.04538	-0.0030	1224	10772	0.03730	0.04554	-0.0035	1225	10835	0.03741	0.04557	-0.0036	1226	10834	0.03688	0.04514	-0.0034
1227	10810	0.03703	0.04525	-0.0036	1228	10835	0.03709	0.04526	-0.0039	1229	10785	0.03701	0.04526	-0.0049	1230	10745	0.03704	0.04523	-0.0038
1231	10797	0.03690	0.04514	-0.0040	1232	10848	0.03696	0.04519	-0.0043	1233	10857	0.03672	0.04486	-0.0039	1234	10861	0.03661	0.04479	-0.0035
1235	10791	0.03655	0.04467	-0.0038	1236	10938	0.03658	0.04479	-0.0049	1237	10847	0.03655	0.04471	-0.0042	1238	10874	0.03644	0.04460	-0.0039
1239	10886	0.03662	0.04475	-0.0046	1240	10784	0.03638	0.04446	-0.0040	1241	10798	0.03640	0.04455	-0.0058	1242	10831	0.03626	0.04436	-0.0051
1243	10866	0.03624	0.04431	-0.0048	1244	10815	0.03653	0.04463	-0.0066	1245	10803	0.03630	0.04439	-0.0048	1246	10833	0.03618	0.04421	-0.0061
1247	10785	0.03613	0.04412	-0.0043	1248	10859	0.03622	0.04423	-0.0057	1249	10841	0.03617	0.04414	-0.0062	1250	10794	0.03600	0.04395	-0.0049
1251	10881	0.03591	0.04387	-0.0059	1252	10941	0.03607	0.04406	-0.0062	1253	10793	0.03604	0.04403	-0.0058	1254	10845	0.03592	0.04392	-0.0068
1255	10828	0.03591	0.04388	-0.0071	1256	10803	0.03582	0.04377	-0.0059	1257	10905	0.03576	0.04377	-0.0073	1258	10817	0.03578	0.04377	-0.0074
1259	10781	0.03588	0.04383	-0.0065	1260	10832	0.03576	0.04362	-0.0078	1261	10809	0.03563	0.04356	-0.0076	1262	10818	0.03568	0.04357	-0.0080
1263	10791	0.03559	0.04343	-0.0091	1264	10897	0.03552	0.04338	-0.0071	1265	10819	0.03548	0.04330	-0.0063	1266	10836	0.03530	0.04317	-0.0066
1267	10848	0.03540	0.04320	-0.0071	1268	10810	0.03529	0.04316	-0.0072	1269	10880	0.03521	0.04301	-0.0080	1270	10816	0.03528	0.04316	-0.0075
1271	10878	0.03516	0.04299	-0.0078	1272	10815	0.03504	0.04290	-0.0087	1273	10824	0.03503	0.04289	-0.0087	1274	10888	0.03506	0.04292	-0.0077
1275	10814	0.03492	0.04272	-0.0084	1276	10858	0.03477	0.04254	-0.0077	1277	10853	0.03490	0.04270	-0.0097	1278	10740	0.03463	0.04247	-0.0101
1279	10910	0.03469	0.04252	-0.0092	1280	10838	0.03443	0.04223	-0.0073	1281	10874	0.03433	0.04210	-0.0088	1282	10808	0.03441	0.04220	-0.0097
1283	10830	0.03445	0.04228	-0.0076	1284	10822	0.03440	0.04224	-0.0092	1285	10840	0.03434	0.04216	-0.0089	1286	10840	0.03405	0.04189	-0.0083
1287	10892	0.03416	0.04188	-0.0095	1288	10803	0.03393	0.04167	-0.0093	1289	10843	0.03394	0.04175	-0.0101	1290	10805	0.03400	0.04183	-0.0098
1291	10772	0.03381	0.04149	-0.0101	1292	10890	0.03362	0.04139	-0.0121	1293	10767	0.03375	0.04150	-0.0110	1294	10886	0.03379	0.04153	-0.0103
1295	10865	0.03353	0.04128	-0.0088	1296	10849	0.03348	0.04117	-0.0095	1297	10808	0.03345	0.04113	-0.0098	1298	10837	0.03328	0.04101	-0.0099
1299	10843	0.03334	0.04101	-0.0102	1300	10813	0.03327	0.04097	-0.0094	1301	10852	0.03313	0.04084	-0.0099	1302	10880	0.03318	0.04085	-0.0108
1303	10821	0.03311	0.04078	-0.0120	1304	10816	0.03303	0.04073	-0.0107	1305	10917	0.03301	0.04067	-0.0115	1306	10771	0.03303	0.04062	-0.0120
1307	10718	0.03279	0.04043	-0.0116	1308	10879	0.03273	0.04031	-0.0120	1309	10874	0.03273	0.04031	-0.0120	1310	10885	0.03290	0.04054	-0.0118
1311	10877	0.03253	0.04012	-0.0097	1312	10808	0.03244	0.03998	-0.0104	1313	10824	0.03248	0.04004	-0.0102	1314	10878	0.03250	0.04007	-0.0118
1315	10832	0.03232	0.03981	-0.0124	1316	10775	0.03239	0.03986	-0.0107	1317	10906	0.03224	0.03977	-0.0112	1318	10839	0.03238	0.03988	-0.0128
1319	10844	0.03218	0.03969	-0.0129	1320	10769	0.03218	0.03972	-0.0151	1321	10810	0.03210	0.03958	-0.0134	1322	10834	0.03205	0.03952	-0.0137
1323	10814	0.03199	0.03952	-0.0137	1324	10888	0.03208	0.03950	-0.0133	1325	10854	0.03208	0.03953	-0.0139	1326	10900	0.03193	0.03935	-0.0138
1327	10827	0.03196	0.03932	-0.0108	1328	10789	0.03172	0.03911	-0.0125	1329	10772	0.03168	0.03906	-0.0129	1330	10876	0.03167	0.03903	-0.0142
1331	10825	0.03158	0.03893	-0.0119	1332	10833	0.03159	0.03890	-0.0124	1333	10841	0.03157	0.03889	-0.0146	1334	10824	0.03141	0.03871	-0.0147
1335	10810	0.03142	0.03883	-0.0146	1336	10799	0.03130	0.03859	-0.0135	1337	10847	0.03138	0.03866	-0.0138	1338	10827	0.03124	0.03841	-0.0149
1339	10830	0.03124	0.03852	-0.0152	1340	10840	0.03124	0.03852	-0.0144	1341	10832	0.03122	0.03845	-0.0144	1342	10839	0.03122	0.03845	-0.0144
1343	10828	0.03098	0.03820	-0.0158	1344	10786	0.03085	0.03802	-0.0125	1345	10863	0.03082	0.03801	-0.0144	1346	10890	0.03088	0.03807	-0.0139
1347	10851	0.03081	0.03796	-0.0158	1348	10820	0.03045	0.03765	-0.0151	1349	10854	0.03071	0.03789	-0.0155	1350	10770	0.03074	0.03789	-0.0159
1351	10777	0.03070	0.03792	-0.0164	1352	10933	0.03058	0.03777	-0.0165	1353	10793	0.03050	0.03766	-0.0170	1354	10810	0.03043	0.03762	-0.0166
1355	10913	0.03038	0.03760	-0.0186	1356	10827	0.03049	0.03774	-0.0180	1357	10800	0.03015	0.03736	-0.0156	1358	10853	0.03017	0.03733	-0.0170
1359	10870	0.03005	0.03721	-0.0160	1360	10836	0.03009	0.03725	-0.0155	1361	10861	0.02995	0.03711	-0.0180	1362	10859	0.02989	0.03700	-0.0184
1363	10803	0.02976	0.03691	-0.0177	1364	10862	0.02985	0.03703	-0.0159	1365	10877	0.02981	0.03699	-0.0175	1366	10793	0.02981	0.03698	-0.0185
1367	10792	0.02972	0.03686	-0.0184	1368	10869	0.02987	0.03698	-0.0168	1369	10868	0.02978	0.03686	-0.0182	1370	10874	0.02981	0.03687	-0.0187
1371	10867	0.02958	0.03673	-0.0199	1372	10866	0.02964	0.03678	-0.0185	1373	10814	0.02925	0.03630	-0.0194	1374	10874	0.02937	0.03641	-0.0159
1375	10767	0.02922	0.03632	-0.0188	1376	10791	0.02919	0.03630	-0.0193	1377	10828	0.02912	0.03607	-0.0197	1378	10803	0.02903	0.03603	-0.0204
1379	10841	0.02906	0.03606	-0.0195	1380	10925	0.02894	0.03592	-0.0184	1381	10795	0.02887	0.03586	-0.0186	1382	10782	0.02901	0.03602	-0.0205
1383	10828	0.02876	0.03577	-0.0228	1384	10795	0.02900	0.03597	-0.0231	1385	10864	0.02870	0.03566	-0.0200	1386	10847	0.02875	0.03576	-0.0200
1387	10931	0.02867	0.03564	-0.0204	1388	10810	0.02851	0.03551	-0.0219	1389	10839	0.02864	0.03553	-0.0198	1390	10862	0.02841	0.03532	-0.0222
1391	10753	0.02854	0.03545	-0.0226	1392	10872	0.02847	0.03542	-0.0227	1393	10767	0.02832	0.03524	-0.0243	1394	10884	0.02816	0.03503	-0.0213
1395	10838	0.02832	0.03528	-0.0243	1396	10839	0.02832	0.03528	-0.0243	1397	10797	0.02819	0.03479	-0.0249	1398	10843	0.02813	0.03515	-0.0247
1399	10834	0.02802	0.03490	-0.0245	1400	10870	0.02809	0.03498	-0.0245	1401	10842	0.02805	0.03493	-0.0254	1402	10949	0.02802	0.03491	-0.0230
1403	10884	0.02790	0.03481	-0.0249	1404	10863	0.02798	0.03484	-0.0245	1405	10842	0.02788	0.03474	-0.0248	1406	10805	0.02771	0.03456	-0.0241
1407	10815	0.02780	0.03467	-0.0263	1408	10830	0.02766	0.03449	-0.0245	1409	10808	0.02754	0.03437	-0.0270	1410	10847	0.02764	0.03435	-0.0248
1411	10841	0.02759	0.03438	-0.0285	1412	10856	0.02747	0.03424	-0.0229	1413	10768	0.02746	0.03422	-0.0283	1414	10842	0.02743	0.03419	-0.0278
1415	10822	0.02739	0.03414	-0.0276	1416	10857	0.02751	0.03432	-0.0270	1417	10835	0.02750	0.03429	-0.0272	1418	10895	0.02733	0.03405	-0.0301
1419	10905	0.02735	0.03412	-0.0280	1420	10846	0.02737	0.03415	-0.0286	1421	10806	0.02727	0.03405	-0.0287	1422	10841	0.02708	0.03387	-0.0306
1423	10780	0.02717	0.03388	-0.0285	1424	10818	0.02691	0.03366	-0.0300	1425	10863	0.02713	0.03377	-0.0287	1426	10828	0.02696	0.03375	-0.0300
1427	10822	0.02701	0.03375	-0.0315	1428	10854	0.02684	0.03355	-0.0314	1429	10874	0.02683	0.03355	-0.0314	1430	10814	0.02677	0.03346	-0.0323
1431	10830	0.02683	0.03357	-0.0321	1432	10902	0.02682	0.03352	-0.0330	1433	10883	0.02688	0.03358	-0.0327	1434	10920	0.02674	0.03338	-0.0310
1435	10820	0.02669	0.03337	-0.0356	1436	10854	0.02670	0.03338	-0.0322	1437	10798	0.02680	0.03353	-0.0345	1438	10767	0.02663	0.03331	-

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
1591	10827	0.02398	0.02969	-0.0593	1592	10809	0.02394	0.02963	-0.0572	1593	10807	0.02389	0.02953	-0.0559	1594	10813	0.02393	0.02955	-0.0587
1595	10797	0.02400	0.02965	-0.0549	1596	10822	0.02389	0.02952	-0.0527	1597	10914	0.02388	0.02953	-0.0566	1598	10859	0.02394	0.02960	-0.0540
1599	10840	0.02401	0.02961	-0.0554	1600	10887	0.02388	0.02954	-0.0548	1601	10833	0.02395	0.02961	-0.0552	1602	10742	0.02388	0.02948	-0.0543
1603	10841	0.02382	0.02940	-0.0562	1604	10817	0.02389	0.02948	-0.0561	1605	10829	0.02389	0.02948	-0.0563	1606	10832	0.02381	0.02939	-0.0520
1607	10802	0.02385	0.02940	-0.0524	1608	10790	0.02370	0.02925	-0.0526	1609	10824	0.02378	0.02928	-0.0540	1610	10858	0.02377	0.02929	-0.0483
1611	10828	0.02374	0.02927	-0.0498	1612	10856	0.02365	0.02915	-0.0487	1613	10782	0.02379	0.02929	-0.0510	1614	10841	0.02366	0.02921	-0.0488
1615	10859	0.02366	0.02919	-0.0477	1616	10812	0.02369	0.02924	-0.0495	1617	10825	0.02362	0.02912	-0.0486	1618	10807	0.02366	0.02914	-0.0477
1619	10814	0.02368	0.02938	-0.0474	1620	10856	0.02372	0.02913	-0.0482	1621	10869	0.02360	0.02903	-0.0474	1622	10859	0.02364	0.02905	-0.0475
1623	10839	0.02348	0.02890	-0.0468	1624	10814	0.02354	0.02893	-0.0477	1625	10847	0.02354	0.02893	-0.0423	1626	10877	0.02361	0.02898	-0.0424
1627	10786	0.02362	0.02904	-0.0432	1628	10772	0.02352	0.02888	-0.0416	1629	10821	0.02342	0.02882	-0.0430	1630	10866	0.02346	0.02882	-0.0422
1631	10839	0.02338	0.02875	-0.0416	1632	10834	0.02345	0.02884	-0.0434	1633	10776	0.02342	0.02877	-0.0429	1634	10883	0.02343	0.02881	-0.0460
1635	10927	0.02337	0.02870	-0.0434	1636	10871	0.02339	0.02874	-0.0440	1637	10838	0.02334	0.02868	-0.0420	1638	10796	0.02326	0.02859	-0.0398
1639	10828	0.02331	0.02866	-0.0396	1640	10783	0.02331	0.02861	-0.0380	1641	10802	0.02342	0.02873	-0.0385	1642	10800	0.02337	0.02866	-0.0367
1643	10822	0.02338	0.02866	-0.0370	1644	10821	0.02314	0.02843	-0.0353	1645	10804	0.02315	0.02843	-0.0354	1646	10892	0.02334	0.02861	-0.0360
1647	10833	0.02320	0.02848	-0.0354	1648	10854	0.02315	0.02845	-0.0363	1649	10832	0.02320	0.02849	-0.0362	1650	10862	0.02321	0.02845	-0.0355
1651	10908	0.02311	0.02842	-0.0308	1652	10895	0.02309	0.02840	-0.0361	1653	10856	0.02329	0.02852	-0.0345	1654	10804	0.02312	0.02826	-0.0304
1655	10838	0.02313	0.02831	-0.0322	1656	10906	0.02314	0.02826	-0.0303	1657	10806	0.02313	0.02827	-0.0314	1658	10856	0.02318	0.02830	-0.0326
1659	10840	0.02317	0.02830	-0.0290	1660	10835	0.02297	0.02807	-0.0282	1661	10795	0.02313	0.02818	-0.0269	1662	10856	0.02308	0.02814	-0.0281
1663	10842	0.02297	0.02808	-0.0286	1664	10803	0.02306	0.02813	-0.0285	1665	10840	0.02292	0.02802	-0.0295	1666	10866	0.02315	0.02819	-0.0290
1667	10846	0.02298	0.02801	-0.0276	1668	10822	0.02311	0.02813	-0.0272	1669	10854	0.02299	0.02804	-0.0278	1670	10745	0.02294	0.02796	-0.0240
1671	10927	0.02296	0.02799	-0.0263	1672	10775	0.02290	0.02790	-0.0239	1673	10839	0.02307	0.02808	-0.0249	1674	10764	0.02293	0.02790	-0.0219
1675	10814	0.02285	0.02786	-0.0217	1676	10769	0.02293	0.02788	-0.0196	1677	10814	0.02277	0.02774	-0.0206	1678	10865	0.02288	0.02785	-0.0207
1679	10866	0.02282	0.02778	-0.0208	1680	10815	0.02294	0.02768	-0.0204	1681	10860	0.02298	0.02773	-0.0222	1682	10784	0.02297	0.02784	-0.0215
1683	10855	0.02283	0.02773	-0.0212	1684	10930	0.02278	0.02769	-0.0201	1685	10859	0.02273	0.02761	-0.0183	1686	10813	0.02271	0.02756	-0.0168
1687	10845	0.02269	0.02756	-0.0162	1688	10803	0.02278	0.02761	-0.0160	1689	10777	0.02275	0.02762	-0.0166	1690	10780	0.02276	0.02760	-0.0153
1691	10773	0.02268	0.02752	-0.0146	1692	10809	0.02260	0.02743	-0.0143	1693	10911	0.02262	0.02746	-0.0133	1694	10913	0.02259	0.02744	-0.0145
1695	10886	0.02257	0.02741	-0.0122	1696	10879	0.02265	0.02749	-0.0140	1697	10845	0.02267	0.02753	-0.0142	1698	10803	0.02262	0.02741	-0.0135
1699	10820	0.02264	0.02743	-0.0135	1700	10828	0.02262	0.02744	-0.0122	1701	10848	0.02259	0.02738	-0.0124	1702	10872	0.02251	0.02729	-0.0115
1703	10829	0.02252	0.02734	-0.0123	1704	10791	0.02249	0.02728	-0.0100	1705	10813	0.02251	0.02733	-0.0095	1706	10754	0.02252	0.02728	-0.0109
1707	10760	0.02252	0.02740	-0.0093	1708	10840	0.02242	0.02717	-0.0085	1709	10847	0.02233	0.02733	-0.0083	1710	10884	0.02235	0.02712	-0.0087
1711	10776	0.02236	0.02712	-0.0084	1712	10873	0.02244	0.02713	-0.0089	1713	10828	0.02237	0.02722	-0.0090	1714	10848	0.02237	0.02727	-0.0091
1715	10842	0.02239	0.02715	-0.0087	1716	10893	0.02227	0.02699	-0.0084	1717	10807	0.02225	0.02697	-0.0074	1718	10858	0.02226	0.02695	-0.0078
1719	10813	0.02230	0.02699	-0.0062	1720	10767	0.02231	0.02701	-0.0057	1721	10796	0.02225	0.02695	-0.0063	1722	10720	0.02217	0.02684	-0.0052
1723	10815	0.02215	0.02681	-0.0048	1724	10796	0.02217	0.02679	-0.0046	1725	10877	0.02219	0.02685	-0.0038	1726	10861	0.02222	0.02681	-0.0043
1727	10832	0.02224	0.02684	-0.0044	1728	10848	0.02218	0.02683	-0.0051	1729	10928	0.02226	0.02689	-0.0042	1730	10836	0.02216	0.02680	-0.0047
1731	10883	0.02215	0.02682	-0.0051	1732	10842	0.02218	0.02683	-0.0039	1733	10870	0.02214	0.02681	-0.0038	1734	10825	0.02215	0.02675	-0.0027
1735	10727	0.02209	0.02674	-0.0028	1736	10804	0.02209	0.02669	-0.0027	1737	10779	0.02208	0.02670	-0.0025	1738	10838	0.02212	0.02673	-0.0018
1739	10821	0.02195	0.02668	-0.0019	1740	10871	0.02204	0.02664	-0.0014	1741	10860	0.02195	0.02659	-0.0015	1742	10863	0.02194	0.02656	-0.0016
1743	10836	0.02200	0.02659	-0.0020	1744	10826	0.02198	0.02654	-0.0017	1745	10851	0.02204	0.02662	-0.0015	1746	10950	0.02197	0.02651	-0.0021
1747	10936	0.02199	0.02654	-0.0007	1748	10851	0.02200	0.02655	-0.0012	1749	10834	0.02199	0.02656	-0.0008	1750	10774	0.02199	0.02651	-0.0005
1751	10770	0.02198	0.02652	-0.0007	1752	10756	0.02190	0.02640	-0.0003	1753	10820	0.02194	0.02646	-0.0002	1754	10829	0.02198	0.02647	-0.0002
1755	10782	0.02196	0.02647	-0.0001	1756	10933	0.02190	0.02637	-0.0001	1757	10885	0.02188	0.02635	-0.0001	1758	10806	0.02199	0.02644	-0.0001
1759	10819	0.02195	0.02641	-0.0001	1760	10826	0.02184	0.02633	-0.0000	1761	10880	0.02192	0.02639	-0.0001	1762	10855	0.02200	0.02650	-0.0000
1763	10852	0.02189	0.02636	-0.0000	1764	10859	0.02183	0.02629	-0.0000	1765	10779	0.02189	0.02637	-0.0000	1766	10750	0.02185	0.02633	-0.0001
1767	10890	0.02183	0.02631	-0.0002	1768	10823	0.02183	0.02628	-0.0003	1769	10849	0.02190	0.02634	-0.0003	1770	10868	0.02198	0.02643	-0.0004
1771	10845	0.02183	0.02629	-0.0005	1772	10820	0.02188	0.02632	-0.0004	1773	10806	0.02188	0.02631	-0.0004	1774	10841	0.02171	0.02618	-0.0004
1775	10795	0.02180	0.02625	-0.0004	1776	10876	0.02175	0.02619	-0.0009	1777	10908	0.02187	0.02632	-0.0010	1778	10813	0.02179	0.02626	-0.0010
1779	10833	0.02179	0.02622	-0.0010	1780	10886	0.02179	0.02619	-0.0013	1781	10819	0.02182	0.02620	-0.0016	1782	10815	0.02177	0.02617	-0.0019
1783	10799	0.02169	0.02608	-0.0022	1784	10852	0.02165	0.02600	-0.0026	1785	10880	0.02172	0.02610	-0.0039	1786	10842	0.02166	0.02601	-0.0030
1787	10840	0.02174	0.02610	-0.0027	1788	10852	0.02171	0.02610	-0.0030	1789	10822	0.02169	0.02606	-0.0026	1790	10873	0.02159	0.02597	-0.0029
1791	10848	0.02157	0.02595	-0.0031	1792	10920	0.02163	0.02598	-0.0037	1793	10832	0.02168	0.02606	-0.0033	1794	10834	0.02166	0.02604	-0.0045
1795	10876	0.02164	0.02601	-0.0031	1796	10752	0.02161	0.02596	-0.0030	1797	10812	0.02171	0.02609	-0.0057	1798	10848	0.02154	0.02586	-0.0053
1799	10797	0.02157	0.02586	-0.0058	1800	10856	0.02162	0.02589	-0.0066	1801	10832	0.02168	0.02585	-0.0065	1802	10774	0.02147	0.02577	-0.0061
1803	10838	0.02157	0.02587	-0.0071	1804	10910	0.02148	0.02580	-0.0069	1805	10862	0.02158	0.02587	-0.0074	1806	10855	0.02154	0.02583	-0.0071
1807	10839	0.02148	0.02581	-0.0076	1808	10821	0.02162	0.02596	-0.0084	1809	10792	0.02154	0.02586	-0.0086	1810	10884	0.02152	0.025	

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2
1963	10816	0.01941	0.02325	-0.0719	1964	10790	0.01943	0.02331	-0.0716	1965	10829	0.01956	0.02341	-0.0736	1966	10865	0.01934	0.02317	-0.0719
1967	10858	0.01948	0.02332	-0.0688	1968	10814	0.01949	0.02329	-0.0747	1969	10850	0.01939	0.02318	-0.0654	1970	10858	0.01929	0.02308	-0.0674
1971	10832	0.01920	0.02298	-0.0667	1972	10839	0.01924	0.02301	-0.0686	1973	10797	0.01917	0.02297	-0.0677	1974	10782	0.01912	0.02290	-0.0709
1975	10769	0.01913	0.02294	-0.0683	1976	10797	0.01910	0.02288	-0.0699	1977	10845	0.01916	0.02292	-0.0661	1978	10791	0.01911	0.02290	-0.0689
1979	10814	0.01917	0.02298	-0.0737	1980	10847	0.01919	0.02295	-0.0708	1981	10903	0.01918	0.02292	-0.0700	1982	10906	0.01915	0.02291	-0.0663
1983	10906	0.01916	0.02293	-0.0657	1984	10816	0.01908	0.02284	-0.0674	1985	10903	0.01899	0.02271	-0.0604	1986	10833	0.01888	0.02260	-0.0599
1987	10854	0.01888	0.02261	-0.0634	1988	10780	0.01880	0.02254	-0.0637	1989	10742	0.01883	0.02257	-0.0664	1990	10752	0.01877	0.02252	-0.0681
1991	10743	0.01880	0.02255	-0.0675	1992	10731	0.01870	0.02252	-0.0637	1993	10818	0.01886	0.02265	-0.0632	1994	10805	0.01878	0.02250	-0.0641
1995	10807	0.01883	0.02255	-0.0681	1996	10857	0.01874	0.02243	-0.0654	1997	10997	0.01868	0.02239	-0.0611	1998	10894	0.01872	0.02244	-0.0587
1999	10926	0.01865	0.02233	-0.0567	2000	10890	0.01857	0.02227	-0.0565	2001	10882	0.01857	0.02225	-0.0568	2002	10872	0.01852	0.02219	-0.0605
2003	10776	0.01843	0.02209	-0.0623	2004	10794	0.01845	0.02210	-0.0641	2005	10703	0.01844	0.02208	-0.0618	2006	10684	0.01852	0.02218	-0.0626
2007	10713	0.01844	0.02207	-0.0604	2008	10738	0.01860	0.02225	-0.0617	2009	10811	0.01849	0.02213	-0.0591	2010	10851	0.01842	0.02204	-0.0562
2011	10862	0.01831	0.02197	-0.0568	2012	10900	0.01839	0.02202	-0.0573	2013	10905	0.01830	0.02189	-0.0504	2014	10968	0.01819	0.02181	-0.0498
2015	10923	0.01819	0.02179	-0.0490	2016	10893	0.01827	0.02187	-0.0531	2017	10863	0.01814	0.02176	-0.0515	2018	10908	0.01819	0.02178	-0.0559
2019	10800	0.01804	0.02163	-0.0568	2020	10791	0.01805	0.02164	-0.0568	2021	10728	0.01799	0.02158	-0.0555	2022	10694	0.01804	0.02164	-0.0535
2023	10779	0.01811	0.02167	-0.0556	2024	10769	0.01810	0.02177	-0.0545	2025	10866	0.01890	0.02161	-0.0500	2026	10820	0.01787	0.02148	-0.0480
2027	10792	0.01784	0.02142	-0.0481	2028	10869	0.01788	0.02144	-0.0481	2029	10944	0.01783	0.02138	-0.0431	2030	10976	0.01782	0.02139	-0.0374
2031	10900	0.01773	0.02128	-0.0436	2032	10969	0.01773	0.02131	-0.0464	2033	10782	0.01771	0.02125	-0.0455	2034	10751	0.01777	0.02129	-0.0493
2035	10866	0.01764	0.02117	-0.0484	2036	10908	0.01763	0.02116	-0.0459	2037	10698	0.01778	0.02132	-0.0476	2038	10708	0.01783	0.02135	-0.0501
2039	10713	0.01776	0.02127	-0.0464	2040	10652	0.01768	0.02119	-0.0428	2041	10739	0.01763	0.02113	-0.0409	2042	10676	0.01766	0.02116	-0.0379
2043	10691	0.01767	0.02113	-0.0385	2044	10753	0.01758	0.02108	-0.0350	2045	10832	0.01762	0.02108	-0.0319	2046	10861	0.01758	0.02105	-0.0330
2047	10938	0.01749	0.02097	-0.0338	2048	10881	0.01752	0.02099	-0.0390	2049	10814	0.01749	0.02095	-0.0379	2050	10773	0.01744	0.02087	-0.0406
2051	10731	0.01748	0.02097	-0.0377	2052	10748	0.01750	0.02105	-0.0377	2053	10825	0.01743	0.02087	-0.0377	2054	10805	0.01744	0.02087	-0.0375
2055	12828	0.01741	0.02087	-0.0420	2056	11020	0.01769	0.02114	-0.0580	2057	8408	0.01802	0.02138	-0.0486	2058	8106	0.01791	0.02119	-0.0180
2059	8689	0.01800	0.02138	-0.0005	2060	7802	0.01825	0.02157	-0.0002	2061	6415	0.01874	0.02196	-0.0171	2062	4922	0.01831	0.02171	-0.2037
2063	4560	0.01819	0.02151	-0.3882	2064	4361	0.01799	0.02138	-0.6264	2065	4257	0.01754	0.02089	-1.0737	2066	4243	0.01732	0.02070	-1.2402
2067	4275	0.01756	0.02096	-1.3076	2068	4341	0.01761	0.02102	-1.3231	2069	4071	0.01728	0.02066	-1.2401	2070	4045	0.01711	0.02048	-1.2676
2071	4073	0.01709	0.02050	-1.2434	2072	4028	0.01697	0.02035	-1.2748	2073	3992	0.01704	0.02042	-1.3021	2074	4006	0.01703	0.02039	-1.2897
2075	3968	0.01700	0.02038	-1.2836	2076	3991	0.01700	0.02035	-1.2710	2077	4274	0.01761	0.02086	-1.4047	2078	5788	0.02000	0.02323	-1.8576
2079	663	0.01748	0.02038	-1.8468	2080	4524	0.01639	0.01969	-1.2637	2081	4274	0.01761	0.02086	-1.4047	2082	2877	0.01351	0.01659	-0.7967
2083	7709	0.01327	0.01633	-0.7583	2084	6268	0.01323	0.01629	-0.7266	2085	2730	0.01629	0.01629	-0.7180	2086	2651	0.01327	0.01632	-0.7512
2087	2637	0.01320	0.01627	-0.7531	2088	2628	0.01312	0.01620	-0.7347	2089	2697	0.01314	0.01617	-0.7212	2090	2673	0.01305	0.01614	-0.7260
2091	2682	0.01306	0.01614	-0.7140	2092	3231	0.01304	0.01627	-0.7619	2093	3810	0.01349	0.01647	-0.8475	2094	3471	0.01337	0.01621	-0.8649
2095	1980	0.01202	0.01498	-0.5779	2096	1245	0.01239	0.01574	-0.2495	2097	1200	0.01180	0.01489	-0.2855	2098	1215	0.01171	0.01478	-0.3110
2099	1239	0.01160	0.01472	-0.2910	2100	1236	0.01152	0.01462	-0.3001	2101	1233	0.01176	0.01487	-0.3031	2102	1236	0.01157	0.01474	-0.2965
2103	1221	0.01147	0.01454	-0.2972	2104	1206	0.01142	0.01450	-0.2864	2105	1218	0.01139	0.01445	-0.2918	2106	1218	0.01135	0.01442	-0.2716
2107	1188	0.01148	0.01452	-0.2881	2108	1236	0.01149	0.01449	-0.2767	2109	1260	0.01158	0.01457	-0.3042	2110	1254	0.01185	0.01488	-0.3189
2111	1218	0.01168	0.01468	-0.2783	2112	1213	0.01137	0.01437	-0.2819	2113	1224	0.01129	0.01417	-0.2847	2114	1230	0.01134	0.01426	-0.2542
2115	1227	0.01136	0.01428	-0.2449	2116	1230	0.01140	0.01435	-0.2563	2117	1224	0.01129	0.01417	-0.2396	2118	1197	0.01151	0.01442	-0.2470
2119	1224	0.01160	0.01465	-0.2454	2120	1218	0.01141	0.01428	-0.2391	2121	1209	0.01111	0.01403	-0.2285	2122	1212	0.01120	0.01413	-0.2219
2123	1230	0.01124	0.01412	-0.2320	2124	1245	0.01136	0.01422	-0.2521	2125	1308	0.01155	0.01443	-0.2451	2126	1230	0.01136	0.01431	-0.2240
2127	1227	0.01115	0.01400	-0.1962	2128	1215	0.01120	0.01412	-0.2163	2129	1206	0.01110	0.01400	-0.2087	2130	1209	0.01110	0.01401	-0.2074
2131	1227	0.01102	0.01387	-0.2031	2132	1224	0.01111	0.01401	-0.1977	2133	1209	0.01109	0.01399	-0.1983	2134	1206	0.01106	0.01390	-0.1960
2135	1212	0.01119	0.01415	-0.1937	2136	1212	0.01103	0.01395	-0.1916	2137	1215	0.01111	0.01396	-0.1764	2138	1215	0.01103	0.01384	-0.1685
2139	1230	0.01113	0.01398	-0.1938	2140	1238	0.01107	0.01397	-0.1948	2141	1218	0.01108	0.01394	-0.1847	2142	1236	0.01084	0.01375	-0.1752
2143	1206	0.01084	0.01371	-0.1689	2144	1224	0.01071	0.01358	-0.1749	2145	1209	0.01080	0.01364	-0.1616	2146	1206	0.01065	0.01352	-0.1618
2147	1218	0.01069	0.01352	-0.1502	2148	1215	0.01059	0.01343	-0.1485	2149	1233	0.01059	0.01340	-0.1591	2150	1230	0.01064	0.01348	-0.1385
2151	1230	0.01074	0.01347	-0.1255	2152	1221	0.01068	0.01355	-0.1399	2153	1257	0.01077	0.01359	-0.1280	2154	1236	0.01081	0.01349	-0.1625
2155	1257	0.01087	0.01351	-0.1715	2156	1257	0.01085	0.01359	-0.1464	2157	1230	0.01054	0.01322	-0.1229	2158	1242	0.01048	0.01315	-0.1263
2159	1242	0.01034	0.01299	-0.1319	2160	1218	0.01049	0.01316	-0.1122	2161	1221	0.01063	0.01331	-0.1107	2162	1221	0.01056	0.01324	-0.1229
2163	1227	0.01048	0.01320	-0.1149	2164	1227	0.01050	0.01320	-0.1138	2165	1221	0.01036	0.01309	-0.1132	2166	1212	0.01066	0.01337	-0.1160
2167	1203	0.01054	0.01325	-0.1022	2168	1252	0.01049	0.01328	-0.1088	2169	1227	0.01053	0.01323	-0.1094	2170	2234	0.01278	0.01559	-0.0645
2171	4894	0.01434	0.01736	-0.4337	2172	4894	0.01434	0.01736	-0.4337	2173	4894	0.01434	0.01736	-0.4337	2174	4894	0.01434	0.01736	-0.4337
2175	5172	0.01415	0.01702	-0.0425	2176	5380	0.01373	0.01665	-0.0319	2177	4997	0.01400	0.01688	-0.0302	2178	4323	0.01462	0.01744	-0.0231
2179	4118	0.01498	0.01774	-0.0237	2180	4468	0.01469	0.01743	-0.0211	2181	4656	0.01455	0.01735	-0.0261	2182	4156	0.01505	0.01785	-0.0227
2183	3674	0.01558	0.01828	-0.0163	2184	3684	0.01555	0.01823	-0.014										

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
2335	4362	0.01342	0.01600	-0.0008	2336	4362	0.01337	0.01593	-0.0007	2337	4408	0.01331	0.01589	-0.0010	2338	4364	0.01335	0.01590	-0.0011
2339	4402	0.01334	0.01589	-0.0012	2340	4456	0.01329	0.01581	-0.0012	2341	4396	0.01326	0.01577	-0.0014	2342	4430	0.01318	0.01569	-0.0026
2343	4408	0.01319	0.01569	-0.0021	2344	4406	0.01314	0.01565	-0.0029	2345	4408	0.01310	0.01560	-0.0030	2346	4414	0.01308	0.01555	-0.0037
2347	4418	0.01306	0.01553	-0.0042	2348	4350	0.01304	0.01546	-0.0034	2349	4414	0.01302	0.01541	-0.0027	2350	4388	0.01302	0.01544	-0.0041
2351	4330	0.01290	0.01535	-0.0035	2352	4338	0.01285	0.01525	-0.0050	2353	4372	0.01282	0.01522	-0.0045	2354	4318	0.01281	0.01521	-0.0047
2355	4388	0.01282	0.01519	-0.0057	2356	4356	0.01282	0.01519	-0.0052	2357	4384	0.01277	0.01510	-0.0077	2358	4344	0.01257	0.01494	-0.0087
2359	4338	0.01256	0.01489	-0.0080	2360	4286	0.01255	0.01486	-0.0111	2361	4276	0.01246	0.01477	-0.0120	2362	4326	0.01239	0.01471	-0.0133
2363	4282	0.01234	0.01467	-0.0125	2364	4320	0.01235	0.01463	-0.0182	2365	4282	0.01231	0.01457	-0.0153	2366	4286	0.01232	0.01458	-0.0176
2367	4460	0.01302	0.01559	-0.0012	2368	4440	0.01312	0.01570	-0.0006	2369	4458	0.01305	0.01564	-0.0004	2370	4470	0.01305	0.01565	-0.0004
2371	4472	0.01308	0.01569	-0.0004	2372	4492	0.01305	0.01565	-0.0006	2373	4490	0.01303	0.01560	-0.0005	2374	4502	0.01295	0.01555	-0.0013
2375	4488	0.01295	0.01556	-0.0003	2376	4454	0.01292	0.01551	-0.0006	2377	4450	0.01297	0.01556	-0.0008	2378	4478	0.01285	0.01545	-0.0008
2379	4488	0.01299	0.01557	-0.0010	2380	4464	0.01299	0.01556	-0.0007	2381	4490	0.01301	0.01559	-0.0005	2382	4464	0.01293	0.01552	-0.0006
2383	4448	0.01297	0.01554	-0.0004	2384	4436	0.01290	0.01550	-0.0004	2385	4506	0.01288	0.01547	-0.0003	2386	4508	0.01295	0.01556	-0.0001
2387	4490	0.01294	0.01553	-0.0001	2388	4478	0.01289	0.01549	-0.0001	2389	4478	0.01283	0.01542	-0.0006	2390	4500	0.01273	0.01532	-0.0004
2391	4508	0.01272	0.01531	-0.0005	2392	4472	0.01271	0.01531	-0.0002	2393	4440	0.01271	0.01530	-0.0002	2394	4500	0.01262	0.01524	-0.0004
2395	4484	0.01277	0.01532	-0.0002	2396	4502	0.01270	0.01530	-0.0002	2397	4464	0.01277	0.01535	-0.0001	2398	4466	0.01276	0.01533	-0.0000
2399	4464	0.01278	0.01534	-0.0002	2400	4456	0.01279	0.01536	-0.0001	2401	4502	0.01279	0.01538	-0.0000	2402	4520	0.01276	0.01531	-0.0000
2403	4482	0.01278	0.01533	-0.0000	2404	4492	0.01272	0.01528	-0.0000	2405	4430	0.01268	0.01522	-0.0001	2406	4508	0.01270	0.01525	-0.0001
2407	4474	0.01260	0.01518	-0.0001	2408	4496	0.01257	0.01515	-0.0001	2409	4442	0.01269	0.01524	-0.0001	2410	4488	0.01268	0.01528	-0.0001
2411	4458	0.01266	0.01523	-0.0000	2412	4436	0.01274	0.01530	-0.0000	2413	4470	0.01273	0.01528	-0.0000	2414	4444	0.01267	0.01522	-0.0001
2415	4472	0.01267	0.01520	-0.0000	2416	4498	0.01274	0.01529	-0.0000	2417	4448	0.01268	0.01522	-0.0000	2418	4462	0.01262	0.01516	-0.0002
2419	4482	0.01261	0.01516	-0.0002	2420	4470	0.01255	0.01510	-0.0000	2421	4434	0.01257	0.01510	-0.0001	2422	4466	0.01248	0.01502	-0.0001
2423	4466	0.01249	0.01497	-0.0004	2424	4458	0.01249	0.01509	-0.0001	2425	4442	0.01243	0.01506	-0.0002	2426	4440	0.01242	0.01494	-0.0003
2427	4434	0.01240	0.01492	-0.0001	2428	4410	0.01247	0.01501	-0.0002	2429	4456	0.01246	0.01499	-0.0003	2430	4460	0.01242	0.01496	-0.0002
2431	4448	0.01242	0.01495	-0.0004	2432	4442	0.01250	0.01502	-0.0002	2433	4458	0.01243	0.01494	-0.0000	2434	4476	0.01238	0.01488	-0.0001
2435	4458	0.01240	0.01493	-0.0005	2436	4458	0.01234	0.01487	-0.0003	2437	4472	0.01232	0.01480	-0.0001	2438	4492	0.01230	0.01477	-0.0000
2439	4480	0.01232	0.01481	-0.0000	2440	4440	0.01228	0.01473	-0.0000	2441	4456	0.01216	0.01460	-0.0002	2442	4404	0.01221	0.01465	-0.0000
2443	4396	0.01212	0.01455	-0.0000	2444	4410	0.01213	0.01458	-0.0001	2445	4384	0.01214	0.01457	-0.0002	2446	4412	0.01212	0.01454	-0.0002
2447	4398	0.01213	0.01452	-0.0001	2448	4362	0.01207	0.01446	-0.0001	2449	4354	0.01204	0.01441	-0.0000	2450	4420	0.01193	0.01428	-0.0000
2451	4392	0.01189	0.01425	-0.0000	2452	4414	0.01196	0.01424	-0.0000	2453	4352	0.01192	0.01438	-0.0000	2454	4368	0.01184	0.01417	-0.0000
2455	4336	0.01180	0.01414	-0.0000	2456	4368	0.01174	0.01424	-0.0003	2457	4384	0.01158	0.01389	-0.0003	2458	4296	0.01166	0.01393	-0.0003
2459	4248	0.01157	0.01385	-0.0006	2460	4250	0.01155	0.01380	-0.0005	2461	4260	0.01153	0.01379	-0.0004	2462	4262	0.01151	0.01376	-0.0007
2463	4226	0.01141	0.01362	-0.0009	2464	4238	0.01141	0.01361	-0.0012	2465	4378	0.01216	0.01466	-0.0022	2466	4434	0.01204	0.01454	-0.0017
2467	4440	0.01201	0.01451	-0.0019	2468	4408	0.01201	0.01451	-0.0023	2469	4442	0.01201	0.01450	-0.0015	2470	4446	0.01202	0.01449	-0.0017
2471	4402	0.01197	0.01447	-0.0019	2472	4376	0.01195	0.01444	-0.0021	2473	4406	0.01189	0.01438	-0.0024	2474	4342	0.01194	0.01446	-0.0040
2475	4348	0.01199	0.01450	-0.0033	2476	4398	0.01190	0.01442	-0.0033	2477	4368	0.01197	0.01450	-0.0047	2478	4400	0.01194	0.01445	-0.0033
2479	4370	0.01191	0.01440	-0.0036	2480	4360	0.01190	0.01441	-0.0033	2481	4388	0.01186	0.01436	-0.0041	2482	4378	0.01187	0.01434	-0.0034
2483	4346	0.01189	0.01439	-0.0039	2484	4374	0.01185	0.01439	-0.0044	2485	4412	0.01181	0.01438	-0.0042	2486	4408	0.01177	0.01433	-0.0043
2487	4368	0.01173	0.01420	-0.0038	2488	4328	0.01178	0.01422	-0.0039	2489	4380	0.01169	0.01417	-0.0057	2490	4352	0.01177	0.01424	-0.0043
2491	4326	0.01174	0.01423	-0.0058	2492	4320	0.01168	0.01416	-0.0053	2493	4368	0.01176	0.01424	-0.0057	2494	4380	0.01166	0.01412	-0.0040
2495	4342	0.01168	0.01415	-0.0048	2496	4352	0.01167	0.01412	-0.0047	2497	4334	0.01165	0.01409	-0.0045	2498	4338	0.01161	0.01406	-0.0047
2499	4324	0.01158	0.01404	-0.0048	2500	4406	0.01154	0.01398	-0.0050	2501	4358	0.01154	0.01399	-0.0053	2502	4416	0.01149	0.01393	-0.0032
2503	4392	0.01155	0.01399	-0.0044	2504	4314	0.01151	0.01395	-0.0052	2505	4374	0.01154	0.01401	-0.0058	2506	4336	0.01152	0.01399	-0.0055
2507	4316	0.01144	0.01391	-0.0077	2508	4330	0.01149	0.01394	-0.0069	2509	4312	0.01149	0.01393	-0.0052	2510	4360	0.01149	0.01392	-0.0066
2511	4352	0.01133	0.01377	-0.0069	2512	4326	0.01145	0.01392	-0.0056	2513	4352	0.01150	0.01392	-0.0052	2514	4318	0.01146	0.01387	-0.0059
2515	4320	0.01138	0.01379	-0.0067	2516	4370	0.01137	0.01379	-0.0068	2517	4356	0.01136	0.01378	-0.0077	2518	4352	0.01134	0.01376	-0.0061
2519	4364	0.01137	0.01376	-0.0063	2520	4340	0.01137	0.01378	-0.0044	2521	4348	0.01140	0.01380	-0.0070	2522	4320	0.01136	0.01375	-0.0061
2523	4322	0.01140	0.01381	-0.0068	2524	4318	0.01135	0.01373	-0.0060	2525	4310	0.01127	0.01366	-0.0069	2526	4356	0.01130	0.01368	-0.0065
2527	4296	0.01129	0.01367	-0.0065	2528	4338	0.01125	0.01359	-0.0061	2529	4326	0.01123	0.01357	-0.0060	2530	4288	0.01123	0.01361	-0.0062
2531	4284	0.01118	0.01353	-0.0078	2532	4288	0.01114	0.01348	-0.0072	2533	4322	0.01109	0.01344	-0.0080	2534	4336	0.01115	0.01350	-0.0070
2535	4332	0.01112	0.01344	-0.0063	2536	4276	0.01103	0.01337	-0.0073	2537	4322	0.01117	0.01350	-0.0064	2538	4320	0.01107	0.01339	-0.0061
2539	4340	0.01120	0.01356	-0.0077	2540	4300	0.01105	0.01337	-0.0073	2541	4298	0.01106	0.01336	-0.0072	2542	4318	0.01108	0.01335	-0.0066
2543	4292	0.01110	0.01347	-0.0058	2544	4296	0.01107	0.01337	-0.0048	2545	4298	0.01093	0.01318	-0.0062	2546	4250	0.01085	0.01318	-0.0053
2547	4242	0.01085	0.01306	-0.0049	2548	4262	0.01078	0.01299	-0.0049	2549	4278	0.01083	0.01305	-0.0053	2550	4296	0.01084	0.01303	-0.0032
2551	4248	0.01079	0.01300	-0.0037	2552	4256	0.01067	0.01287	-0.0047	2553	4264	0.01072	0.01290	-0.0025	2554	4296	0.01072	0.01290	-0.0032
2555	4240	0.01064	0.01280	-0.0026	2556	4222	0.01064	0.01275	-0.0013	2557	4234	0.01046	0.01256	-0.0015	2558	4212	0.01051	0.01259	-0.0006

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2	Depth	<i>n</i>	MAE	RMSE	R^2
2707	4268	0.00967	0.01172	-0.0001	2708	4290	0.00965	0.01168	-0.0000	2709	4238	0.00970	0.01175	-0.0002	2710	4270	0.00963	0.01169	-0.0001
2711	4288	0.00964	0.01168	-0.0000	2712	4284	0.00971	0.01175	-0.0000	2713	4302	0.00966	0.01170	-0.0000	2714	4260	0.00967	0.01171	-0.0001
2715	4280	0.00970	0.01173	-0.0000	2716	4262	0.00958	0.01162	-0.0001	2717	4238	0.00957	0.01161	-0.0001	2718	4286	0.00965	0.01168	-0.0000
2719	4280	0.00959	0.01163	-0.0000	2720	4292	0.00953	0.01156	-0.0000	2721	4310	0.00960	0.01160	-0.0000	2722	4262	0.00957	0.01158	-0.0001
2723	4258	0.00957	0.01158	-0.0000	2724	4300	0.00952	0.01152	-0.0001	2725	4280	0.00952	0.01151	-0.0003	2726	4276	0.00955	0.01154	-0.0002
2727	4284	0.00956	0.01156	-0.0001	2728	4292	0.00952	0.01154	-0.0000	2729	4262	0.00953	0.01153	-0.0002	2730	4264	0.00955	0.01153	-0.0002
2731	4270	0.00949	0.01147	-0.0002	2732	4230	0.00950	0.01151	-0.0001	2733	4248	0.00952	0.01150	-0.0004	2734	4220	0.00946	0.01143	-0.0003
2735	4312	0.00944	0.01141	-0.0005	2736	4266	0.00944	0.01139	-0.0017	2737	4236	0.00944	0.01138	-0.0007	2738	4256	0.00935	0.01133	-0.0003
2739	4216	0.00938	0.01132	-0.0009	2740	4260	0.00931	0.01123	-0.0018	2741	4234	0.00927	0.01120	-0.0019	2742	4240	0.00932	0.01126	-0.0025
2743	4242	0.00934	0.01124	-0.0036	2744	4240	0.00925	0.01117	-0.0022	2745	4208	0.00919	0.01108	-0.0033	2746	4228	0.00926	0.01115	-0.0037
2747	4230	0.00922	0.01108	-0.0038	2748	4200	0.00911	0.01097	-0.0043	2749	4152	0.00912	0.01098	-0.0070	2750	4210	0.00908	0.01094	-0.0063
2751	4222	0.00909	0.01095	-0.0077	2752	4188	0.00902	0.01086	-0.0095	2753	4180	0.00899	0.01082	-0.0086	2754	4198	0.00898	0.01082	-0.0082
2755	4200	0.00900	0.01084	-0.0114	2756	4122	0.00892	0.01074	-0.0104	2757	4104	0.00889	0.01068	-0.0124	2758	4222	0.00921	0.01115	-0.0044
2759	4290	0.00944	0.01144	-0.0003	2760	4292	0.00936	0.01135	-0.0004	2761	4282	0.00945	0.01145	-0.0003	2762	4290	0.00940	0.01139	-0.0003
2763	4244	0.00937	0.01136	-0.0001	2764	4242	0.00936	0.01136	-0.0002	2765	4238	0.00937	0.01137	-0.0007	2766	4254	0.00935	0.01134	-0.0004
2767	4254	0.00940	0.01137	-0.0007	2768	4240	0.00939	0.01140	-0.0001	2769	4252	0.00930	0.01128	-0.0007	2770	4258	0.00927	0.01125	-0.0005
2771	4216	0.00932	0.01130	-0.0003	2772	4238	0.00933	0.01130	-0.0005	2773	4212	0.00932	0.01130	-0.0006	2774	4212	0.00933	0.01131	-0.0007
2775	4234	0.00929	0.01126	-0.0004	2776	4316	0.00924	0.01121	-0.0002	2777	4298	0.00931	0.01128	-0.0009	2778	4286	0.00936	0.01132	-0.0016
2779	4234	0.00927	0.01121	-0.0011	2780	4248	0.00924	0.01120	-0.0007	2781	4280	0.00922	0.01116	-0.0013	2782	4292	0.00918	0.01113	-0.0016
2783	4230	0.00922	0.01117	-0.0009	2784	4260	0.00923	0.01121	-0.0011	2785	4322	0.00921	0.01117	-0.0010	2786	4262	0.00913	0.01106	-0.0013
2787	4296	0.00914	0.01109	-0.0012	2788	4260	0.00916	0.01112	-0.0006	2789	4230	0.00920	0.01115	-0.0019	2790	4246	0.00913	0.01107	-0.0014
2791	4326	0.00920	0.01115	-0.0012	2792	4330	0.00913	0.01107	-0.0018	2793	4296	0.00911	0.01107	-0.0012	2794	4260	0.00918	0.01110	-0.0017
2795	4264	0.00914	0.01111	-0.0017	2796	4298	0.00914	0.01107	-0.0017	2797	4298	0.00913	0.01103	-0.0017	2798	4266	0.00907	0.01100	-0.0015
2799	4226	0.00910	0.01106	-0.0016	2800	4232	0.00909	0.01105	-0.0013	2801	4246	0.00910	0.01105	-0.0006	2802	4220	0.00907	0.01103	-0.0010
2803	4242	0.00908	0.01103	-0.0014	2804	4218	0.00907	0.01100	-0.0011	2805	4192	0.00908	0.01100	-0.0014	2806	4234	0.00903	0.01096	-0.0010
2807	4226	0.00904	0.01096	-0.0010	2808	4266	0.00903	0.01097	-0.0016	2809	4244	0.00900	0.01093	-0.0016	2810	4250	0.00905	0.01096	-0.0023
2811	4246	0.00903	0.01094	-0.0023	2812	4242	0.00898	0.01090	-0.0018	2813	4272	0.00895	0.01086	-0.0014	2814	4230	0.00902	0.01092	-0.0016
2815	4222	0.00895	0.01083	-0.0018	2816	4244	0.00896	0.01087	-0.0021	2817	4276	0.00893	0.01084	-0.0010	2818	4246	0.00892	0.01086	-0.0014
2819	4242	0.00891	0.01081	-0.0025	2820	4248	0.00895	0.01085	-0.0027	2821	4272	0.00891	0.01081	-0.0029	2822	4222	0.00895	0.01086	-0.0027
2823	4256	0.00891	0.01078	-0.0023	2824	4276	0.00895	0.01077	-0.0022	2825	4244	0.00899	0.01083	-0.0029	2826	4278	0.00897	0.01087	-0.0026
2827	4294	0.00885	0.01075	-0.0041	2828	4272	0.00888	0.01073	-0.0050	2829	4272	0.00888	0.01073	-0.0039	2830	4260	0.00880	0.01067	-0.0046
2831	4228	0.00881	0.01068	-0.0057	2832	4220	0.00873	0.01060	-0.0052	2833	4238	0.00864	0.01047	-0.0050	2834	4176	0.00869	0.01052	-0.0042
2835	4154	0.00864	0.01046	-0.0059	2836	4170	0.00860	0.01040	-0.0051	2837	4138	0.00861	0.01041	-0.0058	2838	4128	0.00853	0.01032	-0.0050
2839	4142	0.00856	0.01034	-0.0066	2840	4128	0.00853	0.01032	-0.0052	2841	4152	0.00846	0.01022	-0.0083	2842	4126	0.00844	0.01019	-0.0071
2843	4152	0.00848	0.01023	-0.0077	2844	4156	0.00850	0.01026	-0.0108	2845	4164	0.00847	0.01021	-0.0145	2846	4174	0.00846	0.01021	-0.0148
2847	4142	0.00842	0.01017	-0.0134	2848	4156	0.00842	0.01017	-0.0178	2849	4146	0.00836	0.01010	-0.0186	2850	4176	0.00845	0.01019	-0.0196
2851	4124	0.00841	0.01015	-0.0195	2852	4082	0.00832	0.01004	-0.0215	2853	4066	0.00829	0.01000	-0.0228	2854	4074	0.00829	0.00998	-0.0239
2855	4036	0.00815	0.00986	-0.0236	2856	4036	0.00815	0.00986	-0.0236	2857	4034	0.00815	0.00986	-0.0236	2858	4170	0.00815	0.00986	-0.0239
2859	4170	0.00868	0.01054	-0.0006	2860	4140	0.00864	0.01048	-0.0025	2861	4134	0.00864	0.01049	-0.0009	2862	4130	0.00863	0.01047	-0.0013
2863	4098	0.00858	0.01041	-0.0009	2864	4126	0.00855	0.01040	-0.0017	2865	4124	0.00854	0.01038	-0.0019	2866	4136	0.00859	0.01045	-0.0011
2867	4110	0.00858	0.01042	-0.0011	2868	4112	0.00857	0.01041	-0.0022	2869	4088	0.00854	0.01039	-0.0012	2870	4056	0.00856	0.01041	-0.0009
2871	4058	0.00849	0.01034	-0.0011	2872	4068	0.00849	0.01034	-0.0007	2873	4032	0.00845	0.01031	-0.0008	2874	4054	0.00849	0.01035	-0.0003
2875	4046	0.00841	0.01028	-0.0003	2876	4028	0.00848	0.01035	-0.0005	2877	4036	0.00849	0.01036	-0.0005	2878	3994	0.00834	0.01021	-0.0004
2879	4020	0.00840	0.01025	-0.0005	2880	4020	0.00841	0.01027	-0.0004	2881	4032	0.00838	0.01024	-0.0003	2882	4020	0.00832	0.01018	-0.0003
2883	4040	0.00835	0.01025	-0.0003	2884	4032	0.00835	0.01027	-0.0003	2885	4034	0.00835	0.01027	-0.0003	2886	4016	0.00842	0.01024	-0.0003
2887	4008	0.00839	0.01024	-0.0005	2888	4006	0.00840	0.01023	-0.0004	2889	3996	0.00837	0.01021	-0.0003	2890	4022	0.00838	0.01022	-0.0004
2891	3990	0.00835	0.01018	-0.0002	2892	3990	0.00831	0.01015	-0.0001	2893	3990	0.00830	0.01012	-0.0005	2894	3978	0.00830	0.01012	-0.0001
2895	3966	0.00824	0.01007	-0.0001	2896	4014	0.00824	0.01006	-0.0005	2897	3990	0.00829	0.01010	-0.0002	2898	3990	0.00824	0.01008	-0.0006
2899	3992	0.00827	0.01010	-0.0001	2900	3988	0.00825	0.01006	-0.0001	2901	3956	0.00824	0.01005	-0.0003	2902	3978	0.00827	0.01009	-0.0001
2903	3932	0.00824	0.01004	-0.0003	2904	3944	0.00825	0.01006	-0.0002	2905	3952	0.00819	0.01000	-0.0001	2906	3968	0.00819	0.01001	-0.0001
2907	3956	0.00823	0.01003	-0.0002	2908	3952	0.00817	0.00997	-0.0001	2909	3950	0.00815	0.00995	-0.0006	2910	3980	0.00818	0.00996	-0.0001
2911	3970	0.00818	0.00995	-0.0004	2912	3970	0.00816	0.00994	-0.0005	2913	3946	0.00817	0.00995	-0.0003	2914	3976	0.00815	0.00994	-0.0001
2915	3920	0.00812	0.00990	-0.0002	2916	4002	0.00816	0.00994	-0.0001	2917	3912	0.00812	0.00992	-0.0012	2918	3976	0.00818	0.00993	-0.0003
2919	3920	0.00812	0.00988	-0.0007	2920	3940	0.00811	0.00987	-0.0008	2921	3968	0.00812	0.00988	-0.0011	2922	3942	0.00811	0.00986	-0.0016
2923	3950	0.00809	0.00982	-0.0013	2924	3936	0.00804	0.00980	-0.0016	2925	3952	0.00808	0.00980	-0.0015	2926	3952	0.00804	0.00974	-0.0017
2927	3946	0.00800	0.00971	-0.0021	2928	3930	0.00799	0.00968	-0.0025	2929	3982	0.00796	0.00968	-0.0030	2930	3946	0.00791	0.00961	-0.0

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
3079	3930	0.00725	0.00882	-0.0068	3080	3918	0.00724	0.00879	-0.0051	3081	3974	0.00724	0.00880	-0.0058	3082	3938	0.00726	0.00882	-0.0070
3083	3936	0.00722	0.00877	-0.0066	3084	3926	0.00724	0.00879	-0.0087	3085	3906	0.00724	0.00880	-0.0083	3086	3954	0.00718	0.00873	-0.0089
3087	3936	0.00714	0.00869	-0.0093	3088	3924	0.00715	0.00871	-0.0096	3089	3916	0.00718	0.00872	-0.0081	3090	3938	0.00716	0.00869	-0.0089
3091	3918	0.00712	0.00869	-0.0072	3092	3904	0.00713	0.00868	-0.0075	3093	3918	0.00710	0.00863	-0.0076	3094	3882	0.00713	0.00866	-0.0107
3095	3892	0.00712	0.00864	-0.0084	3096	3894	0.00710	0.00862	-0.0080	3097	3940	0.00711	0.00863	-0.0106	3098	3940	0.00710	0.00861	-0.0131
3099	3886	0.00705	0.00858	-0.0116	3100	3898	0.00706	0.00858	-0.0123	3101	3886	0.00707	0.00860	-0.0125	3102	3930	0.00703	0.00855	-0.0152
3103	3914	0.00692	0.00842	-0.0148	3104	3904	0.00698	0.00848	-0.0142	3105	3904	0.00698	0.00847	-0.0152	3106	3932	0.00696	0.00847	-0.0138
3107	3874	0.00690	0.00844	-0.0143	3108	3896	0.00695	0.00844	-0.0138	3109	3898	0.00695	0.00842	-0.0133	3110	3896	0.00694	0.00841	-0.0172
3111	3860	0.00695	0.00842	-0.0154	3112	3894	0.00687	0.00833	-0.0119	3113	3898	0.00686	0.00830	-0.0199	3114	3882	0.00689	0.00834	-0.0195
3115	3862	0.00684	0.00830	-0.0223	3116	3840	0.00683	0.00827	-0.0233	3117	3858	0.00681	0.00827	-0.0232	3118	3850	0.00676	0.00821	-0.0271
3119	3878	0.00673	0.00817	-0.0284	3120	3850	0.00670	0.00814	-0.0263	3121	3850	0.00671	0.00813	-0.0276	3122	3842	0.00666	0.00809	-0.0298
3123	3798	0.00669	0.00810	-0.0290	3124	3836	0.00666	0.00808	-0.0290	3125	3840	0.00666	0.00807	-0.0301	3126	3804	0.00667	0.00808	-0.0349
3127	3778	0.00659	0.00800	-0.0350	3128	3788	0.00663	0.00803	-0.0357	3129	3788	0.00658	0.00798	-0.0343	3130	3822	0.00659	0.00797	-0.0411
3131	3790	0.00654	0.00793	-0.0443	3132	3768	0.00652	0.00789	-0.0464	3133	3736	0.00651	0.00787	-0.0457	3134	3738	0.00646	0.00782	-0.0480
3135	3752	0.00647	0.00781	-0.0531	3136	3782	0.00643	0.00780	-0.0514	3137	3728	0.00638	0.00773	-0.0594	3138	3704	0.00640	0.00776	-0.0587
3139	3692	0.00635	0.00769	-0.0619	3140	3684	0.00632	0.00767	-0.0619	3141	3676	0.00632	0.00767	-0.0644	3142	3658	0.00628	0.00763	-0.0713
3143	3630	0.00627	0.00761	-0.0751	3144	3630	0.00626	0.00761	-0.0775	3145	3610	0.00620	0.00754	-0.0879	3146	3610	0.00621	0.00755	-0.0889
3147	3582	0.00617	0.00750	-0.0941	3148	3590	0.00620	0.00752	-0.0936	3149	3688	0.00638	0.00780	-0.0570	3150	3892	0.00689	0.00840	-0.0084
3151	3896	0.00687	0.00838	-0.0106	3152	3956	0.00687	0.00837	-0.0131	3153	3914	0.00684	0.00834	-0.0101	3154	3916	0.00684	0.00832	-0.0116
3155	3932	0.00683	0.00832	-0.0102	3156	3942	0.00683	0.00831	-0.0107	3157	3956	0.00681	0.00830	-0.0110	3158	3916	0.00685	0.00835	-0.0111
3159	3880	0.00681	0.00829	-0.0093	3160	3906	0.00678	0.00827	-0.0101	3161	3898	0.00678	0.00826	-0.0121	3162	3896	0.00673	0.00820	-0.0095
3163	3902	0.00679	0.00827	-0.0095	3164	3866	0.00678	0.00825	-0.0110	3165	3878	0.00679	0.00828	-0.0092	3166	3912	0.00674	0.00820	-0.0109
3167	3868	0.00676	0.00825	-0.0102	3168	3894	0.00675	0.00824	-0.0108	3169	3874	0.00675	0.00821	-0.0111	3170	3912	0.00674	0.00822	-0.0111
3171	3900	0.00671	0.00817	-0.0118	3172	3926	0.00673	0.00820	-0.0107	3173	3940	0.00668	0.00814	-0.0115	3174	3900	0.00677	0.00824	-0.0097
3175	3842	0.00670	0.00814	-0.0119	3176	3860	0.00669	0.00815	-0.0122	3177	3862	0.00669	0.00815	-0.0121	3178	3896	0.00671	0.00815	-0.0118
3179	3842	0.00670	0.00813	-0.0132	3180	3888	0.00673	0.00815	-0.0126	3181	3854	0.00667	0.00810	-0.0119	3182	3870	0.00662	0.00804	-0.0139
3183	3872	0.00664	0.00806	-0.0131	3184	3874	0.00665	0.00797	-0.0165	3185	3856	0.00659	0.00799	-0.0160	3186	3872	0.00656	0.00796	-0.0157
3187	3846	0.00656	0.00797	-0.0167	3188	3898	0.00655	0.00795	-0.0135	3189	3902	0.00655	0.00795	-0.0140	3190	3840	0.00658	0.00798	-0.0142
3191	3834	0.00654	0.00793	-0.0164	3192	3838	0.00651	0.00788	-0.0167	3193	3880	0.00652	0.00789	-0.0172	3194	3836	0.00650	0.00785	-0.0170
3195	3846	0.00654	0.00788	-0.0178	3196	3844	0.00652	0.00786	-0.0163	3197	3844	0.00651	0.00786	-0.0170	3198	3870	0.00651	0.00784	-0.0201
3199	3842	0.00653	0.00787	-0.0202	3200	3840	0.00650	0.00786	-0.0202	3201	3856	0.00650	0.00787	-0.0231	3202	3846	0.00648	0.00777	-0.0240
3203	3850	0.00641	0.00773	-0.0253	3204	3832	0.00639	0.00770	-0.0250	3205	3856	0.00638	0.00768	-0.0245	3206	3838	0.00636	0.00767	-0.0258
3207	3812	0.00638	0.00769	-0.0286	3208	3796	0.00632	0.00761	-0.0290	3209	3808	0.00637	0.00764	-0.0297	3210	3760	0.00632	0.00758	-0.0302
3211	3770	0.00633	0.00759	-0.0309	3212	3770	0.00629	0.00755	-0.0347	3213	3752	0.00630	0.00757	-0.0324	3214	3770	0.00623	0.00749	-0.0374
3215	3742	0.00621	0.00745	-0.0354	3216	3746	0.00621	0.00745	-0.0416	3217	3768	0.00619	0.00744	-0.0410	3218	3720	0.00614	0.00737	-0.0428
3219	3746	0.00611	0.00734	-0.0536	3220	3696	0.00611	0.00734	-0.0531	3221	3784	0.00609	0.00731	-0.0524	3222	3744	0.00610	0.00731	-0.0532
3223	3676	0.00604	0.00727	-0.0630	3224	3700	0.00604	0.00725	-0.0650	3225	3686	0.00601	0.00721	-0.0670	3226	3688	0.00605	0.00725	-0.0753
3227	3642	0.00597	0.00719	-0.0719	3228	3650	0.00597	0.00718	-0.0729	3229	3650	0.00597	0.00718	-0.0696	3230	3638	0.00599	0.00716	-0.0829
3231	3640	0.00593	0.00713	-0.0793	3232	3640	0.00588	0.00708	-0.0928	3233	3664	0.00591	0.00712	-0.0998	3234	3628	0.00589	0.00710	-0.1017
3235	3604	0.00584	0.00704	-0.1029	3236	3610	0.00584	0.00705	-0.1096	3237	3634	0.00579	0.00699	-0.1079	3238	3590	0.00577	0.00695	-0.1125
3239	3552	0.00576	0.00696	-0.1220	3240	3604	0.00574	0.00694	-0.1154	3241	3582	0.00576	0.00694	-0.1259	3242	3528	0.00577	0.00694	-0.1209
3243	3514	0.00569	0.00687	-0.1277	3244	3478	0.00567	0.00684	-0.1404	3245	3508	0.00571	0.00688	-0.1389	3246	3526	0.00567	0.00685	-0.1420
3247	3854	0.00637	0.00773	-0.0157	3248	3886	0.00638	0.00772	-0.0168	3249	3882	0.00635	0.00768	-0.0200	3250	3876	0.00635	0.00770	-0.0170
3251	3860	0.00636	0.00770	-0.0171	3252	3880	0.00636	0.00770	-0.0191	3253	3920	0.00630	0.00763	-0.0164	3254	3910	0.00635	0.00767	-0.0185
3255	3878	0.00636	0.00769	-0.0183	3256	3878	0.00636	0.00769	-0.0183	3257	3856	0.00632	0.00761	-0.0164	3258	3856	0.00632	0.00761	-0.0164
3259	3846	0.00631	0.00761	-0.0192	3260	3840	0.00631	0.00761	-0.0173	3261	3842	0.00632	0.00762	-0.0153	3262	3854	0.00627	0.00756	-0.0180
3263	3822	0.00624	0.00754	-0.0188	3264	3862	0.00625	0.00753	-0.0208	3265	3846	0.00628	0.00755	-0.0213	3266	3860	0.00624	0.00752	-0.0241
3267	3876	0.00626	0.00752	-0.0211	3268	3844	0.00622	0.00748	-0.0217	3269	3844	0.00618	0.00743	-0.0254	3270	3830	0.00619	0.00743	-0.0230
3271	3836	0.00620	0.00744	-0.0216	3272	3858	0.00616	0.00741	-0.0231	3273	3884	0.00618	0.00741	-0.0258	3274	3852	0.00616	0.00739	-0.0268
3275	3814	0.00620	0.00744	-0.0229	3276	3822	0.00614	0.00737	-0.0263	3277	3796	0.00615	0.00739	-0.0255	3278	3846	0.00615	0.00738	-0.0257
3279	3850	0.00615	0.00739	-0.0245	3280	3834	0.00613	0.00737	-0.0275	3281	3836	0.00618	0.00741	-0.0277	3282	3846	0.00611	0.00733	-0.0321
3283	3850	0.00612	0.00734	-0.0284	3284	3828	0.00605	0.00727	-0.0289	3285	3806	0.00603	0.00724	-0.0332	3286	3802	0.00602	0.00721	-0.0363
3287	3810	0.00607	0.00727	-0.0290	3288	3790	0.00607	0.00727	-0.0358	3289	3790	0.00607	0.00727	-0.0369	3290	3842	0.00599	0.00717	-0.0369
3291	3768	0.00598	0.00717	-0.0388	3292	3760	0.00595	0.00713	-0.0398	3293	3766	0.00594	0.00712	-0.0405	3294	3782	0.00594	0.00711	-0.0431
3295	3804	0.00592	0.00710	-0.0418	3296	3764	0.00593	0.00708	-0.0474	3297	3754	0.00590	0.00704	-0.0510	3298	3762	0.00590	0.00705	-0.0544
3299	3754	0.00583	0.00697	-0.0556	3300	3760	0.00586	0.00700	-0.0525	3301	3712	0.00584	0.00697	-0.0615	3302	3728	0.00584	0.00697	-0.0621

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
3451	3786	0.00570	0.00664	-0.1649	3452	3722	0.00567	0.00662	-0.1583	3453	3730	0.00567	0.00661	-0.1645	3454	3750	0.00567	0.00661	-0.1726
3455	3706	0.00567	0.00661	-0.1821	3456	3762	0.00562	0.00656	-0.1877	3457	3716	0.00564	0.00657	-0.1884	3458	3750	0.00564	0.00656	-0.1967
3459	3720	0.00562	0.00655	-0.1956	3460	3704	0.00561	0.00652	-0.1937	3461	3714	0.00562	0.00655	-0.2062	3462	3722	0.00558	0.00650	-0.2042
3463	3746	0.00561	0.00652	-0.2101	3464	3728	0.00565	0.00656	-0.2152	3465	3688	0.00560	0.00651	-0.2120	3466	3704	0.00558	0.00649	-0.2178
3467	3664	0.00560	0.00651	-0.2333	3468	3680	0.00561	0.00653	-0.2223	3469	3658	0.00560	0.00652	-0.2334	3470	3660	0.00560	0.00651	-0.2415
3471	3644	0.00558	0.00650	-0.2445	3472	3660	0.00556	0.00649	-0.2522	3473	3636	0.00556	0.00648	-0.2554	3474	3638	0.00558	0.00650	-0.2640
3475	3642	0.00556	0.00648	-0.2506	3476	3634	0.00555	0.00648	-0.2628	3477	3626	0.00554	0.00647	-0.2700	3478	3630	0.00552	0.00644	-0.2618
3479	3644	0.00557	0.00649	-0.2826	3480	3634	0.00553	0.00645	-0.2708	3481	3606	0.00556	0.00649	-0.2836	3482	3630	0.00554	0.00646	-0.2812
3483	3604	0.00552	0.00644	-0.2931	3484	3562	0.00552	0.00645	-0.2993	3485	3600	0.00555	0.00649	-0.3047	3486	3570	0.00552	0.00644	-0.3074
3487	3584	0.00551	0.00643	-0.3261	3488	3548	0.00550	0.00642	-0.3317	3489	3564	0.00550	0.00643	-0.3322	3490	3572	0.00552	0.00643	-0.3456
3491	3554	0.00548	0.00641	-0.3350	3492	3542	0.00551	0.00644	-0.3478	3493	3566	0.00552	0.00645	-0.3584	3494	3572	0.00550	0.00644	-0.3663
3495	3610	0.00552	0.00645	-0.3741	3496	3570	0.00550	0.00643	-0.3742	3497	3516	0.00550	0.00643	-0.3781	3498	3528	0.00548	0.00640	-0.3803
3499	3528	0.00549	0.00641	-0.3894	3500	3452	0.00549	0.00641	-0.4156	3501	3488	0.00549	0.00642	-0.4285	3502	3466	0.00547	0.00640	-0.4331
3503	3470	0.00547	0.00640	-0.4508	3504	3430	0.00545	0.00639	-0.4488	3505	3432	0.00548	0.00642	-0.4668	3506	3444	0.00544	0.00638	-0.4802
3507	3414	0.00545	0.00639	-0.4802	3508	3458	0.00546	0.00641	-0.5032	3509	3432	0.00545	0.00640	-0.4988	3510	3476	0.00547	0.00643	-0.5076
3511	3456	0.00544	0.00639	-0.4892	3512	3512	0.00544	0.00639	-0.4174	3513	3406	0.00544	0.00640	-0.5222	3514	3408	0.00542	0.00638	-0.5210
3515	3422	0.00543	0.00639	-0.5352	3516	3352	0.00546	0.00642	-0.5497	3517	3368	0.00546	0.00642	-0.5745	3518	3394	0.00544	0.00640	-0.5806
3519	3344	0.00546	0.00642	-0.5856	3520	3348	0.00542	0.00639	-0.5932	3521	3350	0.00543	0.00640	-0.6061	3522	3366	0.00545	0.00641	-0.6278
3523	3310	0.00542	0.00639	-0.6317	3524	3356	0.00542	0.00639	-0.6345	3525	3348	0.00540	0.00638	-0.6546	3526	3398	0.00543	0.00641	-0.6466
3527	3394	0.00541	0.00638	-0.6391	3528	3364	0.00541	0.00638	-0.6395	3529	3358	0.00540	0.00637	-0.6501	3530	3366	0.00540	0.00638	-0.6306
3531	3336	0.00541	0.00638	-0.6632	3532	3310	0.00540	0.00637	-0.6524	3533	3310	0.00540	0.00637	-0.6639	3534	3318	0.00544	0.00640	-0.7121
3535	3330	0.00542	0.00640	-0.7086	3536	3278	0.00540	0.00639	-0.7317	3537	3296	0.00542	0.00640	-0.7418	3538	3286	0.00542	0.00640	-0.7421
3539	3308	0.00543	0.00640	-0.6828	3540	3258	0.00539	0.00639	-0.7523	3541	3296	0.00542	0.00640	-0.7623	3542	3268	0.00542	0.00640	-0.7581
3543	3684	0.00568	0.00661	-0.2697	3544	3678	0.00566	0.00659	-0.2709	3545	3630	0.00566	0.00660	-0.2645	3546	3640	0.00567	0.00661	-0.2651
3547	3648	0.00564	0.00657	-0.2714	3548	3644	0.00571	0.00661	-0.2723	3549	3610	0.00568	0.00660	-0.2757	3550	3596	0.00568	0.00660	-0.2780
3551	3608	0.00568	0.00661	-0.2766	3552	3628	0.00564	0.00657	-0.2934	3553	3594	0.00570	0.00662	-0.2946	3554	3592	0.00566	0.00659	-0.2958
3555	3592	0.00567	0.00660	-0.2958	3556	3614	0.00566	0.00659	-0.3171	3557	3624	0.00565	0.00658	-0.3236	3558	3618	0.00563	0.00657	-0.3300
3559	3630	0.00568	0.00660	-0.3128	3560	3616	0.00564	0.00657	-0.3387	3561	3602	0.00560	0.00653	-0.3313	3562	3610	0.00560	0.00655	-0.3218
3563	3592	0.00563	0.00656	-0.3254	3564	3580	0.00564	0.00657	-0.3224	3565	3542	0.00564	0.00658	-0.3568	3566	3562	0.00562	0.00655	-0.3486
3567	3542	0.00562	0.00653	-0.3489	3568	3556	0.00559	0.00653	-0.3547	3569	3494	0.00558	0.00652	-0.3659	3570	3526	0.00558	0.00652	-0.3536
3571	3536	0.00560	0.00654	-0.3628	3572	3536	0.00560	0.00654	-0.3547	3573	3522	0.00558	0.00652	-0.3725	3574	3532	0.00557	0.00652	-0.3916
3575	3572	0.00559	0.00653	-0.3807	3576	3560	0.00559	0.00653	-0.3914	3577	3538	0.00556	0.00650	-0.4027	3578	3610	0.00560	0.00655	-0.4008
3579	3544	0.00553	0.00648	-0.3914	3580	3510	0.00554	0.00648	-0.3852	3581	3522	0.00558	0.00651	-0.3947	3582	3496	0.00555	0.00648	-0.3974
3583	3510	0.00555	0.00649	-0.4095	3584	3508	0.00555	0.00647	-0.4031	3585	3500	0.00553	0.00646	-0.4253	3586	3506	0.00551	0.00646	-0.4079
3587	3480	0.00555	0.00649	-0.4253	3588	3500	0.00550	0.00645	-0.4288	3589	3500	0.00553	0.00648	-0.4408	3590	3520	0.00550	0.00645	-0.4286
3591	3480	0.00550	0.00645	-0.4410	3592	3502	0.00550	0.00645	-0.4638	3593	3466	0.00548	0.00643	-0.4471	3594	3516	0.00548	0.00644	-0.4742
3595	3468	0.00546	0.00643	-0.4427	3596	3452	0.00545	0.00641	-0.4563	3597	3442	0.00548	0.00643	-0.4681	3598	3424	0.00546	0.00640	-0.4430
3599	3474	0.00549	0.00644	-0.4524	3600	3424	0.00547	0.00640	-0.4626	3601	3406	0.00544	0.00637	-0.4826	3602	3434	0.00544	0.00637	-0.5074
3603	3412	0.00547	0.00643	-0.5181	3604	3434	0.00545	0.00640	-0.5269	3605	3404	0.00544	0.00641	-0.5300	3606	3444	0.00544	0.00640	-0.5365
3607	3462	0.00543	0.00640	-0.5419	3608	3460	0.00541	0.00638	-0.5530	3609	3444	0.00539	0.00634	-0.5473	3610	3470	0.00540	0.00636	-0.5515
3611	3418	0.00539	0.00635	-0.5490	3612	3434	0.00540	0.00636	-0.5498	3613	3410	0.00539	0.00636	-0.5643	3614	3400	0.00542	0.00638	-0.5422
3615	3376	0.00539	0.00636	-0.5832	3616	3396	0.00538	0.00634	-0.5831	3617	3366	0.00538	0.00635	-0.5785	3618	3444	0.00538	0.00635	-0.5783
3619	3338	0.00537	0.00636	-0.5972	3620	3332	0.00538	0.00637	-0.5892	3621	3332	0.00534	0.00633	-0.5831	3622	3314	0.00536	0.00634	-0.5835
3623	3346	0.00535	0.00634	-0.5857	3624	3348	0.00533	0.00631	-0.5689	3625	3366	0.00536	0.00634	-0.5786	3626	3370	0.00535	0.00633	-0.5911
3627	3316	0.00535	0.00634	-0.6029	3628	3338	0.00536	0.00632	-0.7026	3629	3362	0.00537	0.00635	-0.6030	3630	3314	0.00536	0.00632	-0.5908
3631	3302	0.00529	0.00628	-0.5894	3632	3322	0.00529	0.00629	-0.5644	3633	3306	0.00526	0.00625	-0.5784	3634	3284	0.00528	0.00627	-0.5989
3635	3292	0.00525	0.00624	-0.6156	3636	3246	0.00524	0.00625	-0.6064	3637	3534	0.00549	0.00644	-0.2563	3638	3600	0.00549	0.00644	-0.2293
3639	3570	0.00545	0.00641	-0.2319	3640	3586	0.00545	0.00640	-0.2433	3641	3604	0.00546	0.00640	-0.2381	3642	3606	0.00545	0.00639	-0.2330
3643	3590	0.00544	0.00638	-0.2393	3644	3602	0.00544	0.00638	-0.2274	3645	3622	0.00545	0.00638	-0.2323	3646	3596	0.00542	0.00638	-0.2268
3647	3618	0.00544	0.00637	-0.2240	3648	3578	0.00544	0.00637	-0.2177	3649	3584	0.00539	0.00632	-0.2245	3650	3534	0.00541	0.00635	-0.2329
3651	3538	0.00539	0.00633	-0.2268	3652	3576	0.00538	0.00633	-0.2289	3653	3542	0.00538	0.00634	-0.2296	3654	3558	0.00538	0.00633	-0.2369
3655	3530	0.00535	0.00630	-0.2266	3656	3550	0.00535	0.00630	-0.2229	3657	3550	0.00535	0.00630	-0.2191	3658	3590	0.00531	0.00625	-0.2200
3659	3604	0.00533	0.00627	-0.2202	3660	3622	0.00532	0.00627	-0.2179	3661	3602	0.00533	0.00627	-0.2179	3662	3570	0.00533	0.00626	-0.2147
3663	3604	0.00530	0.00625	-0.2092	3664	3580	0.00532	0.00626	-0.2152	3665	3562	0.00529	0.00623	-0.2109	3666	3550	0.00528	0.00623	-0.2131
3667	3516	0.00523	0.00619	-0.2118	3668	3540	0.00525	0.00621	-0.2153	3669	3520	0.00527	0.00624	-0.2190	3670	3512	0.00524	0.00620	-0.2157
3671	3486	0.00518	0.00615	-0.2172	3672	3504	0.00519	0.00615	-0.2150	3673	3518	0.00519	0.00615	-0.2142	3674	3550	0.00515	0.00610	-0.2166

Table 17: Complete 1 m depth-resolved reconstruction-accuracy source table continued.

Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2	Depth	n	MAE	RMSE	R^2
3823	3516	0.00399	0.00503	-0.0406	3824	3492	0.00399	0.00504	-0.0497	3825	3480	0.00400	0.00505	-0.0493	3826	3452	0.00397	0.00501	-0.0505
3827	3492	0.00399	0.00504	-0.0600	3828	3492	0.00400	0.00506	-0.0679	3829	3496	0.00399	0.00505	-0.0642	3830	3442	0.00399	0.00505	-0.0747
3831	3380	0.00387	0.00489	-0.0590	3832	3330	0.00371	0.00467	-0.0404	3833	3312	0.00368	0.00464	-0.0488	3834	3314	0.00369	0.00466	-0.0568
3835	3332	0.00369	0.00466	-0.0604	3836	3350	0.00370	0.00467	-0.0643	3837	3290	0.00374	0.00471	-0.0738	3838	3318	0.00375	0.00474	-0.0736
3839	3320	0.00371	0.00469	-0.0791	3840	3318	0.00372	0.00470	-0.0855	3841	3312	0.00375	0.00474	-0.0885	3842	3318	0.00371	0.00470	-0.0881
3843	3312	0.00373	0.00473	-0.1164	3844	3334	0.00373	0.00471	-0.1039	3845	3288	0.00370	0.00472	-0.1178	3846	3238	0.00376	0.00476	-0.1371
3847	3250	0.00375	0.00476	-0.1359	3848	3256	0.00375	0.00477	-0.1431	3849	3268	0.00378	0.00479	-0.1550	3850	3234	0.00378	0.00480	-0.1565
3851	3284	0.00385	0.00487	-0.1656	3852	3304	0.00382	0.00484	-0.1792	3853	3272	0.00383	0.00487	-0.1883	3854	3254	0.00384	0.00487	-0.1954
3855	3302	0.00382	0.00486	-0.1905	3856	3238	0.00382	0.00487	-0.2127	3857	3280	0.00383	0.00489	-0.2074	3858	3284	0.00389	0.00496	-0.2301
3859	3284	0.00384	0.00491	-0.2553	3860	3282	0.00391	0.00500	-0.2681	3861	3250	0.00390	0.00498	-0.2692	3862	3266	0.00393	0.00502	-0.2915
3863	3284	0.00393	0.00500	-0.2941	3864	3288	0.00397	0.00506	-0.3035	3865	3276	0.00398	0.00507	-0.3163	3866	3270	0.00398	0.00507	-0.3327
3867	3268	0.00403	0.00513	-0.3513	3868	3286	0.00404	0.00514	-0.3657	3869	3298	0.00404	0.00515	-0.3813	3870	3282	0.00409	0.00520	-0.3859
3871	3282	0.00410	0.00522	-0.3998	3872	3314	0.00412	0.00527	-0.4059	3873	3312	0.00414	0.00528	-0.3986	3874	3354	0.00421	0.00535	-0.4429
3875	3310	0.00420	0.00534	-0.4415	3876	3322	0.00423	0.00538	-0.4621	3877	3308	0.00426	0.00541	-0.4872	3878	3310	0.00429	0.00543	-0.5003
3879	3344	0.00429	0.00544	-0.5164	3880	3300	0.00432	0.00547	-0.5216	3881	3310	0.00431	0.00547	-0.5370	3882	3334	0.00438	0.00554	-0.5492
3883	3334	0.00438	0.00553	-0.5789	3884	3308	0.00445	0.00563	-0.6002	3885	3344	0.00446	0.00562	-0.6093	3886	3340	0.00446	0.00563	-0.6315
3887	3360	0.00448	0.00564	-0.6427	3888	3378	0.00452	0.00571	-0.6602	3889	3360	0.00454	0.00572	-0.6676	3890	3418	0.00462	0.00581	-0.7013
3891	3378	0.00462	0.00581	-0.7168	3892	3350	0.00461	0.00581	-0.7238	3893	3358	0.00471	0.00590	-0.7793	3894	3376	0.00469	0.00588	-0.7854
3895	3380	0.00472	0.00591	-0.7882	3896	3350	0.00474	0.00592	-0.8197	3897	3340	0.00477	0.00596	-0.8385	3898	3370	0.00479	0.00597	-0.8506
3899	3334	0.00483	0.00602	-0.8934	3900	3354	0.00488	0.00607	-0.9083	3901	3388	0.00489	0.00606	-0.9488	3902	3386	0.00496	0.00614	-0.9741
3903	3394	0.00500	0.00619	-0.9715	3904	3414	0.00502	0.00621	-1.0172	3905	3416	0.00507	0.00625	-1.0241	3906	3428	0.00506	0.00624	-1.0228
3907	3382	0.00510	0.00629	-1.0614	3908	3398	0.00517	0.00636	-1.0640	3909	3408	0.00518	0.00637	-1.1113	3910	3436	0.00518	0.00636	-1.1386
3911	3406	0.00526	0.00643	-1.1732	3912	3368	0.00528	0.00646	-1.1892	3913	3376	0.00529	0.00646	-1.2242	3914	3384	0.00535	0.00650	-1.2489
3915	3390	0.00538	0.00654	-1.2939	3916	3386	0.00540	0.00656	-1.2953	3917	3406	0.00550	0.00665	-1.3561	3918	3394	0.00553	0.00668	-1.3761
3919	3456	0.00563	0.00679	-1.4298	3920	3430	0.00563	0.00679	-1.3920	3921	3440	0.00564	0.00679	-1.4425	3922	3466	0.00567	0.00682	-1.4844
3923	3458	0.00571	0.00687	-1.5016	3924	3438	0.00577	0.00693	-1.5264	3925	3422	0.00578	0.00693	-1.5884	3926	3452	0.00582	0.00695	-1.5833
3927	3442	0.00588	0.00702	-1.6287	3928	3436	0.00593	0.00706	-1.6923	3929	3164	0.00547	0.00647	-1.6950	3930	3202	0.00553	0.00652	-1.7580
3931	3192	0.00555	0.00654	-1.7840	3932	3188	0.00564	0.00663	-1.8145	3933	3214	0.00572	0.00672	-1.8660	3934	3242	0.00580	0.00679	-1.9345
3935	3226	0.00580	0.00681	-1.9200	3936	3238	0.00588	0.00688	-2.0139	3937	3234	0.00591	0.00691	-2.0316	3938	3316	0.00593	0.00694	-2.0348
3939	3294	0.00605	0.00705	-2.0758	3940	3332	0.00598	0.00698	-2.0184	3941	3314	0.00609	0.00707	-2.1735	3942	3280	0.00616	0.00714	-2.2527
3943	3288	0.00618	0.00716	-2.2655	3944	3256	0.00625	0.00723	-2.2862	3945	3266	0.00633	0.00730	-2.3956	3946	3246	0.00638	0.00735	-2.4302
3947	3286	0.00642	0.00739	-2.4657	3948	3264	0.00648	0.00745	-2.5329	3949	3282	0.00656	0.00751	-2.5937	3950	3306	0.00664	0.00759	-2.6151
3951	3308	0.00669	0.00764	-2.6938	3952	3328	0.00676	0.00772	-2.7180	3953	3328	0.00677	0.00771	-2.7019	3954	3396	0.00688	0.00783	-2.8420
3955	3392	0.00690	0.00786	-2.8082	3956	3424	0.00685	0.00781	-2.7706	3957	3386	0.00699	0.00791	-2.9725	3958	3376	0.00706	0.00799	-3.0889
3959	3358	0.00708	0.00800	-3.0832	3960	3330	0.00713	0.00804	-3.2030	3961	3334	0.00722	0.00813	-3.2562	3962	3336	0.00726	0.00815	-3.2802
3963	3308	0.00735	0.00826	-3.3237	3964	3346	0.00741	0.00830	-3.4047	3965	3326	0.00746	0.00835	-3.5466	3966	3350	0.00752	0.00840	-3.6282
3967	3336	0.00759	0.00847	-3.6983	3968	3316	0.00765	0.00852	-3.7917	3969	3376	0.00774	0.00862	-3.8251	3970	3404	0.00777	0.00864	-3.8695
3971	3422	0.00782	0.00869	-3.8963	3972	3458	0.00788	0.00875	-4.0164	3973	3448	0.00789	0.00875	-4.0716	3974	3404	0.00796	0.00881	-4.1504
3975	3402	0.00797	0.00882	-4.1649	3976	3408	0.00808	0.00893	-4.2386	3977	3394	0.00815	0.00897	-4.4843	3978	3388	0.00822	0.00904	-4.4064
3979	3376	0.00829	0.00912	-4.5059	3980	3420	0.00828	0.00910	-4.5424	3981	3700	0.00823	0.00909	-4.2695	3982	4510	0.00806	0.00893	-4.2152
3983	4576	0.00857	0.00940	-4.7771	3984	3890	0.00899	0.00973	-5.7681	3985	2298	0.00899	0.00960	-6.9984	3986	1436	0.00935	0.00979	-9.8341
3987	1118	0.00935	0.00971	-11.8056	3988	992	0.00947	0.00976	-14.9454	3989	978	0.00954	0.00983	-15.1631	3990	974	0.00956	0.00986	-14.8058
3991	966	0.00963	0.00992	-15.2465	3992	966	0.00973	0.01001	-15.9435	3993	978	0.00974	0.01002	-15.0141	3994	990	0.00985	0.01012	-16.8331
3995	994	0.00992	0.01020	-16.9396	3996	996	0.00999	0.01027	-17.1351	3997	994	0.01000	0.01027	-16.9016	3998	996	0.01010	0.01036	-17.7368
3999	1034	0.01016	0.01042	-18.1319	4000	1000	0.01021	0.01047	-18.3876	4001	1012	0.01029	0.01056	-18.2809	4002	1010	0.01031	0.01057	-18.6027
4003	988	0.01036	0.01063	-18.5117	4004	974	0.01049	0.01074	-20.0996	4005	976	0.01054	0.01079	-20.2738	4006	980	0.01058	0.01083	-19.9951
4007	960	0.01067	0.01092	-21.1046	4008	968	0.01071	0.01096	-20.2313	4009	980	0.01080	0.01104	-21.9286	4010	984	0.01086	0.01109	-22.3800
4011	972	0.01090	0.01113	-22.6543	4012	988	0.01098	0.01121	-22.6636	4013	1010	0.01102	0.01125	-23.2020	4014	990	0.01113	0.01135	-23.9191
4015	1028	0.01122	0.01144	-24.9874	4016	1014	0.01125	0.01146	-25.3234	4017	1032	0.01134	0.01155	-25.5518	4018	1008	0.01131	0.01154	-23.9429
4019	994	0.01138	0.01160	-24.4282	4020	980	0.01148	0.01170	-25.6523	4021	972	0.01155	0.01176	-26.6413	4022	970	0.01161	0.01182	-27.4390
4023	968	0.01168	0.01189	-27.6917	4024	978	0.01175	0.01196	-27.5078	4025	978	0.01182	0.01202	-28.8963					

7 Observation-Density Diagnostics

This section appends the internal observation-density diagnostic used to evaluate whether the density of available Station Papa profile records can explain changes in reconstruction accuracy. The term “array density” is treated here as an internal shorthand only; for manuscript reporting, the more accurate wording is observation density, sampling density or gridded depth-bin density, because the Station Papa record is based on repeated moving-profiler casts rather than a static multi-depth sensor array.

Table 18: Observation-density bins and reconstruction-error statistics for the Station Papa diagnostic. Density is summarized by the daily number of 1 m depth bins and the corresponding effective vertical spacing. Stable days are defined by the project threshold $\text{MAE} \leq 0.10^\circ\text{C}$ and $\text{RMSE} \leq 0.20^\circ\text{C}$.

Density bin	Days	Median 1 m bins	Bin-count range	Effective spacing (m)	MAE	RMSE	Stable days
Low density	293	3472	9–3559	1.077	0.0451	0.1117	92.5%
Medium density	292	3564	3559–3639	1.073	0.0701	0.1803	71.9%
High density	293	3705	3639–3716	1.031	0.0287	0.0583	100.0%

The density-binned result indicates that high-density days have the lowest reconstruction error, with MAE decreasing to 0.0287°C and RMSE decreasing to 0.0583°C . However, the relationship is not monotonic: the medium-density bin has higher error than the low-density bin despite containing more 1 m depth bins and a slightly smaller effective vertical spacing. This reversal shows that observation density is not an isolated control on Ocean-TP accuracy. Year-to-year variability, seasonal stratification, depth coverage, platform status and real ocean thermal anomalies can all change simultaneously with sampling density and may dominate the error response.

Therefore, this diagnostic should be used as supporting context rather than as a main mechanistic claim. It is reasonable to state that denser observations increase the information content available to the reconstruction problem, but the present binning analysis supports correlation only and does not by itself prove that increasing observation density necessarily reduces reconstruction error. For the manuscript, the safer interpretation is to keep the main discussion focused on sparse full-depth T–P reconstruction, vertical coherence and cross-scale stability, while treating observation density as a data-quality and error-interpretation factor.

7.1 External Model and Data-Density Context

Table 19 records a working comparison of representative numerical and AI-based ocean prediction or state-estimation systems. The purpose of the table is contextual rather than competitive. The listed systems differ in task definition, variable set, depth range, horizontal grid, forecast lead time, validation protocol and reported metric; therefore the entries should not be interpreted as a direct ranking against Ocean-TP.

Table 19: Representative ocean prediction, reanalysis and AI forecasting systems used as observation-density and grid-density context.

Model/system	Type	Data or grid density	Predicted variables	Forecast scale	Publicly reported accuracy or validation metric	Source
Copernicus GLO12 / PSY4	Numerical forecast	1/12°, 50 levels, 0–5500 m	Temperature, salinity, velocity, SSH, MLD, sea ice	10 d	QUID and Class4 validation, including RMSE-type diagnostics; no single global aggregate RMSE is reported as a universal score.	Copernicus Global Ocean Physics Analysis and Forecast
GLORYS12	Reanalysis / AI training data	1/12°, 50 vertical levels, daily output	Temperature, salinity, velocity, SSH, sea ice	1993 onward reanalysis	Commonly used as a quasi-reference field for AI training and validation rather than as a forecast model with a single headline metric.	Copernicus Global Ocean Physics Reanalysis
HYCOM / GOF3.1 / RTOFS	Numerical forecast	1/12°, 41 hybrid layers	Temperature, salinity, SSH, U/V velocity, sea ice	7–8 d	Class4 and RMSE-type validation are commonly reported; official product pages do not provide one universal global error number.	HYCOM GOF3.1; NOAA RTOFS
GIOPS	Numerical ice-ocean forecast	1/4°, 50 levels	Temperature, salinity, SSH, velocity, sea ice	10 d	Ocean and sea-ice Class4-style validation.	Global Ice Ocean Prediction System
FOAM	Numerical forecast	1/12°, 75 levels	Temperature, salinity, SSH, velocity, sea ice	7 d	The newer FOAM configuration reports about 40% improvement in SLA and subsurface-temperature observation-minus-background RMSD.	Geoscientific Model Development, 2025
OceanMAPS	Numerical forecast	0.1°, near-global 75°S–75°N domain	Temperature, salinity, SSH, velocity	7 d	Regional validation and product-level error estimates are reported rather than a single global aggregate metric.	CSIRO BlueLink OceanMAPS
Copernicus BGC	Biogeochemical forecast	1/4°, 50 levels	Chl, O ₂ , NO ₃ , PO ₄ , pH, pCO ₂ and related variables	10 d	Variable-level quality assessment is provided in the QUID.	Copernicus Global Ocean BGC product
Copernicus Waves	Wave forecast	1/12°, 3 h output	Significant wave height, wave period, direction, Stokes drift	10 d	Validation against altimeters and buoys with RMSE-type diagnostics.	Copernicus Global Ocean Waves product
XiHe	AI global ocean forecast	GLORYS12, 25 years, 1/12°, 23 levels, 96 variables	SST, SSH, temperature, salinity, U/V velocity, wind	60 d	IV-TT and Class4-style evaluation; reported as outperforming PSY4, GIOPS, OceanMAPS and FOAM across tested variables, with 60 d velocity forecasts still exceeding PSY4 10 d performance in the reported comparison.	XiHe arXiv preprint
WenHai	AI upper-ocean forecast	GLORYS12, 1/12°, 23 levels, 0–643 m	Temperature, salinity, SSH, U/V velocity, SST, SLA	1–10 d	At 10 d, reported RMSE reductions relative to GLO12 are 8.94% for SST, 10.67% for SLA, 6.02% for temperature profiles and 5.64% for salinity profiles.	Nature Communications, WenHai
LangYa	AI cross-scale ocean forecast	GLORYS12, 27 years, 1/12°, daily fields	Temperature, salinity, U/V velocity	1–7 d	Reported temperature RMSE is below 0.3°C in most regions, salinity RMSE is below 0.05 psu in most regions, and thermocline-region skill improves by about 50%.	LangYa arXiv preprint
FuXi-Ocean	AI 6 h ocean forecast	HYCOM 1/12°, 6 h, 40 levels; model uses 20 levels to 1500 m	Temperature, salinity, U/V velocity, SSH	6 h–10 d	Latitude-weighted RMSE is the main metric; 6 h tropical/subtropical temperature RMSE is often below 0.3°C, and SST is reported to outperform HYCOM, BLK and FOAM baselines.	FuXi-Ocean arXiv preprint
AI-GOMS	AI foundation-style ocean model	HYCOM 1/12°, 41 levels for training, downsampled to 1/4°, 15 levels	Temperature, salinity, U/V velocity, SSH; extended SWH and Chl tasks	30 d	ACC and RMSE are reported; in the Kuroshio region, 7 d velocity and SSH ACC exceed 0.6 in the reported evaluation.	AI-GOMS arXiv preprint
GLONET	Mercator AI forecast	GLORYS12; reported as global with 21 vertical levels	Temperature, salinity, velocity	10 d	RMSE, SSH and multivariable composite diagnostics are reported; preliminary results are described as promising.	Mercator GLONET; arXiv preprint
AxiomOcean	AI three-dimensional upper-ocean forecast	Global 1/12°, upper ocean to 643 m	Temperature, salinity, three-dimensional velocity	10 d	Day-1 RMSE is reported to decrease by about 20–35%, with higher ACC relative to baselines.	AxiomOcean arXiv preprint
EAM	AI global SST forecast	OISST v2.1, 0.25°, daily fields	SST	1–15 d and 1–12 month scales	Daily RMSD is reported as 0.212–0.932°C; monthly RMSD is reported as 0.666–0.782°C.	Expert Systems with Applications; OISST product
Taylor-ResNet	AI ENSO-index forecast	OISST 0.25° plus NCEP/NCAR 2.5° variables	Niño3.4 ENSO index	6 month lead	Reported COEF is 0.82 and nRMSE is 0.58–0.59.	ScienceDirect paper

Table 20 summarizes major observation or gridded data sources that frequently support ocean prediction, reconstruction, reanalysis and AI forecasting studies. These data sources provide useful context for the density discussion, but they remain different from the local Station Papa full-depth profile-reconstruction task considered by Ocean-TP.

Table 20: Common observation and gridded data sources used in ocean prediction and reconstruction studies.

Data source	Array or grid density	Variables	Common role
Argo	About 4000 floats globally; approximately one float per 3° box; nominal 10 d profiling cycle; Core Argo generally reaches 2000 m.	Temperature, salinity and pressure; Deep Argo extends toward 6000 m; BGC-Argo adds biogeochemical variables.	Core in situ profile source for assimilation and validation in many numerical forecast and reanalysis systems.
OISST v2.1	0.25° , daily, based on satellite, ship and buoy observations.	SST	Common training and validation product for SST-focused AI models.
GLORYS12	$1/12^\circ$, 50 levels, daily output.	Temperature, salinity, SSH, velocity and sea ice.	Core training data for XiHe, WenHai, LangYa, GLONET and related AI ocean-forecasting models.
HYCOM reanalysis	$1/12^\circ$, 3 h output, about 40–41 vertical layers.	Temperature, salinity, U/V velocity and SSH.	Training or testing data source for FuXi-Ocean, AI-GOMS and related AI ocean models.

Together, Tables 19 and 20 show that observation and grid density define the information available to a reconstruction or forecasting system, but the reported accuracy is also controlled by model class, lead time, validation target, vertical range and variable type. For this reason, external systems provide useful context for the observation-density discussion, whereas Ocean-TP should be positioned as a sparse, full-depth vertical T–P reconstruction framework rather than as a direct replacement for global three-dimensional ocean forecast systems.